

ISC CLASS XII · CHEMISTRY PAPER 1 (THEORY)

First Term Complete Revision

All seven units — concepts, proofs and every type of problem

7 units

105 concepts

37 proofs & mechanisms

107 problem types

7 recall sheets

What this is. One document covering everything examinable in the First Term. Each unit runs **concepts → proofs → problem types → recall sheet**, the same shape throughout, so once you know your way round one unit you know your way round all seven.

Not every question is answered — every *kind* of question is. The 107 problem types each carry a recognition cue, a method, and one fully worked example. If you can do those, you can do anything the paper asks.

Chemical Kinetics and Alcohols, Phenols and Ethers are reused unchanged from *Chemistry-Concepts-Proofs-Problem-Types.pdf* — same text, same numbering within the part — so there is nothing new to learn twice.

What the First Term covers

Seven of the ten ISC units. The scope below is taken from the Heritage School First Term paper for 2024–25, which is the only direct evidence of where the year is divided.

Part	ISC unit	Concepts	Proofs	Types	Where it appeared in the First Term paper
I	Unit 1 · Solutions	14	5	14	Q1A(i), Q1B(ii), Q13, Q16
II	Unit 3 · Chemical Kinetics REUSED	18	6	22	Q1C, Q2, Q5, Q10, Q18, Q21
III	Unit 5 · Coordination Compounds	15	4	14	Q1A(ii)(iv), Q1B(v)(vii), Q7, Q15, Q20
IV	Unit 6 · Haloalkanes and Haloarenes	13	5	13	Q1B(iii)(vi), Q12(i)
V	Unit 7 · Alcohols, Phenols and Ethers REUSED	15	8	17	Q3, Q6(ii), Q9, Q19
VI	Unit 8 · Aldehydes, Ketones and Carboxylic Acids	17	6	16	Q1A(iii), Q1B(i), Q4, Q6(i), Q8, Q11, Q12, Q14
VII	Unit 10 · Biomolecules	13	3	11	Q1B(iv), Q17
Total		105	37	107	

Not in the First Term

Three units appear nowhere in the First Term paper and are left out of this document:

Unit 2 Electrochemistry

Unit 4 d- and f-Block Elements

Unit 9 Organic Compounds containing Nitrogen

If your syllabus divides the year differently, treat this list as the thing to check first.

How the marks fall

The First Term paper follows the board format exactly: **70 marks** in four sections — Section A 14 marks of objective, B ten questions of 2, C seven of 3, D three of 5.

By the ISC weightage, Physical Chemistry (units 1 and 3 here) carries the numericals, Organic (6, 7, 8, 10) the largest share of marks, and Inorganic (5) the most compact and most scoring material of the three.

How to use this document

Four passes, and one thing to know about the numbering.

PASS 1 · UNDERSTAND

Read the concepts

Section 1 of each part, in order. They are sequenced so that each concept uses only what came before. Do not memorise yet. The *Hold on to this* boxes are the one sentence each section exists to deliver.

PASS 2 · DERIVE

Work the proofs

Cover each one and rebuild it on paper. There are only a few per unit, and between them they are worth a guaranteed handful of marks in every paper.

PASS 3 · DRILL

Work the problem types

Read the cue and the method, cover the worked example, do it yourself. Any question in a past paper that you cannot start is a *type* you have not yet recognised — come back and find it here.

PASS 4 · RECALL

Test from the recall sheets

Seven pages, one per unit, and the master sheet at the end. Cover a column and reconstruct it. On the morning of the exam, read only those.

Reference numbers are local to each part

Every part numbers its own concepts **C1**, **C2** ..., its proofs and mechanisms **P1** / **M1** / **D1** ... and its problem types **T1**, **T2** ... So **C5** means something different in Part I and in Part V. A cross-reference without a part name always means *within this part*; where a reference crosses parts, the part is named — as in “esterification (Part V, M2)”.

Part I · Solutions

ISC unit 1 · Concentration measures · Henry's law · Raoult's law and deviations · azeotropes · the four colligative properties · abnormal molar mass · the van't Hoff factor. Fourteen concepts, five proofs, fourteen problem types.

1 · The concepts, in the order they build

C1 – C14

C1 Concentration: which measures depend on temperature

Measure	Definition	Temperature dependent?
Molarity, M	moles of solute per litre of solution	yes — volume expands on heating
Molality, m	moles of solute per kilogram of solvent	no — masses do not change
Mole fraction, x	moles of one component ÷ total moles	no
Normality, N	gram equivalents of solute per litre of solution	yes
% w/w	mass of solute per 100 g of solution	no
ppm	parts of solute per 10 ⁶ parts of solution	no

Hold on to this · Whenever a question mentions a **change of temperature**, work in **molality** or mole fraction, never molarity. This is also the answer to the standard one-mark question “what is the effect of temperature on molality and molarity?” — molality is unaffected, molarity decreases as the solution expands.

C2 Henry's law — solubility of a gas in a liquid

$p = K_H \cdot x$ where p is the partial pressure of the gas and x its mole fraction in solution

- K_H is the **Henry's law constant**. It has the unit of pressure.
- **A larger K_H means a lower solubility** — note the inverse sense, which is the commonest confusion here.
- K_H **increases with temperature**, so gases become **less** soluble as the liquid is warmed (dissolution of a gas is exothermic).

Application	Explanation
Soft drinks are bottled under high CO ₂ pressure	high p forces more CO ₂ into solution
Deep-sea divers suffer bends	N ₂ dissolves in blood at depth and bubbles out on rapid ascent; divers use air diluted with helium, which is less soluble
Climbers suffer anoxia at altitude	low partial pressure of O ₂ lowers its concentration in blood
Aquatic life is more comfortable in cold water	O ₂ is more soluble at lower temperature

C3 Raoult's law — both components volatile

For a solution of two volatile liquids A and B, each contributes in proportion to its mole fraction:

$$p_A = p_A^0 x_A \quad p_B = p_B^0 x_B \quad p_{\text{total}} = p_A^0 x_A + p_B^0 x_B$$

The composition of the **vapour** is *not* the composition of the liquid. By Dalton's law the mole fraction of A in the vapour is

$$y_A = \frac{p_A}{p_{\text{total}}}$$

Hold on to this · The vapour is always **richer in the more volatile component** — the one with the higher p^0 . That single sentence is the whole basis of fractional distillation.

C4 Raoult's law — non-volatile solute, and relative lowering

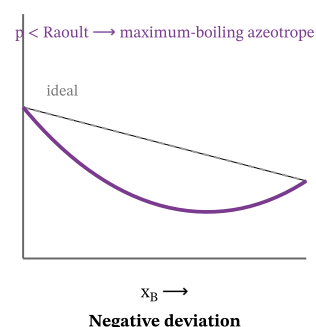
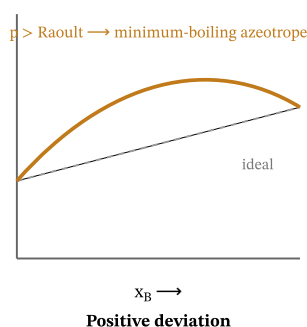
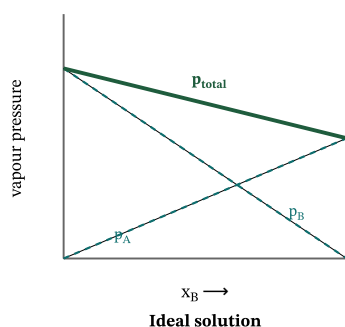
If the solute is non-volatile it contributes nothing to the vapour, so the vapour pressure of the solution is simply that of the solvent, reduced in proportion to its mole fraction:

$$p = p^0 x_1 \Rightarrow \frac{p^0 - p}{p^0} = x_2$$

The quantity $(p^0 - p)/p^0$ is the **relative lowering of vapour pressure**, and it equals the **mole fraction of the solute** — this is Raoult's second law, and it is a colligative property. See proof **P1**.

C5 Ideal and non-ideal solutions

	Ideal	Positive deviation	Negative deviation
Raoult's law	obeyed at all compositions	$p > \text{predicted}$	$p < \text{predicted}$
A–B interaction	same as A–A and B–B	weaker than A–A / B–B	stronger than A–A / B–B
$\Delta_{\text{mix}}H$	0	positive (endothermic)	negative (exothermic)
$\Delta_{\text{mix}}V$	0	positive (expands)	negative (contracts)
Azeotrope	none	minimum boiling	maximum boiling
Example	benzene + toluene; n-hexane + n-heptane	ethanol + water	chloroform + acetone



Hold on to this · Reason it out from the **escaping tendency**. If A and B attract each other *less* than they attract themselves, each escapes more easily than Raoult predicts, so p is higher — positive deviation. Chloroform and acetone form a **hydrogen bond** between them, hold each other back, and give negative deviation.

C6 Azeotropes, and why fractional distillation fails on them

An **azeotrope** is a binary mixture of a definite composition that boils at a constant temperature and **distils over unchanged** — the vapour has the same composition as the liquid, so distillation cannot separate it further.

- **Minimum-boiling azeotrope** — from a solution showing **positive** deviation. Example: ethanol + water, 95% ethanol, boiling at 351.1 K. This is why absolute alcohol cannot be obtained by distillation alone.
- **Maximum-boiling azeotrope** — from a solution showing **negative** deviation. Example: nitric acid + water, 68% HNO₃, boiling at 393.5 K.

C7 The four colligative properties

Colligative properties depend only on the **number of solute particles** present, and not at all on their nature.

Property	Expression	Constant
Relative lowering of vapour pressure	$(p^0 - p)/p^0 = x_2$	—
Elevation of boiling point	$\Delta T_b = K_b m$	K_b , molal elevation constant (ebullioscopic)
Depression of freezing point	$\Delta T_f = K_f m$	K_f , molal depression constant (cryoscopic)
Osmotic pressure	$\pi = CRT$	R , the gas constant

Memory hook · All four are **proportional to the number of particles**. So for the *same* molar concentration, a solute that splits into more ions shows a larger effect: glucose (1 particle) < NaCl (2) < CaCl₂ (3) < K₃[Fe(CN)₆] (4). Every “arrange these solutions” question is this one idea.

C8 The three working formulae for molar mass

Learn these three; they are the whole numerical content of the unit

$$\text{Vapour pressure: } M_2 = \frac{w_2 M_1 p^0}{(p^0 - p) w_1}$$

$$\text{Boiling / freezing point: } M_2 = \frac{1000 K_b w_2}{\Delta T_b w_1} \quad M_2 = \frac{1000 K_f w_2}{\Delta T_f w_1}$$

$$\text{Osmotic pressure: } M_2 = \frac{w_2 R T}{\pi V}$$

w_1 = mass of **solvent** in g, w_2 = mass of **solute** in g, V in litres.

Common slip · The 1000 in the middle pair exists only to convert w_1 from grams to kilograms, because molality is per **kilogram**. Dropping it is the single commonest arithmetic error in this unit — and it makes the answer wrong by a factor of a thousand, so it is easy to spot if you sanity-check the size of your answer.

C9 Osmosis and osmotic pressure

- **Osmosis** — the flow of *solvent* through a semipermeable membrane from the dilute to the concentrated side.
- **Osmotic pressure π** — the excess pressure that must be applied to the solution to *stop* osmosis. $\pi = CRT$, with C in mol L^{-1} and T in kelvin.
- **Isotonic** solutions have equal osmotic pressure; a **hypertonic** solution is more concentrated than the reference and a **hypotonic** one less concentrated.
- **Reverse osmosis** — applying a pressure greater than π drives solvent the other way; this is how sea water is desalinated.

Hold on to this · Osmotic pressure is the colligative property of choice for **macromolecules** such as proteins and polymers: their molar masses are so large that ΔT_f and ΔT_b are too small to measure, but π is measurable at room temperature.

C10 Abnormal molar mass

The molar mass calculated from a colligative property comes out wrong whenever the solute does not stay as the simple molecule you assumed.

Behaviour in solution	Number of particles	Observed colligative property	Observed molar mass
Dissociation (NaCl, KCl, K_2SO_4 in water)	more than expected	higher than calculated	lower than the true value
Association (benzoic acid or phenol in benzene, acetic acid in benzene)	fewer than expected	lower than calculated	higher than the true value

Memory hook · More particles $\Rightarrow$ bigger effect $\Rightarrow$ smaller apparent molar mass. The colligative property counts particles, so if a solute breaks into two, the experiment “sees” twice as many and concludes each must be half as heavy.

C11 The van't Hoff factor

$$i = \frac{\text{observed (experimental) colligative property}}{\text{calculated (theoretical) colligative property}} = \frac{\text{normal (theoretical) molar mass}}{\text{observed molar mass}}$$

$$\text{also } i = \frac{\text{actual number of particles in solution}}{\text{number of particles dissolved}}$$

Situation	Value of i
No association or dissociation (urea, glucose, sucrose in water)	$i = 1$
Dissociation	$i > 1$ (up to n , the number of ions)
Association	$i < 1$ (down to $1/n$)

Every colligative formula is corrected simply by inserting i :

$$\Delta T_b = i K_b m \quad \Delta T_f = i K_f m \quad \pi = i C R T \quad (p^0 - p)/p^0 = i x_2$$

C12 Degree of dissociation and degree of association from i

These two relations are derived in **P5**; both are worth memorising because they turn a three-mark question into two lines.

Dissociation — one particle gives n particles

$$i = 1 + (n - 1)\alpha \Rightarrow \alpha = \frac{i - 1}{n - 1}$$

Association — n particles give one particle

$$i = 1 - \alpha(1 - 1/n) \Rightarrow \alpha = \frac{n(1 - i)}{n - 1}$$

For the common case of **dimerisation** ($n = 2$), the association formula collapses to the very quotable $\alpha = 2(1 - i)$.

C13 Why an electrolyte's i falls short of the ideal value

KCl should give $i = 2$, yet measured values are nearer 1.9. The shortfall is **interionic attraction**: at any instant some K^+ and Cl^- ions are close enough to behave as a single particle (an ion pair), so the solution contains slightly fewer independent particles than complete dissociation would give. The effect shrinks on dilution, so i approaches 2 as the solution becomes very dilute.

C14 Solid solutions and the shape of the answer

- A **solid solution** is a homogeneous mixture of solids — an alloy such as brass (Cu in Zn) or bronze.
- Solutions are classified by the state of solute and solvent: gas in liquid (CO_2 in water), liquid in liquid (ethanol in water), solid in liquid (salt in water), solid in solid (alloys), gas in solid (H_2 in palladium).

Worth one mark, occasionally, in the fill-in-the-blanks.

2 · The proofs

P1 – P5

P1 Relative lowering of vapour pressure gives the molar mass

1. For a non-volatile solute, $p = p^0 x_1$. Since $x_1 + x_2 = 1$:

$$p = p^0(1 - x_2) \Rightarrow \frac{p^0 - p}{p^0} = x_2$$

2. Write x_2 in terms of the masses, with $n_1 = w_1/M_1$ and $n_2 = w_2/M_2$:

$$x_2 = \frac{n_2}{n_1 + n_2} \approx \frac{n_2}{n_1} \text{ for a dilute solution, where } n_2 \ll n_1$$

$$\frac{p^0 - p}{p^0} = \frac{w_2 M_1}{M_2 w_1} \Rightarrow M_2 = \frac{w_2 M_1 p^0}{(p^0 - p) w_1}$$

The approximation in step 2 is what makes this the dilute solution form. ISC expects it; if a question is not dilute, use the exact x_2 .

■ proved

P2 Elevation of boiling point, and the molar mass formula

1. Adding a non-volatile solute lowers the vapour pressure, so the solution must be heated **further** before its vapour pressure reaches atmospheric. Hence the boiling point rises.
2. For a dilute solution the elevation is proportional to the lowering of vapour pressure, and hence to the mole fraction of solute, and hence to the molality:

$$\Delta T_b \propto m \Rightarrow \Delta T_b = K_b m$$

3. Substituting $m = (w_2/M_2) \div (w_1/1000)$:

$$\Delta T_b = \frac{1000 K_b w_2}{M_2 w_1} \Rightarrow M_2 = \frac{1000 K_b w_2}{\Delta T_b w_1}$$

K_b is the elevation produced by a **1 molal** solution; its unit is $K \text{ kg mol}^{-1}$. For water $K_b = 0.52$.

■ proved

P3 Depression of freezing point, and the molar mass formula

1. A liquid freezes when its vapour pressure equals that of the solid. The solute lowers the solution's vapour pressure, so that equality is reached at a **lower** temperature.
2. By the same dilute-solution argument as P2:

$$\Delta T_f = K_f m \Rightarrow \Delta T_f = \frac{1000 K_f w_2}{M_2 w_1}$$

$$M_2 = \frac{1000 K_f w_2}{\Delta T_f w_1}$$

For water $K_f = 1.86 K \text{ kg mol}^{-1}$; for benzene 5.12 . This is why salt or ethylene glycol is used to stop water freezing.

■ proved

P4 Osmotic pressure and the molar mass of a macromolecule

1. van't Hoff showed that a dilute solution obeys $\pi V = n_2 R T$, the same form as the ideal gas equation. Hence

$$\pi = \frac{n_2}{V} R T = C R T$$

2. Putting $n_2 = w_2/M_2$:

$$\pi = \frac{w_2 R T}{M_2 V} \Rightarrow M_2 = \frac{w_2 R T}{\pi V}$$

Use $R = 0.0821 L \text{ atm } K^{-1} \text{ mol}^{-1}$ when π is in atm and V in litres. Two solutions are **isotonic** when their π , and therefore their molar concentrations C , are equal.

■ proved

P5 The van't Hoff factor in terms of the degree of dissociation and of association

Dissociation. Let one mole of a solute give n particles, and let α be the fraction that dissociates.

	undissociated	particles produced	total
at equilibrium	$1 - \alpha$	$n\alpha$	$1 - \alpha + n\alpha$

$$i = \frac{1 - \alpha + n\alpha}{1} = 1 + (n - 1)\alpha \Rightarrow \alpha = \frac{i - 1}{n - 1}$$

Association. Now let n moles combine into one, with α the fraction associating.

	free molecules	associated particles	total
at equilibrium	$1 - \alpha$	α/n	$1 - \alpha + \alpha/n$

$$i = 1 - \alpha + \frac{\alpha}{n} = 1 - \alpha(1 - 1/n) \Rightarrow \alpha = \frac{n(1 - i)}{n - 1}$$

For dimerisation, $n = 2$, and this reduces to $\alpha = 2(1 - i)$.

■ proved

3 · Every type of problem that is set

T1 – T14

T1 Interconverting concentration units

RECOGNISE IT BY

“A solution is x molal / $x\%$ by mass; find its mole fraction or molarity.”

METHOD

Fix a convenient basis. For **molality**, take exactly 1 kg = 1000 g of solvent, so the moles of solvent are $1000/M_1$ and the moles of solute are the molality itself. Then read off whatever is asked.

WORKED EXAMPLE

A solution is 2.5 molal in a solute of molar mass 60 g mol^{-1} , the solvent being water. Find the mole fraction of the solute.

1. basis: 1000 g water = $1000/18 = 55.56 \text{ mol}$
2. solute = 2.5 mol (that is what 2.5 molal means)

$$x_2 = \frac{2.5}{2.5 + 55.56} = \frac{2.5}{58.06} = \mathbf{0.0431}$$

T2 Henry's law

RECOGNISE IT BY

A value of K_H and a pressure are given; the solubility (as a mole fraction) is wanted, or the reverse.

METHOD

Use $p = K_H x$ directly, so $x = p/K_H$. Check that the pressure units of p and K_H match before dividing — mixing mm Hg with atm is the usual slip.

WORKED EXAMPLE

Henry's law constant for methane in benzene at 298 K is 4.27×10^5 mm Hg. Calculate the solubility of methane in benzene at 298 K under 760 mm Hg. (Heritage 2024–25)

$$x = \frac{p}{K_H} = \frac{760}{4.27 \times 10^5} = 1.78 \times 10^{-3}$$

A dimensionless mole fraction is the expected answer; there is no need to convert it further unless asked.

T3 Raoult's law for two volatile liquids

RECOGNISE IT BY

Two pure vapour pressures and the amounts of each liquid are given; the total vapour pressure, or the composition of the vapour, is wanted.

METHOD

Find the mole fractions in the **liquid**, apply Raoult to each component, add for p_{total} . For the **vapour** composition use Dalton: $y_A = p_A/p_{\text{total}}$.

WORKED EXAMPLE

An ideal solution contains 2 mol of A ($p_A^\circ = 80$ mm Hg) and 3 mol of B ($p_B^\circ = 120$ mm Hg). Find the total vapour pressure and the mole fraction of A in the vapour.

1. $x_A = 2/5 = 0.4$, $x_B = 0.6$

2. $p_A = 80 \times 0.4 = 32$ mm ; $p_B = 120 \times 0.6 = 72$ mm

$$p_{\text{total}} = 32 + 72 = 104 \text{ mm Hg} \quad y_A = \frac{32}{104} = 0.308$$

Sanity check: $y_A = 0.308$ is less than $x_A = 0.4$, so the vapour is richer in B — and B is the more volatile component. That is the direction it must come out.

T4 Molar mass from the lowering of vapour pressure

RECOGNISE IT BY

The vapour pressures of the pure solvent and of the solution are both quoted.

METHOD

Use $M_2 = w_2 M_1 p^\circ / (p^\circ - p) w_1$ from P1. Compute $(p^\circ - p)$ first; it is usually a small difference of two similar numbers, so keep the decimals.

WORKED EXAMPLE

A solution of 8.45 g of lactose in 100 g of water has a vapour pressure of 4.559 mm Hg at 0 °C. The vapour pressure of pure water is 4.579 mm Hg. Calculate the molecular weight of lactose. (Heritage 2024–25)

1. $p^\circ - p = 4.579 - 4.559 = 0.020$ mm Hg

$$M_2 = \frac{8.45 \times 18 \times 4.579}{0.020 \times 100} = \frac{696.5}{2.0} = 348 \text{ g mol}^{-1}$$

The true molar mass of lactose is 342; the small excess comes from the dilute-solution approximation in P1. An answer near 348 is correct for the method ISC teaches.

T5 Elevation of boiling point

RECOGNISE IT BY

K_b with masses, asking for ΔT_b , the boiling point, or the molar mass.

METHOD

$\Delta T_b = 1000 K_b w_2 / M_2 w_1$. Remember that the question usually wants the **boiling point**, not the elevation — add ΔT_b to the boiling point of the pure solvent.

WORKED EXAMPLE

Calculate the boiling point of a solution of 8.0 g of urea ($M = 60 \text{ g mol}^{-1}$) in 250 g of water. $K_b = 0.52 \text{ K kg mol}^{-1}$, and pure water boils at 373 K.

$$\Delta T_b = \frac{1000 \times 0.52 \times 8.0}{60 \times 250} = \frac{4160}{15000} = 0.277 \text{ K}$$

$$\text{boiling point} = 373 + 0.277 = 373.28 \text{ K}$$

T6 Depression of freezing point

RECOGNISE IT BY

K_f with masses; ΔT_f , the freezing point, or the mass of antifreeze needed.

METHOD

Same formula with K_f . For an antifreeze question, work **backwards**: get the molality you need from $\Delta T_f / K_f$, multiply by the kilograms of solvent to get moles, then by the molar mass.

WORKED EXAMPLE

What mass of ethylene glycol ($M = 62 \text{ g mol}^{-1}$) must be added to 4 kg of water to prevent it from freezing at -6 °C? $K_f = 1.86 \text{ K kg mol}^{-1}$.

1. $m = \Delta T_f / K_f = 6 / 1.86 = 3.226 \text{ mol kg}^{-1}$

2. moles needed = $3.226 \times 4 = 12.90 \text{ mol}$

$$\text{mass} = 12.90 \times 62 = 800 \text{ g}$$

T7 Molar mass from osmotic pressure

RECOGNISE IT BY

A protein or polymer, a small mass, a volume in cm^3 , and a very small π .

METHOD

$M_2 = w_2RT/\pi V$. **Convert the volume to litres** and the temperature to kelvin; $R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$.

Expect an answer in the tens of thousands — that is what makes osmotic pressure the right property for a macromolecule.

WORKED EXAMPLE

An aqueous solution contains 0.63 g of protein in 300 cm^3 . Its osmotic pressure at 300 K is $1.29 \times 10^{-3} \text{ atm}$. Calculate the molar mass of the protein. (Specimen 2026)

$$M_2 = \frac{0.63 \times 0.0821 \times 300}{1.29 \times 10^{-3} \times 0.300} = \frac{15.52}{3.87 \times 10^{-4}} \\ = 4.01 \times 10^4 \text{ g mol}^{-1}$$

T8 Isotonic solutions

RECOGNISE IT BY

“How much glucose per litre is isotonic with blood?”, or two solutions said to be isotonic.

METHOD

Isotonic means **equal π** , and therefore **equal molar concentration C**. Get C from $\pi = CRT$, then convert to a mass using the molar mass. If two solutions are isotonic, simply set their concentrations equal — you never need R at all.

WORKED EXAMPLE

The osmotic pressure of blood at 37°C is 8.21 atm . How much glucose ($M = 180 \text{ g mol}^{-1}$) should be used per litre of aqueous solution for an intravenous injection that is isotonic with blood? (ISC 2023)

1. $T = 37 + 273 = 310 \text{ K}$

$$C = \frac{\pi}{RT} = \frac{8.21}{0.0821 \times 310} = \frac{8.21}{25.45} = 0.3226 \text{ mol L}^{-1} \\ \text{mass} = 0.3226 \times 180 = 58.1 \text{ g L}^{-1}$$

T9 Finding the van't Hoff factor

RECOGNISE IT BY

An **observed** colligative property is quoted alongside enough data to compute the theoretical one.

METHOD

Compute the property assuming $i = 1$, then divide: $i = \text{observed} \div \text{calculated}$. Sanity-check the answer — an electrolyte must give $i > 1$ and an associating solute $i < 1$. If it comes out the wrong side of 1, you have inverted the fraction.

WORKED EXAMPLE

0.5 g of K_2SO_4 ($M = 174 \text{ g mol}^{-1}$) is dissolved in 100 g of water. The observed depression of freezing point is 0.14 K, and $K_f = 1.86 \text{ K kg mol}^{-1}$. Find i and the degree of dissociation.

1. $m = (0.5/174) \div 0.100 = 0.02874 \text{ mol kg}^{-1}$

2. ΔT_f (calculated, $i = 1$) = $1.86 \times 0.02874 = 0.0534 \text{ K}$

$$i = \frac{0.14}{0.0534} = \mathbf{2.62}$$

1. $K_2SO_4 \rightarrow 2K^+ + SO_4^{2-}$, so $n = 3$

$$\alpha = \frac{i - 1}{n - 1} = \frac{2.62 - 1}{2} = \mathbf{0.81, \text{ i.e. } 81\%}$$

T10 Degree of dissociation

RECOGNISE IT BY

A salt in water, with a colligative measurement; the degree of dissociation is wanted.

METHOD

Find i as in T9, count n from the dissociation equation, then $\alpha = (i - 1)/(n - 1)$. Getting **n** right matters: NaCl gives 2, $CaCl_2$ and K_2SO_4 give 3, $K_3[Fe(CN)_6]$ and $AlCl_3$ give 4.

WORKED EXAMPLE

A solution containing 0.5 g of KCl in 100 g of water freezes at -0.24°C . Calculate the degree of dissociation of the salt. $K_f = 1.86 \text{ K kg mol}^{-1}$. (Heritage 2024–25)

1. $M(KCl) = 39.1 + 35.5 = 74.5 \text{ g mol}^{-1}$

2. $m = (0.5/74.5) \div 0.100 = 0.0671 \text{ mol kg}^{-1}$

3. ΔT_f (calculated) = $1.86 \times 0.0671 = 0.1248 \text{ K}$

$$i = \frac{0.24}{0.1248} = 1.923 \quad \alpha = \frac{1.923 - 1}{2 - 1} = \mathbf{0.923, \text{ i.e. } 92.3\%}$$

T11 Degree of association

RECOGNISE IT BY

An acid or phenol dissolved in **benzene** — the giveaway solvent, because it cannot hydrogen bond to the solute and so lets the solute molecules pair up with each other.

METHOD

Same first step: find i , which will come out **less than 1**. Then use $\alpha = n(1 - i)/(n - 1)$, or for a dimer simply $\alpha = 2(1 - i)$.

WORKED EXAMPLE

Phenol associates in benzene to form a dimer. A solution of 2.5 g of phenol ($M = 94 \text{ g mol}^{-1}$) in 120 g of benzene lowers its freezing point by 0.85 K. Calculate the degree of association. $K_f(\text{benzene}) = 5.12 \text{ K kg mol}^{-1}$. (Specimen 2027)

$$1. m = (2.5/94) \div 0.120 = 0.2216 \text{ mol kg}^{-1}$$

$$2. \Delta T_f (\text{calculated}) = 5.12 \times 0.2216 = 1.135 \text{ K}$$

$$i = \frac{0.85}{1.135} = 0.749$$

$$\alpha = 2(1 - 0.749) = \mathbf{0.50, \text{i.e. } 50\%}$$

A quicker route for the multiple-choice form: if a dimerising solute has $i = 0.54$, then $\alpha = 2(1 - 0.54) = 0.92$ — exactly the Heritage 2024–25 objective question.

T12 Comparing or ordering colligative properties

RECOGNISE IT BY

“Arrange these solutions of equal concentration in increasing order of freezing point / boiling point / osmotic pressure.”

METHOD

Count the **particles**, i.e. find i for each. Then decide the direction: a larger i means a larger ΔT_b , ΔT_f and π — so a **higher** boiling point, a **lower** freezing point, and a higher osmotic pressure.

WORKED EXAMPLE

Arrange in increasing order of freezing point: 0.1 M glucose, 0.1 M NaCl, 0.1 M CaCl₂, 0.1 M K₃[Fe(CN)₆].

$$1. i = 1, 2, 3, 4 \text{ respectively}$$

$$2. \text{ bigger } i \Rightarrow \text{ bigger depression} \Rightarrow \text{ lower freezing point}$$

Increasing freezing point



Common slip · Read the direction the question asks for. **Increasing freezing point** is the reverse of increasing depression, so the answer runs from the most-dissociated solute to the least. Half the marks on this type are lost by giving the right chemistry in the wrong order.

T13 Explaining an abnormal molar mass

RECOGNISE IT BY

“The observed molar mass of acetic acid in benzene is 120 instead of 60. Explain.”

METHOD

Decide whether the observation implies **more** or **fewer** particles than expected, then name association or dissociation and give the reason — hydrogen bonding for association, ionisation for dissociation.

WORKED EXAMPLE

The molar mass of acetic acid determined in benzene by the depression of freezing point comes out as about 120 g mol⁻¹, twice the expected 60. Explain.

Answer · An observed molar mass **twice** the true value means the colligative property was **half** what was expected, so the solution contained only **half** as many particles as were dissolved. Acetic acid molecules pair up in benzene through **two intermolecular hydrogen bonds**, forming a cyclic **dimer**; benzene, being non-polar, cannot hydrogen bond to the acid and so does nothing to prevent this. Here $i = 0.5$ and the degree of association is essentially 1.

T14 Identifying the type of deviation or azeotrope

RECOGNISE IT BY

“What type of azeotrope does a solution of chloroform and acetone form? Justify on the basis of intermolecular interactions.”

METHOD

Ask whether the **A–B** interaction is stronger or weaker than A–A and B–B. Stronger $\Rightarrow$ negative deviation $\Rightarrow$ **maximum**-boiling azeotrope. Weaker $\Rightarrow$ positive deviation $\Rightarrow$ **minimum**-boiling azeotrope.

WORKED EXAMPLE

What type of azeotropic mixture is formed by chloroform and acetone? Justify your answer. (Specimen 2027)

Answer · A **maximum-boiling** azeotrope. Chloroform and acetone form a **hydrogen bond between unlike molecules** — the acidic hydrogen of CHCl₃ to the carbonyl oxygen of acetone — which is **stronger** than either of the pure-liquid interactions. Each component is therefore held back from escaping, the vapour pressure is **lower** than Raoult's law predicts (negative deviation), and a higher temperature is needed to boil the mixture.

Ethanol and water are the standard contrast: hydrogen bonding between ethanol molecules is disrupted by water, so escape becomes easier, giving positive deviation and a minimum-boiling azeotrope at 95% ethanol.

4 · Solutions recall sheet

revise from this page alone

The three molar-mass formulae

$$M = w_2 M_1 p^\circ / [(p^\circ - p) w_1]$$

$$M = 1000 K_b w_2 / (\Delta T_b w_1)$$

$$M = 1000 K_f w_2 / (\Delta T_f w_1)$$

$$M = w_2 RT / (\pi V)$$

Constants for water

$$K_b = 0.52 \text{ K kg mol}^{-1}$$

$$K_f = 1.86 \text{ K kg mol}^{-1}$$

$$K_f(\text{benzene}) = 5.12$$

$$R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$$

van't Hoff factor

$i = \text{observed} \div \text{calculated property}$

$i = \text{normal M} \div \text{observed M}$

Dissociation: $\alpha = (i-1)/(n-1)$

Association: $\alpha = n(1-i)/(n-1)$

Dimer: $\alpha = 2(1-i)$

Values of n

NaCl, KCl $\rightarrow 2$

CaCl₂, K₂SO₄ $\rightarrow 3$

AlCl₃, K₃[Fe(CN)₆] $\rightarrow 4$

glucose, urea, sucrose $\rightarrow 1$

Henry's law

$p = K_H x$

Large $K_H \Rightarrow$ low solubility

K_H rises with T, so gases are less soluble when warm

Deviation $\Rightarrow$ azeotrope

Positive (A-B weaker): p high, $\Delta H > 0$, **minimum**-boiling
— ethanol + water

Negative (A-B stronger): p low, $\Delta H < 0$, **maximum**-boiling
— CHCl₃ + acetone

Abnormal molar mass

Dissociation $\rightarrow$ more particles $\rightarrow$ bigger effect $\rightarrow$ **lower** observed M

Association $\rightarrow$ fewer particles $\rightarrow$ smaller effect $\rightarrow$ **higher** observed M

Temperature dependence

Molarity **changes** with T (volume expands)

Molality, mole fraction, % w/w **do not**

Ordering rule

Bigger $i \Rightarrow$ higher b.p., **lower** f.p., higher π

glucose < NaCl < CaCl₂ < K₃[Fe(CN)₆] in effect size

Why osmotic pressure for proteins

ΔT_f and ΔT_b are immeasurably small for a large M; π is measurable at room temperature.

Part II · Chemical Kinetics

Eighteen concepts, six proofs, twenty-two problem types and a one-page recall sheet.

1 · The concepts, in the order they build

C1 – C18

C1 What kinetics adds to thermodynamics

Thermodynamics tells you **whether** a reaction can happen and how far it goes; it says nothing about **how fast**. Chemical kinetics supplies the missing half: the rate, the factors that control it, and the step-by-step mechanism.

Hold on to this · A reaction can be thermodynamically spontaneous (ΔG negative) and still be immeasurably slow — the rusting of iron, or the conversion of diamond to graphite. Spontaneity and speed are independent questions.

C2 Rate of reaction: average and instantaneous

Rate is the change in concentration of a reactant or product per unit time.

$$\text{average rate} = - \frac{\Delta[R]}{\Delta t} = + \frac{\Delta[P]}{\Delta t} \quad \text{instantaneous rate} = - \frac{d[R]}{dt} = + \frac{d[P]}{dt}$$

- The **minus sign** for a reactant is bookkeeping only — it makes the rate positive, since $[R]$ is falling. **Rate is never negative.**
- **Average rate** is measured over a finite interval; **instantaneous rate** is the slope of the tangent to the concentration–time curve at one instant.
- Unit: **mol L⁻¹ s⁻¹**. For gases measured by pressure it may be atm s⁻¹.

Hold on to this · The rate of a reaction is *not* constant during the reaction — it falls as reactants are used up. What stays constant at a given temperature is **k**, not the rate.

C3 Writing the rate in terms of any species

Different species appear and disappear at different speeds, in the ratio of their stoichiometric coefficients. To get one single number for “the rate of the reaction”, divide each term by its own coefficient.

For $aA + bB \rightarrow cC + dD$:

$$\text{rate} = - \frac{1}{a} \frac{d[A]}{dt} = - \frac{1}{b} \frac{d[B]}{dt} = + \frac{1}{c} \frac{d[C]}{dt} = + \frac{1}{d} \frac{d[D]}{dt}$$

Hold on to this · The species with the **largest coefficient** changes concentration **fastest**. In $2N_2O_5 \rightarrow 4NO_2 + O_2$, NO_2 appears four times as fast as O_2 .

C4 The six factors that affect rate — and the two that affect k

Factor	Why it works	Does k change?
Nature of the reactants	Ionic reactions need no bond breaking, so they are fast; reactions breaking strong covalent bonds are slow	—
Concentration (or pressure, for gases)	More particles per unit volume $\Rightarrow$ more collisions per second	no
Surface area of a solid	Reaction occurs only at the exposed surface	no
Temperature	Enlarges the fraction of molecules with $E \geq E_a$	yes
Catalyst	Offers a path of lower E_a	yes
Radiation of suitable frequency	Supplies the energy to initiate photochemical steps	—

Memory hook · Only Temperature and a Catalyst can change **k**. Everything else changes the *rate* without touching the *rate constant*. Examiners test this distinction more often than the list itself.

C5 Rate law and the rate constant

The **rate law** is the experimentally established equation linking rate to concentrations:

$$\text{rate} = k[A]^x[B]^y$$

- x and y are found **only by experiment**. They are not read off the balanced equation.
- **k, the rate constant**, is numerically the rate of the reaction when the concentration of every reactant is unity. It is also called the *specific reaction rate*.
- k is **independent of concentration** but **depends on temperature and on the catalyst**.

C6 Order of a reaction

The **order** is the sum of the powers of the concentration terms in the experimentally determined rate law.

- It is an **experimental quantity**.
- It may be **zero**, a **whole number**, a **fraction**, or even **negative**.
- A negative order means that species *slows* the reaction — usually a product that reverses an earlier equilibrium.
- The order with respect to one reactant is the power of that reactant alone; the overall order is the sum.

Hold on to this · For an **elementary** reaction — and only then — order and stoichiometry coincide. Most reactions are not elementary, which is exactly why order must be measured.

C7 Molecularity, and how it differs from order

Molecularity is the number of reacting species that must collide simultaneously in a single **elementary** step.

	Order	Molecularity
What it describes	the experimental rate law of the <i>overall</i> reaction	the collision in <i>one elementary</i> step
How obtained	experimentally	theoretically, from the elementary step
Possible values	0, whole number, fraction or negative	1, 2 or rarely 3 — never 0, never fractional
Changes with conditions	yes (a reactant in large excess lowers it)	no
Defined for complex reactions	yes	no — only for each elementary step

Memory hook · **Molecularity is a molecule count; order is a stopwatch reading.** You cannot count half a molecule, which is why molecularity is never fractional — but a stopwatch can certainly report a fractional power.

C8 Units of the rate constant

Because rate always has the unit $\text{mol L}^{-1} \text{s}^{-1}$, the unit of k is forced by the order — see proof **P1**.

$$\text{unit of } k = \text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$$

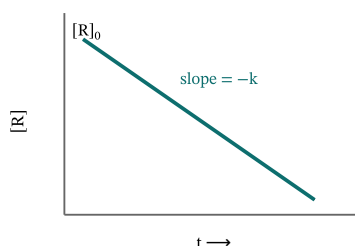
Order n	Unit of k	Read it as
0	$\text{mol L}^{-1} \text{s}^{-1}$	same as the rate
1	s^{-1}	no concentration term at all
2	$\text{L mol}^{-1} \text{s}^{-1}$	one L per mol
3	$\text{L}^2 \text{mol}^{-2} \text{s}^{-1}$	two L per mol squared
$3/2$	$\text{mol}^{-1/2} \text{L}^{1/2} \text{s}^{-1}$	fractional powers are legitimate

Hold on to this · This works **both ways**. Given the order, write the unit; given the unit, read off the order. Both directions are set as one-mark questions.

C9 Zero order reactions

The rate is **independent of the concentration** of the reactant: $\text{rate} = k[\text{R}]^0 = k$.

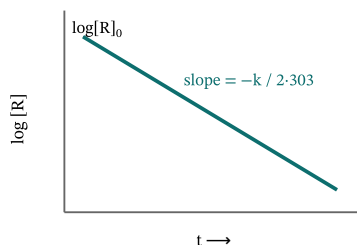
- **Examples:** decomposition of NH_3 on a hot platinum surface; decomposition of HI on a gold surface; some photochemical reactions; enzyme reactions at saturating substrate.
- **Why it happens:** in surface-catalysed reactions the catalyst surface is completely covered, so adding more reactant cannot increase the number of reacting sites.
- The concentration falls **linearly** with time, so a zero order reaction reaches a definite **finishing time**, $t = [\text{R}]_0/k$.



C10 First order reactions

The rate is **directly proportional** to the concentration of one reactant: $\text{rate} = k[R]$.

- **Examples:** all radioactive decay; decomposition of N_2O_5 in CCl_4 ; decomposition of H_2O_2 in aqueous solution.
- $\log[R]$ falls **linearly** with time — the standard graphical test.
- A first order reaction never truly finishes: each equal interval removes an equal *fraction*, not an equal amount.



C11 Half-life period

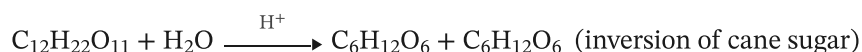
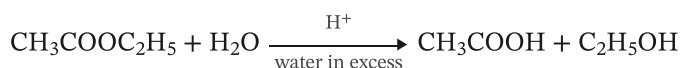
The **half-life** $t_{1/2}$ is the time in which the concentration of the reactant falls to half its initial value.

Order	Expression	Depends on $[R]_0$?
Zero	$t_{1/2} = [R]_0 / 2k$	yes — directly proportional
First	$t_{1/2} = 0.693 / k$	no — independent

Hold on to this • The independence of $t_{1/2}$ from concentration is the **signature of first order**. If a question tells you that successive half-lives are equal, or that the doubling time is constant, it has told you the reaction is first order.

C12 Pseudo first order reactions

A reaction that is genuinely of higher order but **behaves** as first order because one reactant is present in such large excess that its concentration is effectively constant.



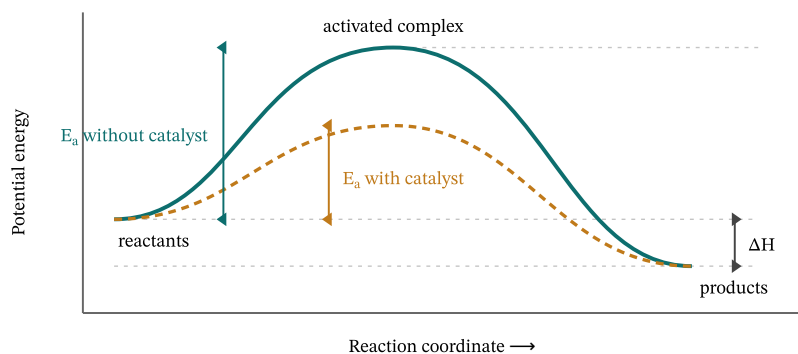
True rate $= k[\text{ester}][\text{H}_2\text{O}]$; but $[\text{H}_2\text{O}]$ hardly changes, so rate $= k'[\text{ester}]$ with $k' = k[\text{H}_2\text{O}]$. The **observed** order is 1, the **true** order 2.

C13 Threshold energy, activation energy, activated complex

Term	Definition
Threshold energy, E_T	The minimum <i>total</i> energy the colliding molecules must possess for reaction to occur.
Activation energy, E_a	The <i>extra</i> energy that reactant molecules must absorb over and above their average energy in order to reach the threshold.
Relation	$E_a = E_T - E_{\text{average, reactants}}$
Activated complex (transition state)	The unstable, highest-energy arrangement at the top of the barrier, in which old bonds are partly broken and new bonds partly formed.

Hold on to this • A high E_a means a **slow reaction**, because the fraction of molecules able to clear the barrier is small. E_a is a property of the reaction path, quite separate from ΔH .

C14 The energy profile, and what a catalyst does to it



- $\Delta H = (\text{energy of products}) - (\text{energy of reactants})$. It is **negative for an exothermic** reaction (products lower, as drawn) and **positive for an endothermic** one.
- A catalyst provides an **alternative path of lower E_a** . It lowers the forward and backward barriers **equally**.
- A catalyst therefore changes **k and the rate**, but **not ΔH , not K_c , and not the position of equilibrium** — only the speed of reaching it.

Common slip · Three sentences that lose marks: “a catalyst supplies energy to the reactants”, “a catalyst increases the yield”, “a catalyst changes ΔH ”. All three are wrong.

C15 Collision theory

For a reaction to occur the particles must collide — but only a tiny fraction of collisions lead to reaction. An **effective collision** must satisfy two independent conditions:

- **Energy condition** — the colliding molecules must together possess energy $\geq$ the threshold energy.
- **Orientation condition** — they must be correctly aligned at the moment of impact, so that the bonds to be broken and formed are properly presented to each other.

Hold on to this · Collision frequency in a gas is of the order of 10^{30} collisions per litre per second, yet most reactions are far slower than that. The gap is explained entirely by these two conditions.

C16 Effect of temperature

A rise of 10 K roughly **doubles** the rate of most reactions — the **temperature coefficient** is about 2.

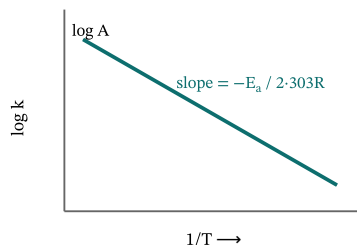
The average kinetic energy rises only slightly over 10 K. What changes dramatically is the **high-energy tail** of the Maxwell–Boltzmann distribution: the number of molecules with $E \geq E_a$ roughly doubles, and only those molecules can react.

Hold on to this · The rate depends **exponentially** on temperature because $k = Ae^{-E_a/RT}$, with T in the exponent. That is why a small change in T produces a large change in k .

C17 The Arrhenius equation

$$k = A e^{-E_a/RT} \quad \log k = \log A - \frac{E_a}{2.303 RT}$$

- **A**, the frequency or pre-exponential factor, is the rate constant the reaction would have if every collision were effective. It has the same unit as k .
- $e^{-E_a/RT}$ is the fraction of molecules possessing energy $\geq E_a$.
- As $T \rightarrow \infty$ the exponent $\rightarrow 0$ and $k \rightarrow A$. As $E_a \rightarrow 0$, again $k \rightarrow A$.



Hold on to this • Slope of $\log k$ against $1/T = -E_a/2.303R$, intercept = $\log A$. So $E_a = -2.303 R \times \text{slope}$, and the slope is negative, which makes E_a positive.

C18 Mechanism and the rate-determining step

Most reactions proceed through a sequence of **elementary steps**. The **slowest** step is the **rate-determining step**, and it alone controls the overall rate — just as a queue moves no faster than its slowest counter.

- The observed rate law reflects the species involved **up to and including** the slow step.
- Reactants consumed only in later, faster steps **do not appear** in the rate law — which is why the rate law often disagrees with the overall stoichiometry.
- A **catalyst can appear in the rate law** if it takes part in the slow step, even though it is absent from the overall equation.

2 • The proofs

P1 – P6

P1 The unit of the rate constant is $\text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$

1. For an n^{th} order reaction, $\text{rate} = k[\text{R}]^n$, so $k = \text{rate}/[\text{R}]^n$.
2. Substituting the units:

$$\text{unit of } k = \frac{\text{mol L}^{-1} \text{s}^{-1}}{(\text{mol L}^{-1})^n} = \text{mol}^{1-n} \text{L}^{-1+n} \text{s}^{-1} = \text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$$

■ proved

P2 Integrated rate equation and half-life for a zero order reaction

1. For $R \rightarrow P$ of zero order, rate $= -d[R]/dt = k[R]^0 = k$.
2. Separating and integrating from $([R]_0, 0)$ to $([R], t)$:

$$-d[R] = k dt \Rightarrow -([R] - [R]_0) = kt \Rightarrow [R] = [R]_0 - kt$$

$$k = \frac{[R]_0 - [R]}{t}$$

3. For the half-life put $[R] = [R]_0/2$ at $t = t_{1/2}$:

$$k = \frac{[R]_0 - [R]_0/2}{t_{1/2}} = \frac{[R]_0}{2 t_{1/2}} \Rightarrow t_{1/2} = [R]_0/2k$$

A plot of $[R]$ against t is a straight line of slope $-k$ and intercept $[R]_0$. Unit of $k = \text{mol L}^{-1} \text{s}^{-1}$.

■ proved

P3 Integrated rate equation and half-life for a first order reaction

1. For $R \rightarrow P$ of first order, rate $= -d[R]/dt = k[R]$.
2. Separate the variables:

$$-\frac{d[R]}{[R]} = k dt$$

3. Integrate between $t = 0, [R] = [R]_0$ and time t , concentration $[R]$:

$$-\ln[R] + \ln[R]_0 = kt \Rightarrow \ln \frac{[R]_0}{[R]} = kt$$

4. Converting to base-ten logarithms ($\ln x = 2.303 \log x$):

$$k = \frac{2.303}{t} \log \frac{[R]_0}{[R]} \quad \text{or} \quad \log[R] = \log[R]_0 - \frac{kt}{2.303}$$

5. For the half-life put $[R] = [R]_0/2$ at $t = t_{1/2}$:

$$k = \frac{2.303}{t_{1/2}} \log 2 = \frac{2.303 \times 0.3010}{t_{1/2}} = \frac{0.693}{t_{1/2}} \Rightarrow t_{1/2} = 0.693/k$$

$t_{1/2}$ contains no concentration term — hence the defining property of first order kinetics. A plot of $\log[R]$ against t is a straight line of slope $-k/2.303$.

■ proved

P4 After n half-lives the fraction remaining is $(1/2)^n$

1. By definition, one half-life halves the amount present, whatever that amount is (first order, by P3).
2. Applying this repeatedly:

$$[R]_0 \xrightarrow{t_{1/2}} \frac{[R]_0}{2} \xrightarrow{t_{1/2}} \frac{[R]_0}{2^2} \xrightarrow{t_{1/2}} \dots \xrightarrow{t_{1/2}} \frac{[R]_0}{2^n}$$

$$\text{fraction left} = (1/2)^n \quad \text{where } n = t/t_{1/2}$$

Useful values: $n = 1 \rightarrow 50\%$, $2 \rightarrow 25\%$, $3 \rightarrow 12.5\%$, $4 \rightarrow 6.25\%$, $5 \rightarrow 3.125\%$, $10 \rightarrow 0.098\%$.

■ proved

P5 $t_{75\%} = 2 t_{50\%}$, and $t_{99\%} = 2 t_{90\%}$

1. From P3, the time for $x\%$ completion is $t_x = (2.303/k) \log[100/(100 - x)]$.
2. Form the ratio, so that k cancels:

$$\frac{t_{75\%}}{t_{50\%}} = \frac{\log(100/25)}{\log(100/50)} = \frac{\log 4}{\log 2} = \frac{2 \log 2}{\log 2} = 2$$

$$\frac{t_{99\%}}{t_{90\%}} = \frac{\log 100}{\log 10} = \frac{2}{1} = 2$$

The same method gives $t_{90\%} = 3.32 t_{1/2}$, $t_{99\%} = 6.64 t_{1/2}$ and $t_{99.9\%} = 9.97 \approx 10 t_{1/2}$. Notice that **no property of the particular reaction survives the division** — these ratios hold for every first order reaction.

■ proved

P6 The two-point form of the Arrhenius equation

1. Take logarithms of $k = Ae^{-E_a/RT}$:

$$\log k = \log A - \frac{E_a}{2.303 RT}$$

2. Write this at two temperatures T_1 and T_2 (E_a and A are assumed constant):

$$\log k_1 = \log A - \frac{E_a}{2.303 R T_1} \quad \log k_2 = \log A - \frac{E_a}{2.303 R T_2}$$

3. Subtract the first from the second; $\log A$ cancels:

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303 R} \frac{T_2 - T_1}{T_1 T_2}$$

Keep $2.303 \times R = 2.303 \times 8.314 = 19.147$ as one number; it shortens every Arrhenius calculation you will ever do.

■ proved

3 · Every type of problem that is set

T1 – T22

T1 Relating the rates of different species**RECOGNISE IT BY**

“If the rate of formation of X is ..., find the rate of disappearance of Y” or “express the rate of the reaction in terms of ...”.

METHOD

Divide every term by its own stoichiometric coefficient (C3). Find the single “rate of reaction” first, then multiply back by the coefficient of the species asked for.

WORKED EXAMPLE

For $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$, the rate of formation of NO is $3.6 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$. Find the rate of the reaction and the rate of consumption of O_2 .

$$\text{rate} = \frac{1}{4} \frac{d[\text{NO}]}{dt} = \frac{3.6 \times 10^{-3}}{4} = 9.0 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$$

$$- \frac{d[\text{O}_2]}{dt} = 5 \times \text{rate} = 4.5 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$$

T2 Effect of changing concentrations, given the rate law

RECOGNISE IT BY

“By how many times will the rate change if [A] is doubled and [B] halved?”

METHOD

Substitute the multiplying factors into the rate law and take the ratio. Each factor is raised to **that reactant's own order**, then all the results are multiplied together. Fractional and negative orders behave exactly the same way.

WORKED EXAMPLE

For rate = $k[A]^{3/2}[B]^{-1}$, what happens to the rate if [A] is quadrupled and [B] is doubled?

$$\frac{\text{rate}'}{\text{rate}} = (4)^{3/2} \times (2)^{-1} = 8 \times \frac{1}{2} = 4$$

Answer • The rate becomes **4 times** the original.

T3 Order from initial-rate data (the commonest 3-mark question)

RECOGNISE IT BY

A table of experiments with concentrations and initial rates.

METHOD

Change one thing at a time. Pick two rows in which one concentration is held fixed; the ratio of rates then isolates the order of the other reactant. If concentration is multiplied by c and rate by r , then $c^x = r$, so $x = \log r / \log c$. Verify with an unused row, then get k by substituting one row — and give k its unit.

WORKED EXAMPLE

For $A + B \rightarrow C$ the data are:

Expt.	[A]	[B]	Rate / $\text{mol L}^{-1} \text{s}^{-1}$
1	0.2	0.2	4.0×10^{-3}
2	0.4	0.2	1.6×10^{-2}
3	0.2	0.4	8.0×10^{-3}

- Rows 1 $\rightarrow$ 2: [B] fixed, $[A] \times 2$, rate $\times 4 \Rightarrow 2^x = 4$, $x = 2$.
- Rows 1 $\rightarrow$ 3: [A] fixed, $[B] \times 2$, rate $\times 2 \Rightarrow 2^y = 2$, $y = 1$.
- rate = $k[A]^2[B]$; overall order 3.

$$k = \frac{4.0 \times 10^{-3}}{(0.2)^2 \times 0.2} = \frac{4.0 \times 10^{-3}}{8.0 \times 10^{-3}} = \mathbf{0.5 \text{ L}^2 \text{ mol}^{-2} \text{ s}^{-1}}$$

T4 Order from the unit of k

RECOGNISE IT BY

“Identify the order of a reaction whose rate constant is ...”.

METHOD

Match the given unit against $\text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$ (P1) and solve for n. The power of L is (n – 1); the power of mol is (1 – n). Either one gives n.

WORKED EXAMPLE

Find the order for $k = 5.2 \times 10^{-2} \text{mol}^{-1/2} \text{L}^{1/2} \text{s}^{-1}$.

$$n - 1 = 1/2 \Rightarrow n = 3/2 \text{ (check: } 1 - n = -1/2 \text{ ✓)}$$

A fractional order is a signal that the reaction has a chain or multi-step mechanism.

T5 Overall order from a given rate law

RECOGNISE IT BY

“What is the overall order of the reaction whose rate law is ...?”

METHOD

Add the exponents algebraically. Include fractions and negatives; a zero exponent contributes nothing.

WORKED EXAMPLE

$$\text{rate} = k[A]^2[B]^{1/2}[C]^{-1}$$

$$2 + 1/2 - 1 = 1.5$$

T6 First order: find k (and $t_{1/2}$) from a percentage completion

RECOGNISE IT BY

“x% of a first order reaction is completed in t minutes. Calculate the rate constant.”

METHOD

Take a = 100 and (a – x) = (100 – x). Substitute in $k = (2.303/t) \log[a/(a - x)]$. If $t_{1/2}$ is also wanted, use $t_{1/2} = 0.693/k$ afterwards.

WORKED EXAMPLE

60% of a first order reaction is complete in 40 minutes. Find k and $t_{1/2}$.

$$k = \frac{2.303}{40} \log \frac{100}{40} = 0.05758 \times 0.3979 = 2.29 \times 10^{-2} \text{ min}^{-1}$$

$$t_{1/2} = \frac{0.693}{2.29 \times 10^{-2}} = 30.2 \text{ minutes}$$

T7 First order: time for a given percentage (the ratio shortcut)

RECOGNISE IT BY

" $x_1\%$ is complete in t_1 minutes. How long for $x_2\%$?" — k is **not** asked for.

METHOD

Do **not** calculate k . Write the equation twice and divide, so that 2.303 and k both cancel:

WORKED EXAMPLE

$$\frac{t_2}{t_1} = \frac{\log [100/(100 - x_2)]}{\log [100/(100 - x_1)]}$$

A first order reaction is 20% complete in 12 minutes. Find the time for 90% completion.

$$\frac{t}{12} = \frac{\log (100/10)}{\log (100/80)} = \frac{1.0000}{0.0969} = 10.32$$

$$t = 12 \times 10.32 = \mathbf{123.8 \text{ minutes}}$$

T8 First order: from one concentration to another

RECOGNISE IT BY

Concentrations, masses, volumes or pressures are given instead of percentages.

METHOD

The equation uses only the **ratio**, so any consistent unit will do — no conversion needed. **Check first whether the ratio is a power of 2:** if it is, just count half-lives and avoid the logarithm altogether.

WORKED EXAMPLE

For a first order reaction $k = 2.0 \times 10^{-3} \text{ s}^{-1}$. Find the time for the concentration to fall from 0.50 M to 0.125 M.

1. $0.50 \rightarrow 0.25 \rightarrow 0.125$ is **2 half-lives**.

2. $t_{1/2} = 0.693 / 2.0 \times 10^{-3} = 346.5 \text{ s}$

$$t = 2 \times 346.5 = \mathbf{693 \text{ s}}$$

The long route confirms it: $t = (2.303 / 2.0 \times 10^{-3}) \log 4 = 1151.5 \times 0.6021 = 693.3 \text{ s}$.

T9 Converting between k and $t_{1/2}$

RECOGNISE IT BY

Either quantity given, the other wanted; often as a stepping stone inside a longer problem.

METHOD

First order: $k = 0.693/t_{1/2}$ and $t_{1/2} = 0.693/k$. Zero order: $t_{1/2} = [R]_0/2k$ — you need $[R]_0$ as well.

WORKED EXAMPLE

A first order reaction has $t_{1/2} = 5$ minutes. Find k and the time for 87.5% completion.

$$k = \frac{0.693}{5} = \mathbf{0.1386 \text{ min}^{-1}}$$

1. 87.5% complete leaves $12.5\% = (1/2)^3$, so 3 half-lives.

$$t = 3 \times 5 = \mathbf{15 \text{ minutes}}$$

T10 Counting half-lives (and working backwards)**RECOGNISE IT BY**

Radioactive decay, or any first order process, where the fraction is a power of $\frac{1}{2}$.

METHOD

Use fraction = $(\frac{1}{2})^n$ with $n = t/t_{1/2}$ (P4). Read the question carefully: it may give you the fraction and ask for $t_{1/2}$, which is the same equation run backwards.

WORKED EXAMPLE

A radioactive sample decays to $\frac{1}{32}$ of its initial amount in 100 days. Calculate its half-life.

$$(\frac{1}{2})^n = \frac{1}{32} = (\frac{1}{2})^5 \Rightarrow n = 5 \text{ half-lives}$$

$$t_{1/2} = \frac{100}{5} = \mathbf{20 \text{ days}}$$

T11 Awkward fractions — radiocarbon dating and the like**RECOGNISE IT BY**

The fraction remaining is **not** a power of $\frac{1}{2}$ (80%, 60%, 15% ...).

METHOD

Two steps, always in this order: (1) $k = 0.693/t_{1/2}$; (2) $t = (2.303/k) \log[a/(a-x)]$. Take care over which figure is the *remaining* amount.

WORKED EXAMPLE

The half-life of ^{14}C is 5730 years. A wooden relic has 60% of the ^{14}C content of living wood. Find its age.

$$k = \frac{0.693}{5730} = 1.209 \times 10^{-4} \text{ year}^{-1}$$

$$t = \frac{2.303}{1.209 \times 10^{-4}} \log \frac{100}{60} = 19042 \times 0.2218 = \mathbf{4224 \text{ years}}$$

Common slip · The percentage quoted is what **remains**: $(a-x) = 60$ and $a = 100$, not the reverse.

T12 “Show that ...” ratio proofs**RECOGNISE IT BY**

“Show that the time for 99% completion is twice that for 90%”, or “show that $t_{90\%} = 3.32 t_{1/2}$ ”.

METHOD

Write each time from the integrated equation, then divide. Both 2.303 and k cancel, leaving a ratio of logarithms — see P5. Never substitute numbers for k .

WORKED EXAMPLE

Show that $t_{90\%} = 3.32 \times t_{1/2}$ for a first order reaction.

$$t_{90\%} = \frac{2.303}{k} \log 10 = \frac{2.303}{k} \quad t_{1/2} = \frac{0.693}{k}$$

$$\frac{t_{90\%}}{t_{1/2}} = \frac{2.303}{0.693} = \mathbf{3.32}$$

T13 E_a from two rate constants**RECOGNISE IT BY**

Two values of k at two temperatures.

METHOD

Substitute directly into P6. Use $2.303 R = 19.147$, and remember the temperature factor is $(T_2 - T_1)/T_1 T_2$ — the *product* in the denominator, not the sum.

WORKED EXAMPLE

$k = 1.0 \times 10^{-4} \text{ s}^{-1}$ at 300 K and $8.0 \times 10^{-4} \text{ s}^{-1}$ at 330 K. Find E_a .

$$\log 8 = 0.9031 = \frac{E_a}{19.147} \times \frac{30}{300 \times 330}$$
$$E_a = \frac{0.9031 \times 19.147}{3.030 \times 10^{-4}} = 57\,060 \text{ J mol}^{-1} = \mathbf{57.06 \text{ kJ mol}^{-1}}$$

T14 E_a from two half-lives (or two completion times)**RECOGNISE IT BY**

“50% complete in t_1 at T_1 and in t_2 at T_2 .”

METHOD

Convert times into a ratio of rate constants first. Because $t_{1/2} = 0.693/k$, $k \propto 1/t_{1/2}$, so $k_2/k_1 = t_1/t_2$ — the times invert. Then proceed as T13.

WORKED EXAMPLE

A first order reaction has $t_{1/2} = 100 \text{ s}$ at 300 K and 25 s at 320 K. Find E_a .

$$\frac{k_2}{k_1} = \frac{t_1}{t_2} = \frac{100}{25} = 4 \quad \log 4 = 0.6021$$
$$E_a = \frac{0.6021 \times 19.147}{20/(300 \times 320)} = \frac{11.527}{2.083 \times 10^{-4}} = \mathbf{55.33 \text{ kJ mol}^{-1}}$$

Common slip · Forgetting to invert the times is the single commonest error in this chapter. The *shorter* time belongs to the *larger* k .

T15 E_a when “the rate becomes n times”, or from the temperature coefficient

RECOGNISE IT BY

“The rate doubles / triples / becomes four times when the temperature rises from ...”, or “the **temperature coefficient** of the reaction is ...”.

METHOD

At fixed concentrations, rate $\propto k$, so $k_2/k_1 = n$ directly; then proceed as T13. The **temperature coefficient** is defined as k_{308}/k_{298} — the factor by which the rate constant rises for a **10 K rise, conventionally from 298 K to 308 K**. So if you are given the temperature coefficient, you have been given k_2/k_1 and both temperatures at once.

WORKED EXAMPLE

The rate of a reaction triples when the temperature rises from 300 K to 310 K. Calculate E_a .

$$\log 3 = 0.4771 = \frac{E_a}{19.147} \times \frac{10}{300 \times 310}$$

$$E_a = \frac{0.4771 \times 19.147}{1.0753 \times 10^{-4}} = \mathbf{84.96 \text{ kJ mol}^{-1}}$$

Variant (Heritage 2016–17). “The temperature coefficient for the hydrolysis of ethyl acetate by NaOH is 1.75. Calculate E_a .” Here $k_2/k_1 = 1.75$ with $T_1 = 298 \text{ K}$ and $T_2 = 308 \text{ K}$, so $\log 1.75 = 0.2430$ and $E_a = 0.2430 \times 19.147 / 1.0895 \times 10^{-4} = \mathbf{42.71 \text{ kJ mol}^{-1}}$. Note that a temperature coefficient of exactly 2 is the case worked as the “rate doubles per 10 K” question.

T16 The reverse Arrhenius problem — find k_2 given E_a

RECOGNISE IT BY

E_a and one rate constant are given; a rate constant at another temperature is wanted.

METHOD

Same equation, rearranged. Compute the right-hand side, then take the **antilogarithm** to get k_2/k_1 , and multiply by k_1 . Convert E_a to J mol^{-1} before substituting.

WORKED EXAMPLE

For a reaction $E_a = 50 \text{ kJ mol}^{-1}$ and $k = 2.0 \times 10^{-3} \text{ s}^{-1}$ at 300 K. Find k at 320 K.

$$\log \frac{k_2}{k_1} = \frac{50\,000}{19.147} \times \frac{20}{300 \times 320} = 2611 \times 2.083 \times 10^{-4} = 0.5440$$

$$\frac{k_2}{k_1} = \text{antilog } 0.5440 = 3.50 \Rightarrow k_2 = 2.0 \times 10^{-3} \times 3.50 = \mathbf{7.0 \times 10^{-3} \text{ s}^{-1}}$$

T17 Graphical Arrhenius questions

RECOGNISE IT BY

“The plot of $\log k$ against $1/T$ has slope ...”, or “find k as $T \rightarrow \infty$ ”.

METHOD

Compare $\log k = \log A - E_a/2.303RT$ with $y = mx + c$. Slope = $-E_a/2.303R$, intercept = $\log A$. Hence $E_a = \mathbf{-2.303 R \times \text{slope}}$. As $T \rightarrow \infty$ the exponent vanishes and $k = A$.

WORKED EXAMPLE

For a reaction the plot of $\log k$ against $1/T$ is a straight line of slope -5000 K . Calculate E_a .

$$E_a = -2.303 R \times \text{slope} = -19.147 \times (-5000) = 95\,735 \text{ J mol}^{-1} = \mathbf{95.74 \text{ kJ mol}^{-1}}$$

T18 Zero order calculations**RECOGNISE IT BY**

The order is stated as zero, or the unit of k is $\text{mol L}^{-1} \text{s}^{-1}$.

METHOD

Use $[R] = [R]_0 - kt$ and $t_{1/2} = [R]_0/2k$ (P2). A zero order reaction has a genuine **finishing time**, $t_{\text{complete}} = [R]_0/k$, unlike a first order one.

WORKED EXAMPLE

A zero order reaction has $k = 2.0 \times 10^{-2} \text{ mol L}^{-1} \text{s}^{-1}$ and $[R]_0 = 0.50 \text{ mol L}^{-1}$. Find $t_{1/2}$ and the time for the reaction to be complete.

$$t_{1/2} = \frac{0.50}{2 \times 2.0 \times 10^{-2}} = \mathbf{12.5 \text{ s}} \quad t_{\text{complete}} = \frac{0.50}{2.0 \times 10^{-2}} = \mathbf{25 \text{ s}}$$

T19 Change of volume or pressure of the vessel**RECOGNISE IT BY**

“The volume of the container is halved / doubled.”

METHOD

Concentration = moles/volume, so **halving the volume doubles every concentration** and doubling the volume halves them. Then apply T2.

WORKED EXAMPLE

For rate = $k[A][B]^2$, what happens to the rate if the volume of the vessel is doubled?

$$\text{each concentration} \times 1/2 \Rightarrow \frac{\text{rate}'}{\text{rate}} = (1/2)^1 \times (1/2)^2 = \frac{1}{8}$$

Answer · The rate falls to **one-eighth** of its original value.

T20 A reactant in large excess (pseudo order)**RECOGNISE IT BY**

“... if $[X]$ is present in large excess”, or the solvent is one of the reactants.

METHOD

The excess species has an essentially constant concentration, so absorb it into the constant: $k' = k[X]^p$. The **observed** order is what remains.

WORKED EXAMPLE

For rate = $k[A]^2[B]$, what is the observed order if B is taken in large excess?

$$\text{rate} = (k[B]) [A]^2 = k'[A]^2$$

Answer · Observed order = **2** (pseudo second order); the true order remains 3.

T21 From a mechanism to the rate law

RECOGNISE IT BY

A sequence of steps is given, one marked “slow”.

METHOD

The rate law comes from the **slow step alone**. If the slow step contains an intermediate, replace it using the equilibrium of the preceding fast step. Species that appear only after the slow step do not enter the rate law.

WORKED EXAMPLE

The iodide-catalysed decomposition of hydrogen peroxide, $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$, proceeds as: Step 1 (slow): $\text{H}_2\text{O}_2 + \text{I}^- \rightarrow \text{H}_2\text{O} + \text{IO}^-$; Step 2 (fast): $\text{H}_2\text{O}_2 + \text{IO}^- \rightarrow \text{H}_2\text{O} + \text{O}_2 + \text{I}^-$. Write the rate law and the order.

$$\text{rate} = k[\text{H}_2\text{O}_2][\text{I}^-] \quad \text{overall order} = 2$$

Two lessons in one example: the overall equation shows a coefficient of 2 for H_2O_2 yet the order in it is 1; and I^- , a **catalyst absent from the overall equation**, appears in the rate law because it takes part in the slow step.

T22 Identifying the order from a graph or from a stated behaviour

RECOGNISE IT BY

“Which plot is linear?”, or “the half-life is found to be independent of the initial concentration”.

METHOD

Learn the four diagnostic plots and the half-life test. Any one of them identifies the order on its own.

WORKED EXAMPLE

Observation	Order
[R] against t is a straight line	zero
$\log[\text{R}]$ against t is a straight line	first
rate against [R] is a horizontal line	zero
rate against [R] is a straight line through the origin	first
$t_{1/2}$ independent of $[\text{R}]_0$	first
$t_{1/2}$ directly proportional to $[\text{R}]_0$	zero
successive half-lives are equal / constant doubling time	first

4 • Kinetics recall sheet

revise from this page alone

Four numbers to know cold

- $2.303 \times R = 19.147$
- $\log 2 = 0.3010$, $\log 3 = 0.4771$
- $\log 5 = 0.6990$, $\log 7 = 0.8451$
- $0.693 = 2.303 \times \log 2$

Units of k

$$n = 0 : \text{mol L}^{-1}\text{s}^{-1}$$

$$n = 1 : \text{s}^{-1}$$

$$n = 2 : \text{L mol}^{-1}\text{s}^{-1}$$

$$n = 3 : \text{L}^2\text{mol}^{-2}\text{s}^{-1}$$

$$\text{general} : \text{mol}^{1-n}\text{L}^{n-1}\text{s}^{-1}$$

First order — the whole chapter in three lines

$$k = (2.303/t) \log[a/(a-x)]$$

$$t_{1/2} = 0.693/k$$

$$\text{fraction left after } n \text{ half-lives} = (1/2)^n$$

Zero order

$$[\text{R}] = [\text{R}]_0 - kt$$

$$t_{1/2} = [\text{R}]_0/2k$$

$$t_{\text{complete}} = [\text{R}]_0/k$$

Time ratios (first order, universal)

$$t_{50} : t_{75} : t_{87.5} = 1 : 2 : 3$$

$$t_{90} = 3.32 t_{1/2}$$

$$t_{99} = 6.64 t_{1/2} = 2 t_{90}$$

$$t_{99.9} \approx 10 t_{1/2}$$

Arrhenius

$$k = Ae^{-E_a/RT}$$

$$\log(k_2/k_1) = [E_a/19.147][(T_2 - T_1)/T_1 T_2]$$

$$\text{slope of } \log k \text{ vs } 1/T = -E_a/2.303R$$

$$T \rightarrow \infty \Rightarrow k \rightarrow A$$

$$k \propto 1/t_{1/2}$$

Half-lives given instead of rate constants? **Invert them.**

$$k_2/k_1 = t_1/t_2.$$

Order vs molecularity

Order — experimental; 0, fraction or negative allowed; overall reaction.

Molecularity — theoretical; 1, 2, 3 only; one elementary step.

A catalyst

Does: lower E_a , raise k , speed up forward and backward equally.

Does not: change ΔH , K_c , the equilibrium position or the yield.

Effective collision

(1) energy $\geq$ threshold energy

(2) correct orientation

$$E_a = E_T - E_{\text{average}}$$

Part III · Coordination Compounds

ISC unit 5 · Ligands and coordination number · IUPAC nomenclature · Werner's theory and ionisation · structural and stereo isomerism · valence bond theory · crystal field theory · colour and magnetism. Fifteen concepts, four standard determinations, fourteen problem types.

1 · The concepts, in the order they build

C1 – C15

C1 Complex compound or double salt?

	Double salt	Complex compound
Behaviour in water	dissociates completely into all its simple ions	the coordination sphere stays intact ; only the ions outside it are released
Identity of the metal ion	retained, so it gives its usual tests	lost, so it does not give its usual tests
Example	Mohr's salt, $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$	$[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$

Hold on to this · This is the standard two-mark question, and it is always set the same way. FeSO_4 mixed with $(\text{NH}_4)_2\text{SO}_4$ in a 1:1 ratio gives Mohr's salt, a **double salt**, which dissociates fully and so **does** give the test for Fe^{2+} . CuSO_4 with aqueous ammonia in a 1:4 ratio gives the **complex** $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$, in which Cu^{2+} is locked inside the coordination sphere and so gives **no** test for Cu^{2+} .

C2 Ligands, denticity and chelation

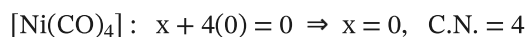
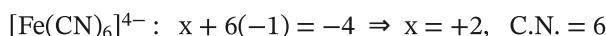
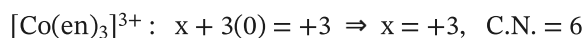
Denticity	Donor sites	Examples
Monodentate	1	Cl^- , CN^- , NH_3 , H_2O , OH^- , NO_2^- , CO
Bidentate	2	ethane-1,2-diamine (en), oxalate ($\text{C}_2\text{O}_4^{2-}$), glycinate
Tridentate	3	diethylenetriamine (dien)
Tetradentate	4	triethylenetetramine (trien)
Pentadentate	5	EDTA acting through five sites
Hexadentate	6	EDTA ⁴⁻ — its usual behaviour

- A ligand that binds through **two or more** sites forms a ring with the metal and is called a **chelating** ligand; the ring is a chelate. Chelation makes a complex markedly **more stable**.
- An **ambidentate** ligand can bind through *either* of two different atoms — NO_2^- through N (nitro) or through O (nitrito), and SCN^- through S (thiocyanato) or N (isothiocyanato). This is what gives rise to linkage isomerism (C7).

Hold on to this · **EDTA** is hexadentate and forms a very stable **octahedral** chelate. That is why it is used in the treatment of **lead poisoning** — it wraps around Pb^{2+} , and the resulting complex is soluble, unreactive and excreted by the kidneys. It is also used to estimate Ca^{2+} and Mg^{2+} in hard water.

C3 Oxidation number and coordination number

- **Coordination number** — the number of ligand **donor atoms** bonded to the central metal. Count donor *sites*, not ligand molecules: in $[\text{Co}(\text{en})_3]^{3+}$ there are three ligands but the coordination number is **6**.
- **Oxidation number** — let it be x , add the charges of all the ligands, and set the total equal to the charge on the complex ion.



C4 IUPAC nomenclature — the rules in order

1. **Cation first, anion second**, as in any salt.
2. Within the complex, name the **ligands first, in alphabetical order**, then the metal. Alphabetise by the **ligand name**, ignoring the multiplying prefix.
3. Anionic ligands end in **-o**: chlorido, cyanido, hydroxido, oxalato, nitrito. Neutral ligands keep their names, except **aqua** (H_2O), **ammine** (NH_3), **carbonyl** (CO) and **nitrosyl** (NO).
4. Use di-, tri-, tetra- for simple ligands; use **bis-**, **tris-**, **tetrakis-** with brackets for ligands whose names already contain a multiplier, e.g. bis(ethane-1,2-diamine).
5. Give the **oxidation state of the metal in Roman numerals** in brackets, immediately after the metal, with no space.
6. If the complex is an **anion**, the metal takes the suffix **-ate** — and often its Latin stem: ferrate, cuprate, argentate, aurate, plumbate, stannate.

Formula	IUPAC name
$[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Br}_3$	tetraamminediaquacobalt(III) bromide
$\text{K}_4[\text{Fe}(\text{CN})_6]$	potassium hexacyanidoferrate(II)
$[\text{Pt}(\text{NH}_3)_4\text{Cl}_2][\text{PtCl}_4]$	tetraamminedichloridoplatinum(IV) tetrachloridoplatinate(II)
$[\text{CoCl}(\text{ONO})(\text{en})_2]^+$	chloridobis(ethane-1,2-diamine)(nitrito-O)cobalt(III) ion
$[\text{Ni}(\text{CO})_4]$	tetracarbonylnickel(0)

Common slip · Alphabetical order is by the ligand name, so **ammine comes before aqua** (m before q) — hence *tetraamminediaqua*, never the reverse. And in the fourth example, *chlorido* (c) precedes *ethane-1,2-diamine* (e), which precedes *nitrito* (n).

C5 Werner's theory and the AgNO₃ test

- Werner proposed two kinds of valency: **primary** (ionisable, satisfied by negative ions, shown outside the square brackets) and **secondary** (non-ionisable, satisfied by ligands, inside the brackets).
- Only the ions **outside** the coordination sphere are free in solution. So the number of moles of AgCl precipitated by excess AgNO₃ counts the chloride ions **outside** the brackets.

Complex	Ions in solution	mol AgCl per mol complex
[Co(NH ₃) ₆]Cl ₃	4 (1:3 electrolyte)	3
[Co(NH ₃) ₅ Cl]Cl ₂	3 (1:2 electrolyte)	2
[Co(NH ₃) ₄ Cl ₂]Cl	2 (1:1 electrolyte)	1
[Co(NH ₃) ₃ Cl ₃]	0 — a non-electrolyte	0

Hold on to this · More ions ⇒ higher molar conductivity. So the conductivity order is exactly the order of the table read upwards: [Co(NH₃)₃Cl₃] < [Co(NH₃)₄Cl₂]Cl < [Co(NH₃)₅Cl]Cl₂ < [Co(en)₃]Cl₃.

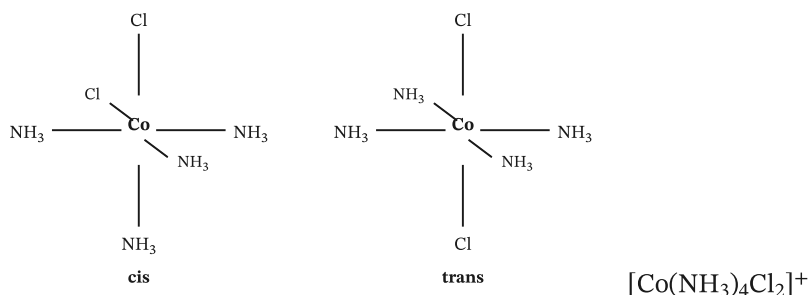
Careful with K₂[PtCl₄]: it ionises to 2K⁺ and [PtCl₄]²⁻. All four chlorides are inside the sphere, so it gives **no** precipitate with AgNO₃.

C6 Structural isomerism — the four kinds

Type	What differs	Example pair
Ionisation	which ion is inside the coordination sphere and which is the counter-ion	[Co(NH ₃) ₅ Cl]SO ₄ and [Co(NH ₃) ₅ SO ₄]Cl
Hydrate	whether water is a ligand or lattice water	[Cr(H ₂ O) ₆]Cl ₃ and [Cr(H ₂ O) ₅ Cl]Cl ₂ ·H ₂ O
Linkage	which atom of an ambidentate ligand is donating	[Co(NH ₃) ₅ (NO ₂)] ²⁺ (nitro, through N) and [Co(NH ₃) ₅ (ONO)] ²⁺ (nitrito, through O)
Coordination	how the ligands are shared between a complex cation and a complex anion	[Co(NH ₃) ₆][Cr(CN) ₆] and [Cr(NH ₃) ₆][Co(CN) ₆]

C7 Stereoisomerism — geometrical and optical

Geometrical (cis–trans) isomerism needs at least two different kinds of ligand, arranged so that two placements are genuinely distinct.



Shape and type	Geometrical isomerism?
Octahedral MA_4B_2	yes — cis and trans
Octahedral MA_3B_3	yes — facial (fac) and meridional (mer)
Octahedral $\text{M}(\text{AA})_2\text{B}_2$, e.g. $[\text{Co}(\text{en})_2\text{Cl}_2]^+$	yes — cis and trans
Octahedral MA_5B	no — only one arrangement exists
Square planar MA_2B_2 , e.g. $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$	yes — cis and trans
Tetrahedral	no — all four positions are adjacent

- **Optical** isomerism requires a non-superimposable mirror image. It is shown chiefly by octahedral complexes with **bidentate** ligands: $[\text{Co}(\text{en})_3]^{3+}$ exists as a d and an l form, and *cis*- $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ is optically active while the *trans* form is not (it has a plane of symmetry).

Common slip · The one that is set as a “which does **not** show geometrical isomerism” distractor is the **MA₅B** type, such as $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$. With five identical ligands, every position for the sixth is equivalent.

C8 Valence bond theory — the procedure

1. Find the oxidation state of the metal and write the d-electron count.
2. Decide whether the ligands are **strong field** (they force pairing) or **weak field** (they do not) — see C10.
3. If pairing occurs, inner (3d) orbitals become available and the complex uses **d²sp³** — an **inner orbital** or **low spin** complex. If not, outer 4d orbitals are used and the hybridisation is **sp³d²** — an **outer orbital** or **high spin** complex.
4. Count the remaining unpaired electrons and get the magnetic moment.

Hybridisation	Shape	Coordination number
sp ³	tetrahedral	4
dsp ²	square planar	4
d ² sp ³	octahedral (inner orbital, low spin)	6
sp ³ d ²	octahedral (outer orbital, high spin)	6

Hold on to this · **Outer orbital complexes involve sp³d² hybridisation and are high spin**; inner orbital complexes are d²sp³ and low spin. That sentence is set verbatim as a fill-in-the-blanks.

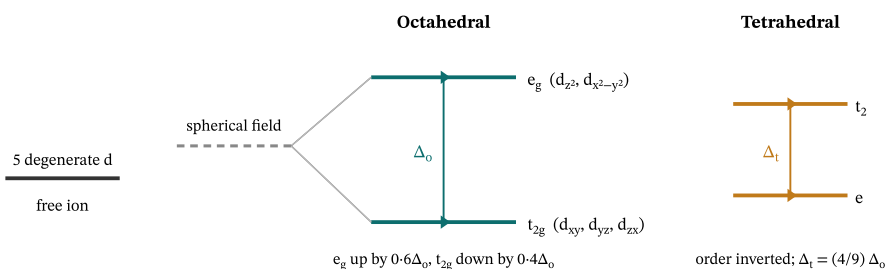
C9 Magnetic moment

$$\mu = \sqrt{n(n+2)} \text{ BM} \quad \text{where } n \text{ is the number of unpaired electrons}$$

n	0	1	2	3	4	5
μ / BM	0	1.73	2.83	3.87	4.90	5.92

- $n = 0 \Rightarrow$ **diamagnetic**; $n \geq 1 \Rightarrow$ **paramagnetic**, and more strongly so as n rises.
- The question is often set backwards: given μ , find n , and hence deduce the geometry or the spin state.

C10 Crystal field theory and the spectrochemical series



- In an **octahedral** field the five d orbitals split into a lower **t_{2g}** set of three and an upper **e_g** set of two, separated by the crystal field splitting energy Δ_o .
- In a **tetrahedral** field the order is **inverted** — e below t₂ — and the splitting is much smaller, $\Delta_t = (4/9)\Delta_o$. Tetrahedral complexes are therefore almost always **high spin**.
- Whether electrons pair up depends on Δ_o against the pairing energy **P**.

The rule that answers every CFT question

$\Delta_o > P$ (strong field): electrons prefer to pair in t_{2g} → **low spin**, fewer unpaired electrons.

$\Delta_o < P$ (weak field): electrons prefer to occupy e_g → **high spin**, more unpaired electrons.

Spectrochemical series (weak field → strong field)



Memory hook • Only the two ends really need memorising: the **halides and water are weak field** (high spin, often paramagnetic), while **CN[−] and CO are strong field** (low spin, often diamagnetic). NH₃ sits in the middle and is treated as strong field for cobalt(III) and nickel(II).

C11 The d⁴ case, which is the one always set

Condition	Configuration	Unpaired electrons	Spin state
$\Delta_o > P$ (strong field)	$t_{2g}^4 e_g^0$	2	low spin
$\Delta_o < P$ (weak field)	$t_{2g}^3 e_g^1$	4	high spin

The same reasoning gives the d⁵, d⁶ and d⁷ cases. d⁶ is worth knowing separately: strong field gives $t_{2g}^6 e_g^0$, **zero** unpaired electrons and a diamagnetic complex — which is why $[\text{Fe}(\text{CN})_6]^{4-}$ and $[\text{Co}(\text{NH}_3)_6]^{3+}$ are both diamagnetic.

C12 Colour — the d-d transition

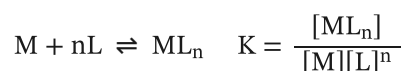
A transition metal complex is coloured because Δ_o happens to correspond to the energy of **visible light**. An electron absorbs a photon and is promoted from t_{2g} to e_g — a **d-d transition** — and the colour seen is the **complement** of the colour absorbed.

- A larger Δ_o means a higher absorbed frequency, so changing the ligand changes the colour. $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ is pale blue; $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is deep blue.
- A complex with **no d electrons** (d^0 , e.g. Sc^{3+} , Ti^{4+}) or a **full d subshell** (d^{10} , e.g. Zn^{2+} , Cu^+) has no possible d-d transition and is therefore **colourless**.

Hold on to this · The two-part answer that earns full marks: **the colour is due to unpaired electrons in the d subshell and a d-d transition**. Both halves are needed — this is set as a fill-in-the-blanks in the specimen papers.

C13 Stability of complexes

Stability is measured by the **stability constant K** for



- **A larger K means a more stable complex.**
- Stability rises with the **charge on the metal ion**, with a **smaller ionic radius**, and strongly with **chelation** — which is why EDTA complexes are so stable.

C14 Two complexes worth knowing individually

Complex	Oxidation state	Hybridisation	Shape	Magnetism
$[\text{Ni}(\text{CO})_4]$	0	sp^3	tetrahedral	diamagnetic
$[\text{Ni}(\text{CN})_4]^{2-}$	+2	dsp^2	square planar	diamagnetic
$[\text{NiCl}_4]^{2-}$	+2	sp^3	tetrahedral	paramagnetic (2 unpaired)
$[\text{Fe}(\text{CN})_6]^{4-}$	+2	d^2sp^3	octahedral	diamagnetic

Hold on to this · The $[\text{Ni}(\text{CN})_4]^{2-}$ against $[\text{NiCl}_4]^{2-}$ pair is the single most-set comparison in this unit. Same metal, same oxidation state, same coordination number — yet different shapes and different magnetism, entirely because CN^- is a strong field ligand that forces the d^8 electrons to pair, while Cl^- is weak field and does not.

C15 Importance and uses

- **Qualitative analysis** — Fe^{3+} with SCN^- gives a blood-red complex; Cu^{2+} with excess NH_3 gives the deep blue $[\text{Cu}(\text{NH}_3)_4]^{2+}$; Ni^{2+} with dimethylglyoxime gives a red precipitate.
- **Extraction of metals** — silver and gold are leached as their cyanide complexes, $[\text{Ag}(\text{CN})_2]^-$.
- **Purification** — nickel by Mond's process, through the volatile $[\text{Ni}(\text{CO})_4]$.
- **Biological systems** — haemoglobin (Fe), chlorophyll (Mg), vitamin B_{12} (Co), carboxypeptidase (Zn).
- **Medicine** — *cis*-platin, *cis*- $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$, in cancer chemotherapy; EDTA in lead poisoning.

D1 Oxidation number, coordination number and the number of ions

Worked on $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$:

1. **Coordination number** — count donor atoms inside the brackets: five NH_3 plus one Cl = **6**.
2. **Charge on the complex ion** — two chlorides sit outside, so the ion must be $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$.
3. **Oxidation number** — $x + 5(0) + (-1) = +2$, so $x = +3$.
4. **Ions in solution** — one complex cation and two chlorides = **3 ions**, a 1:2 electrolyte, giving **2 mol AgCl** with excess AgNO_3 .

■ **determined**

D2 Valence bond treatment of $[\text{Fe}(\text{CN})_6]^{4-}$

1. $x + 6(-1) = -4$, so Fe is **+2**; Fe^{2+} is **$3d^6$** .
2. CN^- is a **strong field** ligand, so it forces the six d electrons to pair up in the three t_{2g} -type orbitals.
3. Two 3d orbitals are thereby vacated, so the metal hybridises as **d^2sp^3** — an **inner orbital** complex.
4. Shape **octahedral**; unpaired electrons **$n = 0$** ; $\mu = 0$, so the complex is **diamagnetic**.

Contrast $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$: water is weak field, so the d^6 configuration stays high spin with $n = 4$, giving sp^3d^2 hybridisation, an outer orbital complex and $\mu = 4.90 \text{ BM}$.

■ **determined**

D3 Crystal field treatment of a d^4 ion

1. Place the four electrons in the split levels, following the rule in C10.

$$\Delta_o > P: t_{2g}^4 e_g^0 \Rightarrow \text{two electrons paired, } n = 2, \text{ low spin, } \mu = 2.83 \text{ BM}$$

$$\Delta_o < P: t_{2g}^3 e_g^1 \Rightarrow \text{all four unpaired, } n = 4, \text{ high spin, } \mu = 4.90 \text{ BM}$$

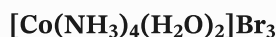
2. Read the answer off: the strong-field complex is the less paramagnetic of the two, and would be the more deeply coloured, since a larger Δ_o absorbs at higher frequency.

■ **determined**

D4 Deducing a molecular formula from conductivity data

“CoBr₃·4NH₃·2H₂O has a molar conductivity corresponding to a 1:3 electrolyte. Write its molecular formula and IUPAC name.”

1. A **1:3 electrolyte** gives one cation and three anions, so **four ions** in all.
2. Three of these must be the three Br[−], so **all three bromides lie outside** the coordination sphere and the complex ion carries a charge of **+3**.
3. Everything else — four NH₃ and two H₂O — must therefore be **inside**, giving a coordination number of 6, which is what cobalt(III) prefers.



Naming it: ligands alphabetically, ammine before aqua, then the metal with its oxidation state, then the counter-ion —

tetraamminediaquacobalt(III) bromide

■ **determined**

3 · Every type of problem that is set

T1 – T14

T1 Oxidation number and coordination number

RECOGNISE IT BY

“What is the oxidation number of the metal / the coordination number?”

METHOD

Follow D1. Count **donor atoms**, not ligand molecules — a bidentate ligand contributes 2 and EDTA contributes 6.

WORKED EXAMPLE

For $[\text{Cr}(\text{en})_2(\text{H}_2\text{O})\text{Cl}]^{2+}$, find the oxidation number of chromium and its coordination number.

1. donor atoms: $2 \text{ en} \times 2 = 4$, plus 1 H₂O, plus 1 Cl = 6
2. $x + 2(0) + 0 + (-1) = +2$

Cr is +3; coordination number 6

T2 Write the IUPAC name from the formula

RECOGNISE IT BY

A formula is given; the name is wanted.

METHOD

Apply C4 in order: ligands alphabetically (by ligand name, ignoring prefixes), anionic ligands in -o, oxidation state in Roman numerals, -ate if the complex is an anion.

WORKED EXAMPLE

Name (a) $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2][\text{PtCl}_4]$ and (b) $[\text{CoCl}(\text{ONO})(\text{en})_2]^+$. (Heritage 2024–25)

- (a) The cation is $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]^{2+}$, so Pt is +4; the anion is $[\text{PtCl}_4]^{2-}$, so Pt is +2. Name: **tetraamminedichloridoplatinum(IV) tetrachloridoplatinate(II)**.
- (b) $x + (-1) + (-1) + 0 = +1$, so Co is +3. Alphabetical order is chlorido, ethane-1,2-diamine, nitrito. Name: **chloridobis(ethane-1,2-diamine)(nitrito-O)cobalt(III) ion**.

T3 Write the formula from the name

RECOGNISE IT BY

The reverse direction, or “deduce the formula from the conductivity”.

METHOD

Work out the charge on the complex ion from the metal's oxidation state and the ligand charges, then add counter-ions to balance. For a conductivity question, follow D4.

WORKED EXAMPLE

$\text{CoBr}_3 \cdot 4\text{NH}_3 \cdot 2\text{H}_2\text{O}$ has a molar conductivity corresponding to a 1:3 electrolyte. Write the molecular formula and the IUPAC name. (Heritage 2024–25)



Fully worked in D4.

T4 Identify the type of isomerism

RECOGNISE IT BY

A pair of formulae, or a single complex, with “state the type of isomerism”.

METHOD

Ask what has moved. Across the brackets $\Rightarrow$ **ionisation** (or hydrate, if it is water). A different donor atom of the same ligand $\Rightarrow$ **linkage**. Ligands swapped between two complex ions $\Rightarrow$ **coordination**. Nothing moved, only the arrangement in space $\Rightarrow$ **geometrical** or **optical**.

WORKED EXAMPLE

State the type of isomerism shown by (a) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{SO}_4$ and (b) $[\text{Co}(\text{en})_3]^{3+}$. (Heritage 2024–25)

- (a) **Ionisation isomerism** — its isomer is $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Cl}$, in which the sulphate is the ligand and chloride the counter-ion. The two give different precipitation tests: the first gives a white precipitate with BaCl_2 , the second with AgNO_3 .
- (b) **Optical isomerism** — three bidentate ligands about an octahedral centre give a propeller-shaped ion with no plane of symmetry, so d and l forms exist.

T5 Draw the geometrical isomers

RECOGNISE IT BY

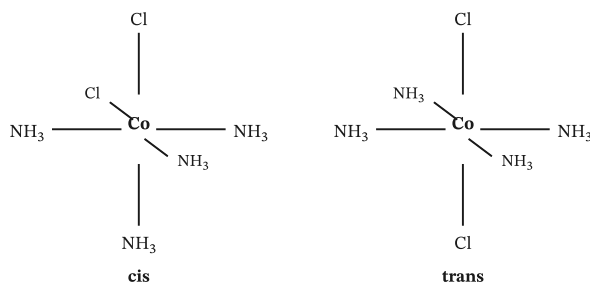
“Draw the geometrical isomers of ...”

METHOD

First check the type is capable of it (C7). Draw the octahedron as a square plane with two axial positions, then put the two odd ligands **adjacent** (cis) and **opposite** (trans).

WORKED EXAMPLE

Draw the geometrical isomers of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$. (Heritage 2024–25)



Neither isomer is optically active here: the trans form has a plane of symmetry, and so does the cis form of this particular complex.

T6 Which complex does not show geometrical isomerism?

RECOGNISE IT BY

A list of four complexes, one of which is the exception.

METHOD

Classify each by type against the table in C7. The answer is almost always the **MA₅B** member of the list, or a tetrahedral one.

WORKED EXAMPLE

Which of the following will **not** show geometrical isomerism? (a) $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (b) $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ (c) $[\text{Co}(\text{NH}_3)_5\text{NO}_2]\text{Cl}_2$ (d) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ (Heritage 2024–25)

Answer • (c). It is of the MA₅B type — five identical ammine ligands and one nitro — so every possible position for the odd ligand is equivalent and only one structure exists. (a) and (b) are MA₄B₂ types and (d) is square planar MA₂B₂; all three give cis and trans forms.

T7 Valence bond theory: hybridisation, shape and magnetism

RECOGNISE IT BY

“State the hybridisation, shape and magnetic behaviour of ...”

METHOD

Follow D2: oxidation state → d-count → strong or weak field → pairing → hybridisation → n → μ . Give **all** the parts the question asks for; each carries a mark.

WORKED EXAMPLE

For $[\text{Fe}(\text{CN})_6]^{4-}$: (a) the oxidation number of iron, (b) the number of unpaired electrons, (c) the hybridisation and magnetic behaviour. (Heritage 2024–25)

- (a) $x + 6(-1) = -4$, so **Fe is +2**, and Fe^{2+} is $3d^6$.
- (b) CN^- is strong field, so all six d electrons pair in t_{2g} : **n = 0**.
- (c) **d^2sp^3** , octahedral, inner orbital, and **diamagnetic** ($\mu = 0$).

T8 Why two complexes of the same metal differ in shape

RECOGNISE IT BY

The $[\text{Ni}(\text{CN})_4]^{2-}$ against $[\text{NiCl}_4]^{2-}$ comparison, or its equivalents.

METHOD

Both have Ni^{2+} , d^8 , coordination number 4. The **ligand field strength** alone decides: a strong field ligand forces pairing, freeing a 3d orbital for dsp^2 and square planar geometry; a weak field ligand does not, leaving sp^3 and tetrahedral.

WORKED EXAMPLE

The geometry of $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ are _____ and _____ respectively. (Heritage 2024–25)

Complex	Ligand	Hybridisation	Geometry	Magnetism
$[\text{Ni}(\text{CN})_4]^{2-}$	strong field	dsp^2	square planar	diamagnetic
$[\text{NiCl}_4]^{2-}$	weak field	sp^3	tetrahedral	paramagnetic, n = 2

T9 Crystal field configuration

RECOGNISE IT BY

“On the basis of CFT, write the electronic configuration of a d^4 ion if $\Delta_o > P$ and if $\Delta_o < P$.”

METHOD

Use the one rule from C10 and fill the levels. Then, if asked, count unpaired electrons and give μ .

WORKED EXAMPLE

Write the CFT electronic configuration of a d^4 ion when (a) $\Delta_o > P$ and (b) $\Delta_o < P$. (Heritage 2024–25)

$$(a) \ t_{2g}^4 e_g^0 \text{ — low spin, } n = 2 \quad (b) \ t_{2g}^3 e_g^1 \text{ — high spin, } n = 4$$

T10 Conductivity order and number of ions

RECOGNISE IT BY

“Arrange the following complexes in increasing order of conductivity in solution.”

METHOD

Count the ions each formula releases — that is, count what lies **outside** the square brackets. More ions means higher conductivity; a complex with nothing outside the brackets is a **non-electrolyte** and conducts least.

WORKED EXAMPLE

Arrange in increasing order of conductivity: $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$, $[\text{Co}(\text{en})_3]\text{Cl}_3$, $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$, $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$ (Heritage 2024–25)

1. ions released: 0, 4, 3, 2 respectively

Increasing conductivity



T11 Precipitate with silver nitrate

RECOGNISE IT BY

“Will it give a precipitate with AgNO_3 ? How does it ionise?”

METHOD

Only **ionisable** halide — outside the brackets — precipitates. Count it, and say how many moles of AgCl or AgBr result per mole of complex.

WORKED EXAMPLE

How does $\text{K}_2[\text{PtCl}_4]$ ionise in water? Will it give a precipitate with AgNO_3 ? (Heritage 2024–25)



Answer · **No precipitate.** All four chlorides are inside the coordination sphere, held by non-ionisable secondary valencies, so there are **no free Cl^- ions** in solution for the silver ion to react with.

The related assertion–reason question is worth watching: $[\text{Cr}(\text{NH}_3)_3\text{Cl}_3]$ indeed gives no precipitate, but the reason “it contains neutral ligands only” is **false** — the three chlorides are anionic ligands. Assertion true, reason false.

T12 Complex or double salt?

RECOGNISE IT BY

Two mixtures in stated molar ratios, one giving the metal ion test and one not.

METHOD

A **1:1** ratio of two simple salts usually means a **double salt**, which dissociates completely and gives the normal tests. A ratio matching a coordination number (1:4, 1:6) means a **complex**, in which the metal ion is not free.

WORKED EXAMPLE

FeSO_4 mixed with $(\text{NH}_4)_2\text{SO}_4$ in a 1:1 molar ratio gives the test for Fe^{2+} , but CuSO_4 with aqueous ammonia in a 1:4 molar ratio does not give the test for Cu^{2+} . Explain. (Heritage 2024–25)

Answer · The first gives **Mohr’s salt**, $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$, a **double salt**. In water it dissociates completely into Fe^{2+} , NH_4^+ and SO_4^{2-} , so free Fe^{2+} is present and gives its usual test. The second gives the **complex** $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$, in which Cu^{2+} is bound inside the coordination sphere by non-ionisable secondary valencies. It does not ionise, so no free Cu^{2+} is available and the test fails.

T13 EDTA and chelation

RECOGNISE IT BY

A passage about EDTA in medicine or water treatment, then short parts.

METHOD

Four facts answer nearly every version: EDTA is **hexadentate**; it forms a **very stable octahedral chelate**; in lead poisoning it **sequesters Pb^{2+}** into a soluble complex that is excreted; and in water analysis it is used to **estimate Ca^{2+} and Mg^{2+}** .

WORKED EXAMPLE

(i) What type of ligand is EDTA? (ii) Why is EDTA used in the treatment of lead poisoning? (iii) What is the geometry of the EDTA complex? (Heritage 2024–25)

- (i) A **hexadentate chelating** ligand — it donates through two nitrogen atoms and four carboxylate oxygens.
- (ii) It **chelates the Pb^{2+} ion**, forming a complex of very high stability constant. The lead is thereby removed from the tissues, and the water-soluble complex is excreted through the kidneys.
- (iii) **Octahedral**, since the coordination number is 6.

T14 Explaining colour or magnetism

RECOGNISE IT BY

“Why is $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ coloured but $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ colourless?”

METHOD

Both answers rest on the **d-electron count**. Colour requires unpaired electrons in the d subshell and a possible **d–d transition**; a d^0 or d^{10} ion has neither. Magnetism requires unpaired electrons, so count n and use $\mu = \sqrt{n(n+2)}$.

WORKED EXAMPLE

Why is $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ violet while $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ is colourless?

Answer · Ti^{3+} is **$3d^1$** . Its single d electron can absorb visible light and be promoted from t_{2g} to e_g — a **d–d transition** — and the complex appears in the complementary colour. Sc^{3+} is **$3d^0$** : with no d electron at all, no d–d transition is possible, so it absorbs nothing in the visible region and is colourless. Zn^{2+} (d^{10}) is colourless for the opposite reason — the subshell is full, so there is no vacancy to be promoted into.

4 · Coordination compounds recall sheet

revise from this page alone

Hybridisation $\Rightarrow$ shape

$sp^3 \rightarrow$ tetrahedral

$dsp^2 \rightarrow$ square planar

$d^2sp^3 \rightarrow$ octahedral, **inner**, low spin

$sp^3d^2 \rightarrow$ octahedral, **outer**, high spin

Magnetic moment

$\mu = \sqrt{n(n+2)}$ BM

$n = 0, 1, 2, 3, 4, 5 \rightarrow 0, 1.73, 2.83, 3.87, 4.90, 5.92$

$n = 0$ diamagnetic; $n \geq 1$ paramagnetic

CFT in one line

Octahedral: t_{2g} below e_g

Tetrahedral: **inverted**, $\Delta_t = (4/9)\Delta_o$

$\Delta_o > P \rightarrow$ low spin; $\Delta_o < P \rightarrow$ high spin

d^4 : t_{2g}^4 ($n=2$) or $t_{2g}^3e_g^1$ ($n=4$)

Field strength

Weak: I^- , Br^- , Cl^- , F^- , H_2O

Strong: NH_3 , en, NO_2^- , CN^- , CO

The four Ni / Fe complexes

$[\text{Ni}(\text{CO})_4]$ — 0, sp^3 , tetrahedral, diamagnetic

$[\text{Ni}(\text{CN})_4]^{2-}$ — dsp^2 , square planar, diamagnetic

$[\text{NiCl}_4]^{2-}$ — sp^3 , tetrahedral, paramagnetic

$[\text{Fe}(\text{CN})_6]^{4-}$ — d^2sp^3 , diamagnetic

Ions and conductivity

$[\text{M}(\text{NH}_3)_6]\text{Cl}_3 \rightarrow 4 \text{ ions, } 3 \text{ AgCl}$

$[\text{M}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2 \rightarrow 3 \text{ ions, } 2 \text{ AgCl}$

$[\text{M}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \rightarrow 2 \text{ ions, } 1 \text{ AgCl}$

$[\text{M}(\text{NH}_3)_3\text{Cl}_3] \rightarrow 0 \text{ ions, no ppt.}$

Isomerism, four structural kinds

Ionisation — ion swaps across the brackets

Hydrate — water inside or as lattice water

Linkage — NO_2^- / SCN^- donor atom

Coordination — ligands swapped between two complex ions

Geometrical isomerism

Yes: MA_4B_2 , MA_3B_3 , $\text{M}(\text{AA})_2\text{B}_2$, square planar MA_2B_2

No: MA_5B , and all tetrahedral

Optical: $[\text{M}(\text{en})_3]$, cis- $[\text{M}(\text{en})_2\text{Cl}_2]$

Nomenclature traps

Alphabetise by **ligand name**, ignore prefixes

ammine before aqua

bis/tris for en, not di/tri

anionic complex $\rightarrow$ metal takes **-ate**

Colour

Due to **unpaired d electrons** and a **d-d transition**

d^0 and d^{10} are **colourless**

Double salt vs complex

Double salt (Mohr's) — dissociates fully, **gives** the metal test

Complex ($[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$) — sphere intact, **no** metal test

EDTA

Hexadentate chelating ligand, octahedral

Lead poisoning; estimation of $\text{Ca}^{2+}/\text{Mg}^{2+}$

Part IV · Haloalkanes and Haloarenes

ISC unit 6 · Nature of the C–X bond · preparation including the named exchange reactions · S_N1 and S_N2 · the full list of nucleophiles and the ambident ones · elimination and Saytzeff · haloarenes and the directive influence of halogen · chloroform, iodoform, freons and DDT. Thirteen concepts, five mechanisms, thirteen problem types.

1 · The concepts, in the order they build

C1 – C13

C1 Nature of the C–X bond

- The bond is **polar**, because the halogen is more electronegative than carbon: C^{δ+}—X^{δ-}. The carbon therefore carries a partial positive charge and is open to attack by a **nucleophile**. That single fact drives the whole unit.
- Bond **strength** falls and bond **length** rises down the group: C–F > C–Cl > C–Br > C–I. So **iodides are the most reactive** and fluorides the least — the weakest bond breaks most easily, and I[–] is also the best leaving group.
- In a **haloarene** the halogen is attached to an sp² carbon and its lone pair conjugates with the ring, giving the C–X bond **partial double bond character**. It is therefore **shorter and stronger**, and very much harder to break.

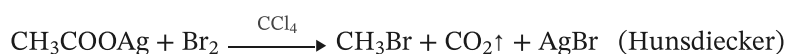
Hold on to this · Two consequences of that last point run through the whole of the haloarene section: aryl halides **resist nucleophilic substitution** (C11), and the halogen is **deactivating but ortho/para directing** (C12).

C2 Physical properties

- Boiling point** rises with the mass of the halogen and the length of the chain: RF < RCl < RBr < RI. Among **isomers**, branching lowers the boiling point, as always.
- Haloalkanes boil **above** the parent alkanes (greater dipole–dipole and van der Waals forces) but **below** alcohols of comparable mass, since they cannot hydrogen bond.
- They are **insoluble in water** — they cannot form hydrogen bonds with it, and the energy needed to break the water–water bonds is not recovered — but are freely soluble in organic solvents.
- They are **denser than water**, and the density increases with the number and mass of halogen atoms.

C3 Preparation — the six routes

Route	Reagents and conditions	Note
From an alkane	X ₂ in sunlight or UV	free radical; gives a mixture, so of little use preparatively
From an alkene	HX	Markovnikov addition; with peroxide and HBr, anti-Markovnikov
From an alcohol	HX, or PCl ₃ , PCl ₅ , SOCl ₂	SOCl₂ is best — both by-products are gases
Finkelstein reaction	R–Cl + NaI in dry acetone	gives the iodide; NaCl precipitates and drives it forward
Swarts reaction	R–Cl + AgF (or Hg ₂ F ₂ , CoF ₂ , SbF ₃)	the only practical way to a fluoride
Hunsdiecker reaction	RCOOAg + Br ₂ in CCl ₄	loses CO ₂ , so the product has one carbon fewer



C4 S_N2 and S_N1 — the two mechanisms compared

	S _N 2	S _N 1
Steps	one , concerted	two , through a carbocation
Rate law	rate = k[RX][nucleophile] — second order	rate = k[RX] — first order , independent of the nucleophile
Molecularity	bimolecular	unimolecular in the slow step
Favoured by	1° halides, a strong nucleophile, a polar aprotic solvent	3° halides, a weak nucleophile, a polar protic solvent
Reactivity order	1° > 2° > 3° (steric)	3° > 2° > 1° (carbocation stability)
Stereochemistry	inversion of configuration (Walden inversion)	racemisation — a mixture of both forms
Intermediate / transition state	a five-coordinate transition state	a planar carbocation

Hold on to this · The two reactivity orders are **exact opposites**, and that is not a coincidence: S_N2 needs room at the back of the carbon, which branching removes, while S_N1 needs a stable carbocation, which branching provides.

For a **primary** haloalkane, reactivity in S_N2 also falls as the chain lengthens, because the approach becomes more hindered: CH₃X > CH₃CH₂X > CH₃CH₂CH₂X.

C5 Optical activity, inversion and racemisation

- A haloalkane whose halogen-bearing carbon has four different groups is **chiral** and optically active — e.g. 2-chlorobutane.
- An S_N2 reaction attacks from the side opposite the leaving group, so the configuration is turned inside out like an umbrella in the wind: the product has the **opposite** configuration. This is **Walden inversion**.
- An S_N1 reaction goes through a **planar** carbocation, which the nucleophile can attack equally from either face. Equal amounts of the two forms are produced, so the product is a **racemic mixture** and is optically inactive.

Hold on to this · So the stereochemical outcome **identifies the mechanism**: complete inversion means S_N2, racemisation means S_N1. That is how the mechanism was established experimentally.

C6 The nucleophiles, and what each gives

Reagent	Product	Name of the product class
aqueous KOH or NaOH	R-OH	alcohol
H ₂ O	R-OH	alcohol (slow)
NaI in dry acetone	R-I	iodoalkane (Finkelstein)
NH ₃ (excess)	R-NH ₂	primary amine
a 1° amine	R ₂ NH	secondary amine
a 2° amine	R ₃ N	tertiary amine
KCN (alcoholic)	R-CN	alkyl cyanide (nitrile) — one carbon more
AgCN	R-NC	alkyl isocyanide
KNO ₂	R-O-N=O	alkyl nitrite
AgNO ₂	R-NO ₂	nitroalkane
silver salt of a fatty acid, RCOOAg	RCOOR'	ester
LiAlH ₄	R-H	alkane (reduction)
alcoholic KOH	alkene	elimination , not substitution

Common slip • **Aqueous** KOH substitutes and gives the alcohol; **alcoholic** KOH eliminates and gives the alkene. One word in the question decides the answer.

C7 Ambident nucleophiles — the K salt against the Ag salt

An **ambident** nucleophile has two different atoms able to donate. Which one attacks depends on how ionic the reagent is.

Reagent	Bonding	Attacks through	Product
KCN	ionic, so free CN ⁻	carbon (the more nucleophilic end)	R-CN, alkyl cyanide
AgCN	largely covalent	nitrogen	R-NC, alkyl isocyanide
KNO ₂	ionic, free NO ₂ ⁻	oxygen	R-O-N=O, alkyl nitrite
AgNO ₂	largely covalent	nitrogen	R-NO ₂ , nitroalkane

Memory hook • **Silver sends the nitrogen in.** With both CN⁻ and NO₂⁻, the silver salt gives the product bonded through N — isocyanide and nitroalkane — while the potassium salt gives the other one.

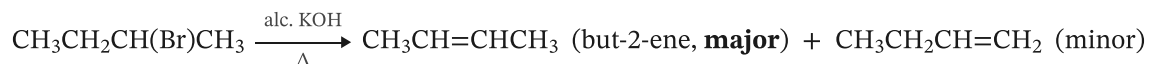
A favourite assertion–reason: “AgCN reacts with ethyl bromide to give ethyl cyanide” is **false** (it gives ethyl isocyanide), while “CN⁻ is an ambident nucleophile” is **true**.

C8 Elimination, and Saytzeff's rule

Alcoholic KOH removes HX across two adjacent carbons — a **β -elimination** — to give an alkene. Where two different alkenes are possible:

Saytzeff's rule

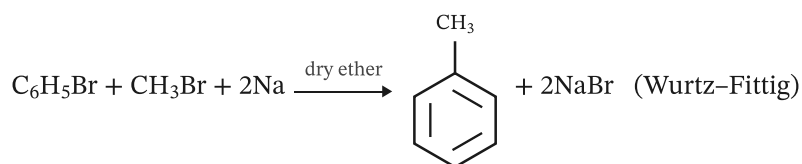
The alkene formed in larger amount is the **more highly substituted one** — the one obtained by removing the hydrogen from the β -carbon bearing the **fewest** hydrogens. The more substituted alkene is more stable, because of hyperconjugation.



Hold on to this • Substitution and elimination **compete**. Elimination is favoured by a **tertiary** halide, a strong bulky base, a high temperature and an alcoholic medium; substitution by a primary halide and an aqueous medium.

C9 Reactions with metals

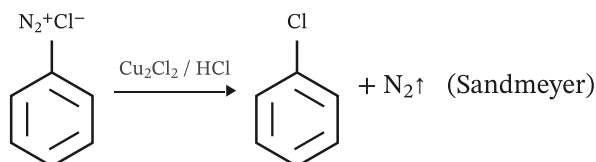
Reaction	Reagents	Product
Wurtz	$2\text{R-X} + 2\text{Na}$ in dry ether	R-R, a symmetrical alkane
Grignard reagent	$\text{R-X} + \text{Mg}$ in dry ether	R-MgX, an alkyl magnesium halide
Wurtz-Fittig	aryl halide + alkyl halide + 2Na, dry ether	an alkylbenzene
Fittig	2 aryl halide + 2Na, dry ether	biphenyl



The Grignard reagent must be made and used in **perfectly dry** ether: any water or alcohol destroys it, giving the alkane. That connects directly to Part VI, where Grignard reagents build alcohols and acids.

C10 Preparation of haloarenes

Route	Reagents	Note
Electrophilic substitution	Cl_2 or Br_2 with anhydrous $\text{AlCl}_3/\text{FeCl}_3$	gives o- and p-products
Sandmeyer's reaction	diazonium salt + $\text{Cu}_2\text{Cl}_2/\text{HCl}$	the general and higher-yielding route
Gattermann reaction	diazonium salt + Cu powder/HCl	a variant of the above
Balz-Schiemann	diazonium fluoroborate, heated	the route to fluorobenzene

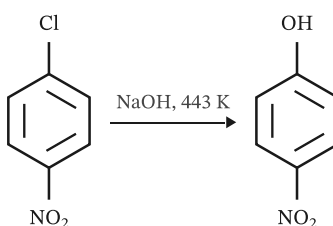


Hold on to this • Direct iodination and fluorination of benzene are impractical, which is why the **diazonium route** exists at all. Remember it as the general gateway: almost any group can be put on a ring through the diazonium salt.

C11 Why haloarenes resist nucleophilic substitution

- The C–X bond has **partial double bond character** from resonance with the ring, so it is **shorter and stronger** and much harder to break (C1).
- The carbon is **sp²** hybridised and therefore more electronegative, holding the electron pair more tightly.
- The **π electron cloud** of the ring repels the approaching nucleophile.
- The **phenyl cation** that an S_N1 route would need is very unstable, and backside S_N2 attack on a flat sp² carbon is geometrically impossible.

So forcing conditions are needed — chlorobenzene with NaOH at **623 K and 300 atm** (Dow's process, Part V). But an **electron-withdrawing group at the ortho or para position** makes it far easier, because it stabilises the intermediate carbanion:



*With two or three nitro groups, hydrolysis proceeds at room temperature. A group at the **meta** position gives no such help, because its resonance withdrawal cannot reach the carbon under attack.*

C12 Directive influence of halogen — deactivating but o/p directing

A halogen on a benzene ring does two opposing things:

- By the **–I effect** it withdraws electron density from the whole ring, making the ring **less reactive** than benzene towards electrophiles — so halogen is **deactivating**.
- By the **+R effect** its lone pair is donated into the ring, and that donation reaches the **ortho and para** positions specifically — so halogen is **ortho/para directing**.

Hold on to this • Halogen is therefore the standard example of a group that is **deactivating yet o/p directing**. The –I effect dominates for *how fast*; the +R effect dominates for *where*.

C13 The named compounds: chloroform, iodoform, freons, DDT

Compound	Preparation / structure	Uses and hazards
Chloroform , CHCl ₃	ethanol or acetone with bleaching powder (haloform reaction)	solvent; formerly an anaesthetic. In air and light it is oxidised to the poisonous gas phosgene , COCl ₂ , so it is kept in dark bottles filled to the brim with 1% ethanol added
Iodoform , CHI ₃	ethanol or acetone with I ₂ and NaOH	a yellow solid used as an antiseptic — the antiseptic action is due to the iodine it liberates , not to the compound itself
Freons , e.g. CCl ₂ F ₂ (Freon-12)	Swarts reaction on CCl ₄	refrigerants and propellants; they diffuse to the stratosphere and deplete the ozone layer , so they are being phased out
DDT	chlorobenzene + chloral (trichloroacetaldehyde) with conc. H ₂ SO ₄	the first modern insecticide; non-biodegradable , so it accumulates in fatty tissue up the food chain, and is now banned in most countries
Dichloromethane , CH ₂ Cl ₂	—	paint remover, degreaser; harms the central nervous system
Tetrachloromethane , CCl ₄	—	solvent and former fire extinguisher; damages the liver, and also depletes ozone

2 · The mechanisms

M1 – M5

M1 S_N2 — one step, backside attack, inversion

1. The nucleophile approaches the carbon from the side **directly opposite** the leaving group — the only side not shielded by the halogen.
2. Bond making and bond breaking happen **together**, passing through a single transition state in which the carbon is **five-coordinate** and the three unchanged groups lie in a plane.
3. As the halide leaves, those three groups fold through, and the configuration is **inverted**.



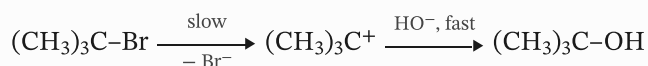
rate = k[CH₃Br][HO[−]] — **second order, and the nucleophile appears in the rate law**

Because the attack needs room at the back, reactivity falls sharply with branching: 1° > 2° > 3°. A tertiary halide is essentially inert to S_N2.

■ **mechanism complete**

M2 S_N1 — two steps, carbocation, racemisation

1. **Slow, rate determining**: the C–X bond ionises on its own, giving a **planar carbocation** and the halide ion.
2. **Fast**: the nucleophile attacks the carbocation.



rate = k[(CH₃)₃CBr] — **first order; the nucleophile does not appear**, because it enters after the slow step

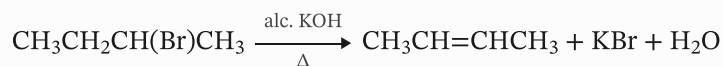
The carbocation is **flat**, so the nucleophile is equally likely to arrive at either face. Equal amounts of the two configurations result, and the product is a **racemic mixture**.

Reactivity follows carbocation stability, 3° > 2° > 1°. This is also the mechanism behind the **Lucas test** in Part V.

■ **mechanism complete**

M3 β -elimination (E2) and Saytzeff's rule

1. The base removes a hydrogen from a carbon **adjacent** to the one bearing the halogen (the β -carbon).
2. Simultaneously the halide leaves and the released electron pair forms the π **bond**.
3. Where two β -carbons are available, the **more substituted alkene** predominates (Saytzeff), because it is stabilised by hyperconjugation.

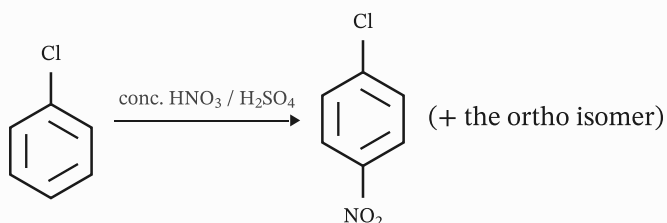


Count the hydrogens on each β -carbon and take the H from the one with **fewest** — that is the practical form of Saytzeff's rule.

■ mechanism complete

M4 Electrophilic substitution in a haloarene

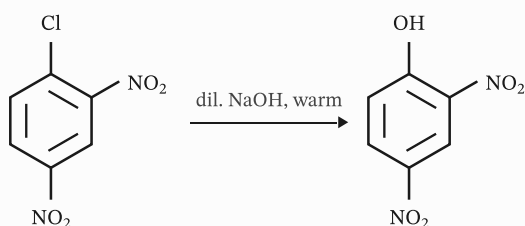
1. The electrophile is generated as usual — Cl^+ from $\text{Cl}_2/\text{AlCl}_3$, NO_2^+ from conc. $\text{HNO}_3/\text{H}_2\text{SO}_4$.
2. It attacks the ring to give a resonance-stabilised carbocation (the arenium ion), and loss of a proton restores aromaticity.
3. **Rate:** the halogen's $-\text{I}$ effect has drained the ring, so the reaction is **slower than with benzene**.
4. **Position:** the halogen's lone pair can be donated by resonance, and that donation places extra electron density specifically at the **ortho and para** carbons, so those positions are attacked.



■ mechanism complete

M5 Nucleophilic substitution in a haloarene, and why nitro groups help

1. The nucleophile adds to the carbon bearing the halogen, giving a **carbanion** intermediate in which the negative charge is delocalised round the ring.
2. The halide is then expelled, restoring aromaticity.
3. Step 1 is very difficult for chlorobenzene itself, because the carbanion is poorly stabilised — hence 623 K and 300 atm.
4. An **electron-withdrawing group at the ortho or para position** can accept that negative charge by resonance, stabilising the intermediate and making the substitution far easier. A **meta** group cannot, because its resonance withdrawal does not reach the carbon under attack.



So 2,4-dinitrochlorobenzene is hydrolysed by warm dilute alkali, while chlorobenzene needs a pressure vessel. The contrast is a standard three-mark question.

■ mechanism complete

3 · Every type of problem that is set

T1 – T13

T1 Nomenclature and classification

RECOGNISE IT BY

“Give the IUPAC name”, or classify as 1°, 2° or 3°.

METHOD

Count the alkyl groups on the carbon carrying the halogen. Number the chain to give the halogen the lowest locant consistent with the principal chain.

WORKED EXAMPLE

Name and classify (a) $(\text{CH}_3)_3\text{C}-\text{Br}$ (b) $\text{CH}_3\text{CH}(\text{Cl})\text{CH}_2\text{CH}_3$

- (a) **2-bromo-2-methylpropane**, tertiary
- (b) **2-chlorobutane**, secondary — and it is chiral, so it is optically active

T2 Deciding the mechanism, and hence the stereochemistry

RECOGNISE IT BY

“Which mechanism does it follow?”, or a rate law is quoted and the structure is wanted.

METHOD

Use the rate law if given: if the nucleophile appears in it, the mechanism is S_N2 and the halide is primary; if not, it is S_N1 and the halide is tertiary. Then read off the stereochemistry — inversion for S_N2 , racemisation for S_N1 .

WORKED EXAMPLE

An organic compound [A], C_4H_9Br , reacts with aqueous KOH at a rate that depends only on the concentration of [A]. Its structural isomer [B] reacts at a rate depending on the concentrations of both [B] and KOH. Identify [A] and [B]. (ISC 2025)

- [A]: rate independent of KOH $\Rightarrow S_N1 \Rightarrow$ tertiary halide
- [B]: rate depends on both $\Rightarrow S_N2 \Rightarrow$ primary halide

[A] = **2-bromo-2-methylpropane**, $(CH_3)_3C-Br$ [B] = **1-bromobutane**, $CH_3CH_2CH_2CH_2Br$

T3 Arranging in order of reactivity

RECOGNISE IT BY

“Arrange in increasing order of reactivity towards S_N1 / S_N2 / a given nucleophile.”

METHOD

Decide which mechanism is in play, then apply the matching order — they are opposites. Across the halogens, reactivity always runs $R-I > R-Br > R-Cl > R-F$, because the weakest bond breaks most easily.

WORKED EXAMPLE

The reactivity order of primary alkyl halides towards nucleophilic substitution is: (Heritage 2024–25)

Increasing reactivity



All three are primary, so the mechanism is S_N2 throughout and the only variable is **steric hindrance** at the carbon under attack. Methyl is the least hindered and so the most reactive.

T4 Ambident nucleophile — which product?

RECOGNISE IT BY

KCN against AgCN, or KNO_2 against $AgNO_2$.

METHOD

Apply the rule in C7: the **silver** salt is covalent and attacks through **nitrogen**, giving the isocyanide or the nitroalkane; the potassium salt is ionic and gives the cyanide or the nitrite.

WORKED EXAMPLE

Write the product when ethyl bromide is treated with (a) alcoholic KCN (b) AgCN (c) KNO_2 (d) $AgNO_2$.

Reagent	Product	Name
alc. KCN	C_2H_5CN	ethyl cyanide (propanenitrile)
AgCN	C_2H_5NC	ethyl isocyanide
KNO_2	$C_2H_5-O-N=O$	ethyl nitrite
$AgNO_2$	$C_2H_5-NO_2$	nitroethane

T5 Substitution or elimination?

RECOGNISE IT BY

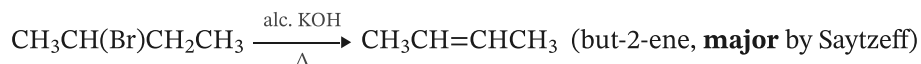
The reagent is KOH, and you must decide which.

METHOD

One word decides it: **aqueous** KOH substitutes and gives the alcohol; **alcoholic** KOH eliminates and gives the alkene. If it eliminates and two alkenes are possible, apply Saytzeff.

WORKED EXAMPLE

What is the product when 2-bromobutane is treated with (a) aqueous KOH (b) alcoholic KOH?



T6 Conversions

RECOGNISE IT BY

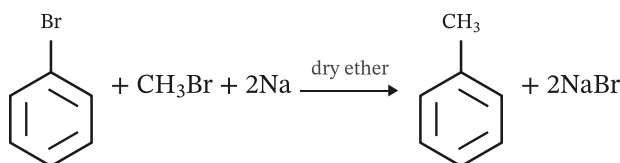
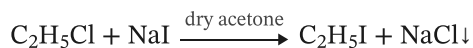
“How will you bring about the following conversion?”

METHOD

Ask what must change. To go **up** one carbon use KCN then hydrolysis, or a Grignard reagent; to go **down** one use Hunsdiecker or decarboxylation; to swap halogens use Finkelstein or Swarts; to join two chains use Wurtz.

WORKED EXAMPLE

Convert (a) ethyl chloride to ethyl iodide (b) bromobenzene to toluene (c) ethyl bromide to propanoic acid.



T7 Identify [A], [B], [C] in a sequence

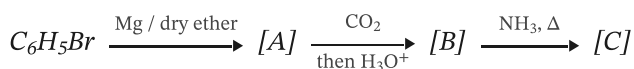
RECOGNISE IT BY

A chain of arrows starting from a halide.

METHOD

Track the **carbon count** at each step, as in Part V. KCN and Grignard add a carbon; Hunsdiecker and decarboxylation remove one; Wurtz doubles the chain.

WORKED EXAMPLE



- [A] = **C₆H₅MgBr**, phenyl magnesium bromide
- [B] = **benzoic acid** — a Grignard reagent with CO₂ then acid gives the carboxylic acid with one carbon more
- [C] = **benzamide**, C₆H₅CONH₂, via ammonium benzoate

T8 Why haloarenes are less reactive than haloalkanes

RECOGNISE IT BY

“Chlorobenzene is far less reactive than ethyl chloride towards nucleophilic substitution. Explain.”

METHOD

Give **at least two** of the four reasons in C11 — the marks are usually two or three, one per distinct reason. The resonance argument is the one that must appear.

WORKED EXAMPLE

Why does chlorobenzene not undergo nucleophilic substitution readily?

- The C–Cl bond has **partial double bond character** from resonance with the ring, so it is shorter and stronger and hard to break.
- The carbon is **sp²** hybridised and more electronegative, holding the bonding pair more tightly.
- The ring's **π electron cloud repels** the incoming nucleophile.
- Neither mechanism is available: the **phenyl cation** is very unstable, and backside S_N2 attack on a planar sp² carbon is geometrically impossible.

T9 Directive influence of halogen

RECOGNISE IT BY

“Halogen is deactivating yet ortho/para directing. Explain.”

METHOD

Separate the two questions the examiner is really asking. **How fast?** — the –I effect withdraws from the whole ring, so deactivating. **Where?** — the +R effect donates the lone pair specifically to the o and p positions, so o/p directing.

WORKED EXAMPLE

Explain why chlorine deactivates the benzene ring but still directs substitution to the ortho and para positions.

Answer · Chlorine exerts two opposing effects. Its strong **–I effect** pulls electron density out of the ring as a whole, so the ring is less nucleophilic than benzene and reacts more slowly — chlorine is **deactivating**. But chlorine also has lone pairs, and by the **+R effect** one of these is donated into the ring; the resonance structures that result place the extra electron density specifically at the **ortho and para** carbons. Those positions are therefore the ones attacked. The –I effect governs the *rate*; the +R effect governs the *orientation*.

T10 Named reaction equations

RECOGNISE IT BY

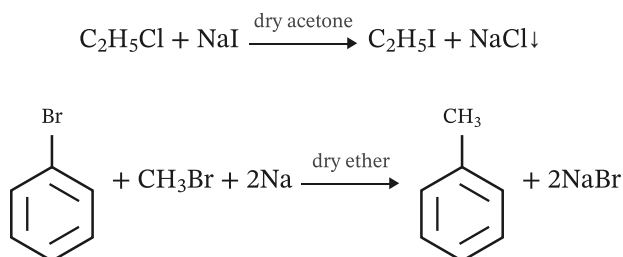
“Give balanced equations for the following named reactions.”

METHOD

Learn the eight of this unit as name ↔ equation pairs: Finkelstein, Swarts, Hunsdiecker, Wurtz, Wurtz–Fittig, Fittig, Sandmeyer, Gattermann. Conditions over the arrow carry marks.

WORKED EXAMPLE

Write equations for the Finkelstein reaction and the Wurtz–Fittig reaction.



T11 Distinguish between two halides

RECOGNISE IT BY

“Give a chemical test to distinguish ...”

METHOD

Two workhorse tests. **Alcoholic AgNO₃**: a haloalkane gives a precipitate of AgX (fastest for the iodide and for a tertiary halide), while a **haloarene gives none**, because it has no ionisable halogen. The **Lucas-type rate** distinguishes 1°, 2° and 3°.

WORKED EXAMPLE

Distinguish between chlorobenzene and benzyl chloride.

Answer · Warm each with **alcoholic silver nitrate**. **Benzyl chloride** (C₆H₅CH₂Cl) has its chlorine on an sp³ side-chain carbon and ionises readily, giving a **white precipitate of AgCl**. **Chlorobenzene** has its chlorine on the ring, held by a bond with partial double bond character, and gives **no precipitate**.

T12 Optical activity and the fate of configuration

RECOGNISE IT BY

A chiral halide reacting by a stated or implied mechanism.

METHOD

Check first that the carbon has four different groups. Then: S_N2 gives a single product of **inverted** configuration; S_N1 gives a **racemic mixture**, which is optically inactive.

WORKED EXAMPLE

(–)-2-Bromooctane is hydrolysed by aqueous KOH to give (+)-octan-2-ol as the only product. What does this tell you about the mechanism?

Answer · A single product with the **opposite** sign of rotation means complete **inversion of configuration** — Walden inversion — and no racemisation. The reaction must therefore be S_N2: the hydroxide attacked from the side opposite the bromine in one concerted step. Had it been S_N1, the planar carbocation would have been attacked from both faces and the product would have been optically inactive.

T13 Uses and environmental effects

RECOGNISE IT BY

“State one use of ...”, or “why is chloroform stored in dark bottles?”

METHOD

Learn the table in C13. The two most-asked are chloroform → **phosgene** and freons → **ozone depletion**.

WORKED EXAMPLE

Why is chloroform stored in dark coloured bottles, filled to the brim?

Answer · In the presence of **air and light** chloroform is slowly oxidised to **phosgene**, COCl_2 , which is extremely poisonous. Dark bottles exclude the light and filling them to the brim excludes the air; about 1% of ethanol is also added, which converts any phosgene formed into harmless diethyl carbonate.



4 · Haloalkanes and haloarenes recall sheet

revise from this page alone

$\text{S}_{\text{N}}1$ vs $\text{S}_{\text{N}}2$

$\text{S}_{\text{N}}2$: one step, rate = $k[\text{RX}][\text{Nu}]$, $1^\circ > 2^\circ > 3^\circ$, **inversion**

$\text{S}_{\text{N}}1$: two steps, rate = $k[\text{RX}]$, $3^\circ > 2^\circ > 1^\circ$, **racemisation**

Halogen reactivity

$\text{R-I} > \text{R-Br} > \text{R-Cl} > \text{R-F}$

weakest bond breaks first; I^- is the best leaving group

One word decides

aqueous KOH → alcohol (substitution)

alcoholic KOH → alkene (elimination)

Silver sends the nitrogen in

$\text{KCN} \rightarrow \text{R-CN}$ cyanide | $\text{AgCN} \rightarrow \text{R-NC}$ isocyanide

$\text{KNO}_2 \rightarrow \text{R-ONO}$ nitrite | $\text{AgNO}_2 \rightarrow \text{R-NO}_2$ nitroalkane

The named reactions

Finkelstein $\text{RCl} + \text{NaI}$ / dry acetone

Swarts $\text{RCl} + \text{AgF}$

Hunsdiecker $\text{RCOOAg} + \text{Br}_2$, one C fewer

Wurtz $2\text{RX} + 2\text{Na} \rightarrow \text{R-R}$

Wurtz-Fittig $\text{ArX} + \text{RX} + 2\text{Na}$

Fittig $2\text{ArX} + 2\text{Na} \rightarrow$ biphenyl

Sandmeyer diazonium + $\text{Cu}_2\text{Cl}_2/\text{HCl}$

Gattermann diazonium + Cu/HCl

Saytzeff

Take the H from the β -carbon with the **fewest** hydrogens
⇒ the **more substituted** alkene is the major product

Haloarenes resist S_{N}

partial double bond character (resonance)

sp^2 carbon

π cloud repels the nucleophile

phenyl cation unstable; no backside attack

Halogen on a ring

-I ⇒ deactivating (how fast)

+R ⇒ ortho/para directing (where)

Nitro groups help

o/p $-\text{NO}_2$ stabilises the carbanion ⇒ easy hydrolysis

meta $-\text{NO}_2$ does not help

Named compounds

$\text{CHCl}_3 \rightarrow$ **phosgene** in air/light; dark bottles + 1% ethanol

CHI_3 antiseptic via liberated I_2

Freons → **ozone depletion**

DDT → non-biodegradable, bioaccumulates

Grignard

$\text{RX} + \text{Mg}$ / **dry** ether → RMgX

water or alcohol destroys it, giving the alkane

Part V · Alcohols, Phenols and Ethers

Fifteen concepts, eight mechanisms, nine reaction tables built for self-testing, seventeen problem types and a one-page recall sheet.

1 · The concepts, in the order they build

C1 – C15

C1 Classification and nomenclature

Class	Basis	Examples
Monohydric / dihydric / polyhydric	number of –OH groups	ethanol / ethylene glycol / glycerol
Primary (1°)	–OH on a carbon bearing one alkyl group	CH ₃ CH ₂ OH
Secondary (2°)	–OH on a carbon bearing two alkyl groups	CH ₃ CH(OH)CH ₃
Tertiary (3°)	–OH on a carbon bearing three alkyl groups	(CH ₃) ₃ COH
Phenol	–OH attached directly to the aromatic ring	C ₆ H ₅ OH
Ether	R–O–R', symmetrical or unsymmetrical	C ₂ H ₅ OC ₂ H ₅ , CH ₃ OC ₂ H ₅

Common slip · Benzyl alcohol, C₆H₅CH₂OH, is **not** a phenol. Its –OH is on the side chain, not on the ring, so it behaves as an ordinary primary alcohol — no violet colour with FeCl₃, no reaction with NaOH.

Hold on to this · Ethers are named as **alkoxy-alkanes**: the smaller group becomes the alkoxy prefix. CH₃–O–C₂H₅ is methoxyethane; C₆H₅–O–CH₃ is methoxybenzene (anisole).

C2 Structure — why phenol is different

- In an alcohol and in an ether the oxygen is **sp³ hybridised** and the C–O–H / C–O–C angle is slightly more than tetrahedral (about 109° in alcohols, 111.7° in ethers) because of the bulk of the alkyl groups and the lone pairs.
- In phenol the –OH is attached to an **sp² carbon**. The oxygen lone pair conjugates with the ring, giving the C–O bond **partial double bond character** — it is shorter (136 pm) than the C–O bond in methanol (142 pm).

Hold on to this · Two consequences follow from that conjugation, and between them they explain nearly all of phenol's chemistry: the **O–H bond is weakened** (so phenol is acidic), and the **ring is activated** (so electrophilic substitution is easy, at the ortho and para positions). A third follows too: the **C–O bond becomes very hard to break**, which is why phenol resists nucleophilic substitution and why aryl ethers are never cleaved on the aryl side.

C3 Isomerism

- Chain and position isomerism** among alcohols: butan-1-ol and butan-2-ol.
- Functional isomerism** between an alcohol and an ether of the same formula: C₂H₆O is ethanol *or* methoxymethane.
- Metamerism** among ethers — unequal distribution of carbons about the oxygen: C₄H₁₀O gives C₂H₅OC₂H₅ and CH₃OC₃H₇.
- Optical isomerism** where the carbinol carbon carries four different groups: butan-2-ol is chiral, butan-1-ol is not.

C4 Physical properties

Boiling point. Alcohols and phenols form **intermolecular hydrogen bonds** and so boil far above ethers, alkyl halides and hydrocarbons of comparable mass. Ethers have no O–H hydrogen and cannot hydrogen bond to themselves.

Boiling point

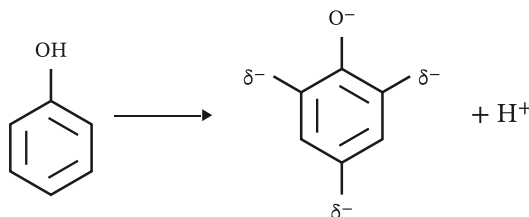
butane 273 K < diethyl ether 308 K < 2-methylpropan-2-ol 356 K < butan-2-ol 371 K < butan-1-ol 391 K

- Among **isomeric alcohols the order is $1^\circ > 2^\circ > 3^\circ$** : branching makes the molecule compact, reduces the surface area available for van der Waals contact, and crowds the –OH group.
- **More –OH groups $\Rightarrow$ much higher boiling point** (ethylene glycol 470 K against ethanol 351 K).
- **Solubility** in water also depends on hydrogen bonding, but falls as the hydrocarbon part grows: methanol and ethanol are fully miscible, butan-1-ol only partly, hexan-1-ol almost insoluble.
- **Phenol is only partly soluble** (about 8 g per 100 g water) — one –OH group cannot compensate for a whole benzene ring.

Hold on to this · Ethers are excellent solvents precisely because they cannot hydrogen bond to themselves but their oxygen lone pairs can accept hydrogen bonds and coordinate to metals — hence dry ether for Grignard reagents.

C5 Acidic character — the central idea of the chapter

All three classes have an O–H or C–O–C oxygen, but only alcohols and phenols ionise. Decide acidity by asking one question: **how stable is the anion left behind?**



In the **phenoxide ion** the negative charge is delocalised by resonance on to the **two ortho and the para carbons**, as marked. This spreads the charge and stabilises the ion, so phenol gives up its proton readily. In an **alkoxide ion** the charge stays on one oxygen, and the electron-releasing (+I) alkyl group pushes more charge on to it, destabilising the ion.

Acidic strength



- **Alkyl groups weaken the acid** (+I): CH₃OH > 1° > 2° > 3°.
- **Electron-withdrawing groups strengthen it** (–I): CH₃CH₂OH < CCl₃CH₂OH < CF₃CH₂OH (F pulls harder than Cl).
- On a phenol ring, **–NO₂ at o- or p- strengthens** the acid strongly (it can delocalise the charge by resonance); at **m-** only the weaker inductive effect operates. **–CH₃ weakens** it, most when closest to the –OH.

Hold on to this · Phenol is acidic enough to dissolve in **NaOH** but not enough to liberate CO₂ from **NaHCO₃** — it sits between water and carbonic acid. That single fact is the standard test separating phenol from a carboxylic acid.

- **Litmus.** Phenol is weakly acidic and turns **blue litmus red** (faintly); an alcohol such as ethanol is **neutral to litmus**. This is set as a fill-in-the-blank.
- **The exception worth knowing. Picric acid** (2,4,6-trinitrophenol) *does* dissolve in NaHCO₃ with effervescence, unlike ordinary phenols. Its three –NO₂ groups withdraw so much electron density that the anion becomes extremely stable, making it a **stronger acid than carbonic acid** — comparable to a carboxylic acid.

C6 The four handles on an alcohol — the frame for all its reactions

Every reaction of an alcohol attacks one of four sites. Sorting them this way turns a list of twenty reactions into four short lists.

HANDLE 1

The O–H bond

The hydrogen is acidic. Na, and acylating agents such as CH₃COCl, attack here. Carbon skeleton untouched.

HANDLE 2

The C–O bond

The –OH is replaced by another group: HX, PCl₃, PCl₅, SOCl₂. Needs the –OH protonated first.

HANDLE 3

The β C–H bond

Dehydration. Conc. H₂SO₄ at 443 K, or Al₂O₃ at 623 K, gives an alkene.

HANDLE 4

The carbinol C–H

Oxidation and dehydrogenation. 1° → aldehyde → acid; 2° → ketone; 3° has no such H, so it resists.

Memory hook · Whenever you are asked “what happens when ...”, first identify which of the four handles the reagent grabs. Sodium and acetyl chloride grab handle 1; anything with a halogen grabs handle 2; heat with strong acid grabs handle 3; anything oxidising grabs handle 4.

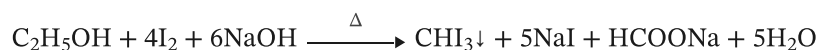
C7 Distinguishing 1°, 2° and 3° alcohols

Test	1° alcohol	2° alcohol	3° alcohol
Lucas reagent (conc. HCl + anhyd. ZnCl ₂), room temp.	no turbidity	turbidity in ~5 min	turbidity at once
Oxidation (acidified K ₂ Cr ₂ O ₇)	aldehyde, then acid	ketone	no reaction (mild conditions)
Cu, 573 K	aldehyde + H ₂	ketone + H ₂	alkene (dehydration)
Victor Meyer (colour of the nitrolic product)	blood red	blue	colourless

Hold on to this · The Lucas test works because the reaction is S_N1. The rate of turbidity is a direct read-out of **carbocation stability**, 3° > 2° > 1°. The cloudiness is the insoluble alkyl chloride separating from the aqueous reagent.

C8 The iodoform test — one structural requirement

A yellow crystalline precipitate of iodoform (CHI₃) on warming with I₂ and NaOH is given **only** by a compound containing a **CH₃–CO–** group, or a **CH₃–CH(OH)–** group that can be oxidised to it.



Gives iodoform	Does not
ethanol, propan-2-ol, butan-2-ol	methanol, propan-1-ol, butan-1-ol
acetaldehyde, acetone, acetophenone	formaldehyde, benzaldehyde, propanal

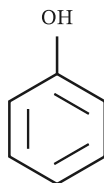
Memory hook · Look for the **CH₃ next to the C bearing the O**. Methanol fails because it has only one carbon; propan-1-ol fails because its carbinol carbon carries a CH₂, not a CH₃.

C9 The two faces of phenol

Phenol's reactions divide cleanly in two. Deciding which face a reagent attacks tells you the answer before you write anything.

Face 1 — at the –OH group

- NaOH → sodium phenoxide (acidic character)
- Na → sodium phenoxide + ½H₂
- Zn dust, distil → benzene (–OH removed)
- CH₃COCl / (CH₃CO)₂O → phenyl acetate
- PCl₅ → chlorobenzene (poor yield)
- CH₃I on the phenoxide → anisole
- neutral FeCl₃ → violet complex

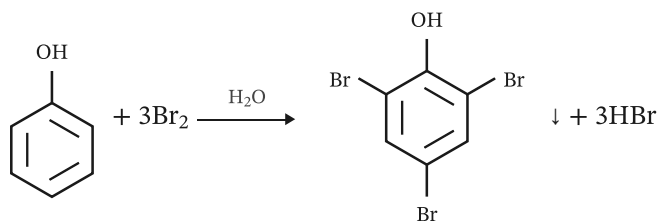
**Face 2 — on the benzene ring**

- Br₂/CS₂ → o- and p-bromophenol
- Br₂ water → 2,4,6-tribromophenol (white ppt)
- dil. HNO₃ → o- and p-nitrophenol
- conc. H₂SO₄ then conc. HNO₃ → picric acid
- CO₂, 400 K, 4–7 atm → salicylic acid (Kolbe)
- CHCl₃ + NaOH → salicylaldehyde (Reimer–Tiemann)
- diazonium salt → azo dye

Hold on to this · Every ring reaction goes to the **ortho and para** positions, because –OH is an **activating, o/p-directing** group. And every ring reaction is **easier than with benzene** — phenol needs no catalyst for bromination, and gives trisubstitution in water at room temperature.

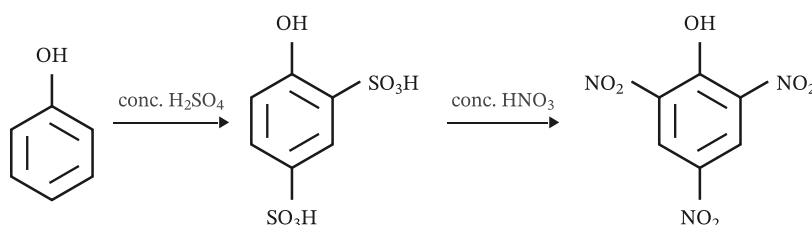
C10 Why the solvent decides how far bromination goes

- In a **non-polar solvent** (CS₂, 273 K) phenol stays un-ionised. It is activated, but only moderately, and gives **monobromination** — o- and p-bromophenol.
- In **water** phenol ionises slightly to **phenoxide**, which carries a full negative charge and is a far stronger activator. Substitution then runs to completion, giving **2,4,6-tribromophenol** as a white precipitate.



C11 Why phenol must be sulphonated before nitration

Concentrated nitric acid is a strong **oxidising** agent, and phenol — being highly activated — is oxidised to dark tarry products rather than nitrated. The remedy is to sulphonate first: the two –SO₃H groups **deactivate the ring** just enough for it to survive the nitric acid, and are then themselves **displaced** by nitro groups.



Hold on to this · The product, **2,4,6-trinitrophenol**, is **picric acid** — a strong acid (all three nitro groups withdrawing) and an explosive.

C12 Ethers — the three things that happen to them

- Cleavage of the C–O bond by HI** (or HBr). This is the reaction that matters most; see C13 for the rule about which fragment takes the iodine.
- Substitution at the α-carbon by chlorine.** In the dark, $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5 + 2\text{Cl}_2 \rightarrow \text{CH}_3\text{CHCl-O-CHClCH}_3 + 2\text{HCl}$; in sunlight chlorination goes all the way to the perchloro ether.
- Peroxide formation on standing in air and light.** The peroxides are explosive and less volatile than the ether, so they concentrate in the residue on distillation — which is why an **ether must never be distilled to dryness**.

With PCl₅, unusually, **both** C–O bonds break: $\text{R-O-R} + \text{PCl}_5 \rightarrow 2\text{R-Cl} + \text{POCl}_3$.

Peroxides are detected with acidified KI (brown colour of liberated iodine) and destroyed by shaking with ferrous sulphate solution.

C13 Ether cleavage by HI — which fragment gets the iodine

HI first protonates the ether oxygen, turning it into a good leaving group. What happens next depends entirely on the **type of carbon** the iodide has to attack.

Type of ether	Mechanism	Iodide formed from	Example
Both groups 1° (or methyl)	S _N 2	the smaller, less hindered group	$\text{CH}_3\text{OC}_2\text{H}_5 \rightarrow \text{CH}_3\text{I} + \text{C}_2\text{H}_5\text{OH}$
One group 3°	S _N 1	the tertiary group (stable carbocation)	$(\text{CH}_3)_3\text{COCH}_3 \rightarrow (\text{CH}_3)_3\text{CI} + \text{CH}_3\text{OH}$
Aryl alkyl (anisole)	S _N 2 at the alkyl carbon only	the alkyl group; phenol is always the other product	$\text{C}_6\text{H}_5\text{OCH}_3 \rightarrow \text{C}_6\text{H}_5\text{OH} + \text{CH}_3\text{I}$

Hold on to this · The aryl case is absolute: the **sp² C–O bond is never broken**, because it has partial double bond character (C2) and an sp² carbon cannot undergo backside attack. Anisole with HI gives phenol and methyl iodide — **never** iodobenzene.

Memory hook · S_N2 — **the small one gets the iodine**. S_N1 — **the big one gets the iodine**. With excess hot HI, everything ends up as the iodide anyway.

C14 Dehydration, and the order of ease

Dehydration is an **E1** elimination through a carbocation (M1), so its ease follows carbocation stability.

Ease of dehydration



- The **temperature decides the product**: conc. H₂SO₄ at **413 K** gives the ether (intermolecular), at **443 K** the alkene (intramolecular).
- Where two alkenes are possible, the **more substituted** one predominates (Saytzeff rule).

Hold on to this · Note the contrast: dehydration and the Lucas test both go **3° > 2° > 1°** because both make a carbocation, but **esterification** goes **1° > 2° > 3°**, because there the alcohol attacks as a nucleophile and bulk gets in the way.

C15 Uses worth remembering

Compound	Uses
Methanol	solvent; making formaldehyde; antifreeze; denaturing ethanol. Highly toxic — causes blindness then death.
Ethanol	alcoholic drinks; solvent; sanitiser; a petrol substitute and additive; making ether, chloroform and acetic acid.
Ethylene glycol	antifreeze; making terylene.
Phenol	0.2% solution as an antiseptic, 1% as a disinfectant ; making bakelite, picric acid, salicylic acid, drugs and dyes.
Diethyl ether	solvent, especially for Grignard reagents; formerly an anaesthetic.

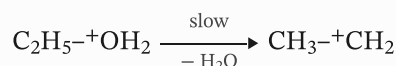
M1 Acid-catalysed dehydration of ethanol to ethene (E1)

The problem the acid solves: **–OH is a very poor leaving group**. Protonating it converts it into water, which is an excellent one.

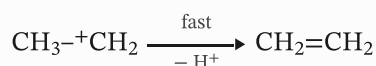
1. **Protonation (fast)**. The oxygen lone pair takes a proton:



2. **Loss of water (slow — rate determining)**. The C–O bond breaks heterolytically, giving a carbocation:

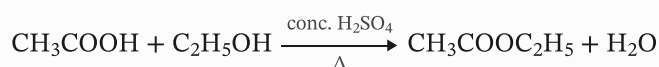


3. **Loss of a β -proton (fast)**. A base — HSO_4^- or another alcohol molecule — removes a hydrogen from the carbon next to the positive centre, and the released electron pair forms the π bond:



The proton is regenerated in step 3, confirming that the acid is a catalyst. Because the slow step forms a carbocation, ease of dehydration follows $3^\circ > 2^\circ > 1^\circ$.

■ mechanism complete

M2 Esterification (nucleophilic acyl substitution)

1. **Protonation** of the carbonyl oxygen of the acid makes the carbonyl carbon strongly electrophilic.
2. **Nucleophilic attack**: the oxygen of ethanol attacks that carbon, giving a tetrahedral intermediate.
3. **Proton transfer** converts one of the intermediate's –OH groups into $\text{--}^+\text{OH}_2$.
4. **Elimination of water** from the intermediate.
5. **Loss of a proton** gives the ester and regenerates the catalyst.

Hold on to this • Labelling with H_2^{18}O shows that the water formed takes its **–OH from the acid and its –H from the alcohol**. Conc. H_2SO_4 serves twice over — as catalyst, and as a dehydrating agent that pulls this reversible reaction to the right.

Rate of esterification: $1^\circ > 2^\circ > 3^\circ$ alcohol, because the alcohol acts as a nucleophile and steric bulk hinders its approach.

■ mechanism complete

M3 Reaction of an alcohol with HX, and the basis of the Lucas test

Again the –OH must be protonated first. What follows depends on the class:

1. **3° and 2° alcohols** — **S_N1**. The protonated alcohol loses water to give a carbocation, which the halide then captures. The rate depends on how stable that carbocation is.
2. **1° alcohols** — **S_N2**. No stable carbocation is available, so the halide attacks the carbon directly while water leaves, in one step.



Hold on to this · This is the Lucas test. The alkyl chloride is insoluble in the aqueous reagent and appears as turbidity; the speed with which it appears measures carbocation stability, so 3° is instant, 2° takes about five minutes, and 1° needs heating.

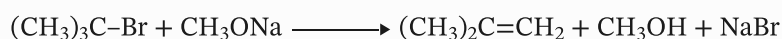
■ mechanism complete

M4 Williamson's synthesis, and why a 3° halide fails

Williamson's synthesis is a straightforward **S_N2** displacement of halide by alkoxide:



An alkoxide is a strong base as well as a nucleophile. If the halide is **tertiary**, its carbon is too crowded for backside attack, so the alkoxide does the other thing available to it — abstracts a β-hydrogen — and **E2 elimination** gives an alkene instead of the ether:

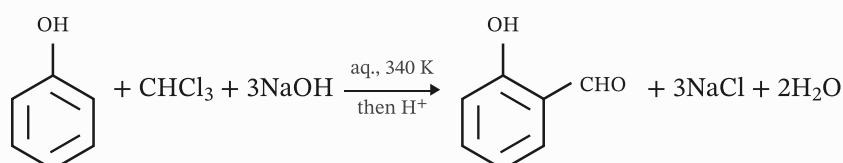


Hold on to this · The rule for planning an unsymmetrical ether: **make the bulky group the alkoxide and the small group the halide**. For tert-butyl methyl ether use $(\text{CH}_3)_3\text{C}-\text{ONa} + \text{CH}_3\text{Br}$ — never $(\text{CH}_3)_3\text{C}-\text{Br} + \text{CH}_3\text{ONa}$.

■ mechanism complete

M5 Reimer-Tiemann reaction (via dichlorocarbene)

1. Alkali removes a proton from chloroform, and the resulting anion loses chloride to give **dichlorocarbene**, an electrophile: $\text{CHCl}_3 + \text{OH}^- \rightarrow \text{CCl}_3^- + \text{H}_2\text{O} \rightarrow \text{:CCl}_2 + \text{Cl}^-$
2. Alkali also converts phenol into the strongly activated **phenoxide** ion, which attacks the carbene at the **ortho** position.
3. The –CHCl₂ group so introduced is **hydrolysed** by the alkali to –CHO, and acidification gives the product.

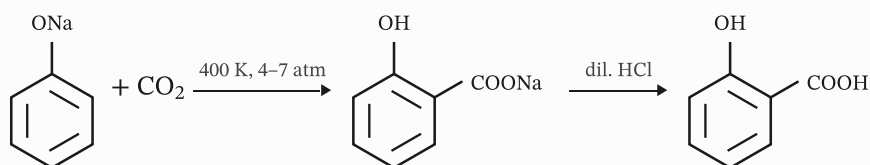


Product: **salicylaldehyde** (2-hydroxybenzaldehyde). With CCl_4 in place of CHCl_3 the same chemistry gives salicylic acid.

■ mechanism complete

M6 Kolbe's reaction (Kolbe-Schmidt)

Phenol itself is not nucleophilic enough to attack so weak an electrophile as CO_2 . It must first be converted by NaOH into the **phenoxide** ion, whose full negative charge makes the ring strongly nucleophilic. CO_2 is then attacked at the **ortho** position, and acidification gives the acid.



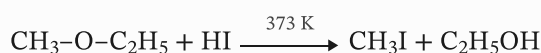
Hold on to this · Product: **salicylic acid**. Acetylating its phenolic $-\text{OH}$ with acetic anhydride gives **aspirin** — the $-\text{COOH}$ is left untouched.

■ mechanism complete

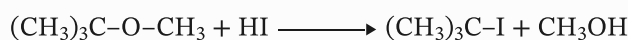
M7 Cleavage of an ether by HI — the two pathways

Common first step in both: **protonation of the ether oxygen**, converting it into a good leaving group.

1. **$\text{S}_{\text{N}}2$ route** (both groups 1° or methyl). I^- attacks the **less hindered** carbon from behind while the alcohol leaves:



2. **$\text{S}_{\text{N}}1$ route** (one group 3°). The protonated ether ionises to the well-stabilised tertiary carbocation, which then captures I^- :



So the products swap over depending on the mechanism — see the table in C13. For an aryl alkyl ether only the alkyl side can be attacked, so phenol is always one product.

■ mechanism complete

M8 Why phenol is acidic — the resonance argument set out formally

1. Phenol ionises to give the **phenoxide** ion and H^+ .
2. In phenoxide, the lone pair on oxygen overlaps with the π system of the ring, so the negative charge is **delocalised on to the two ortho and the para carbons**. Five resonance contributors can be written.
3. Delocalisation spreads charge, and spread charge means **lower energy**. The phenoxide ion is therefore stabilised, and its formation is favourable.
4. By contrast, in an **alkoxide** ion the charge is confined to one oxygen atom, and the $+\text{I}$ effect of the alkyl group pushes still more electron density on to it, **raising** its energy.
5. A more stable conjugate base means a stronger acid. Hence phenol ($K_{\text{a}} \approx 10^{-10}$) is about a million times more acidic than ethanol ($K_{\text{a}} \approx 10^{-16}$).

A second, smaller effect points the same way: the $-\text{OH}$ oxygen in phenol is attached to an **sp^2** carbon, which is more electronegative than an sp^3 carbon and so pulls the O--H electrons towards itself, weakening the bond.

■ proved

3 · Reaction tables — cover the right-hand column and test yourself

R1 – R9

These nine tables are the memorisation core of the chapter. Read across once, then cover the product column and work down from the reagents; then cover the reagent column and work back from the products. Both directions are examined.

R1 · Ethanol — every reaction, grouped by the handle it uses (C6)

Reagent and conditions	Product	Handle / note
Na (or K)	$\text{C}_2\text{H}_5\text{ONa} + \frac{1}{2}\text{H}_2\uparrow$	1 — acidic O–H; the test for an alcohol
CH_3COCl or $(\text{CH}_3\text{CO})_2\text{O}$	$\text{CH}_3\text{COOC}_2\text{H}_5$	1 — acylation
CH_3COOH / conc. H_2SO_4	$\text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$	1 — esterification (M2)
CH_3MgBr	$\text{CH}_4\uparrow + \text{C}_2\text{H}_5\text{OMgBr}$	1 — the acidic H destroys the Grignard reagent
conc. HCl + anhyd. ZnCl_2 (Lucas)	$\text{C}_2\text{H}_5\text{Cl}$	2 — 1°, so needs heating (M3)
PCl_5	$\text{C}_2\text{H}_5\text{Cl} + \text{POCl}_3 + \text{HCl}$	2
PCl_3 (3 mol alcohol)	$3\text{C}_2\text{H}_5\text{Cl} + \text{H}_3\text{PO}_3$	2
SOCl_2 / pyridine	$\text{C}_2\text{H}_5\text{Cl} + \text{SO}_2\uparrow + \text{HCl}\uparrow$	2 — best method : both by-products are gases
conc. H_2SO_4 , 413 K	$\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$	3 — intermolecular dehydration
conc. H_2SO_4 , 443 K	$\text{CH}_2=\text{CH}_2$	3 — intramolecular dehydration (M1)
Al_2O_3 , 623 K	$\text{CH}_2=\text{CH}_2$	3 — catalytic dehydration
Cu , 573 K	$\text{CH}_3\text{CHO} + \text{H}_2\uparrow$	4 — dehydrogenation (1° → aldehyde)
acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or KMnO_4	CH_3CHO , then CH_3COOH	4 — oxidation
I_2 + NaOH , warm	$\text{CHI}_3\downarrow$ (yellow) + HCOONa	4 — iodoform test (C8)

R2 · Phenol — reactions at the –OH group

Reagent and conditions	Product	Note
NaOH	sodium phenoxide + H_2O	acidic character; no reaction with NaHCO_3
Na	sodium phenoxide + $\frac{1}{2}\text{H}_2\uparrow$	
Zn dust, distil	benzene + ZnO	the –OH is removed altogether
CH_3COCl or $(\text{CH}_3\text{CO})_2\text{O}$	phenyl acetate	acylation; a base helps
PCl_5	chlorobenzene + $\text{POCl}_3 + \text{HCl}$	poor yield — the C–O bond is strong (C2)
CH_3I on sodium phenoxide	anisole + NaI	Williamson's synthesis
neutral FeCl_3	violet-coloured complex	the test for a phenolic group

R3 · Phenol — electrophilic substitution on the ring (all o/p)

Reagent and conditions	Product	Note
Br_2 in CS_2 , 273 K	o- and p-bromophenol	monobromination; para predominates
Br_2 in water	2,4,6-tribromophenol ↓ (white)	phenoxide is far more activated (C10)
dil. HNO_3 , 293 K	o- and p-nitrophenol	
conc. H_2SO_4 , then conc. HNO_3	picric acid (2,4,6-trinitrophenol)	sulphonate first, or the phenol is oxidised (C11)
conc. H_2SO_4 , 288 K / 373 K	o- / p-phenolsulphonic acid	temperature selects the isomer
CO_2 , NaOH , 400 K, 4–7 atm, then H^+	salicylic acid	Kolbe's reaction (M6)
CHCl_3 + aq. NaOH , 340 K, then H^+	salicylaldehyde	Reimer–Tiemann (M5)
benzene diazonium chloride, dil. NaOH , 273–278 K	p-hydroxyazobenzene (orange dye)	azo dye (coupling) test

R4 · Ethers

Reagent and conditions	Product	Note
HI, cold and dilute	$R-OH + R'-I$	the smaller group takes the I (S_N2) — C13
HI, hot and in excess	$2 R-I$	cleavage goes all the way
PCl_5	$2 R-Cl + POCl_3$	both C–O bonds break
Cl_2 , in the dark	$CH_3CHCl-O-CHClCH_3 + 2HCl$	substitution at the α -carbon
Cl_2 , in sunlight	perchloro ether	complete chlorination
O_2 / air, on standing	explosive peroxides	never distil an ether to dryness
Anisole + Br_2 / CH_3COOH	p-bromoanisole	$-OCH_3$ is activating and o/p-directing
Anisole + conc. HNO_3/H_2SO_4	o- and p-nitroanisole	
Anisole + CH_3COCl / anhyd. $AlCl_3$	p-methoxyacetophenone	Friedel–Crafts acylation

R5 · Preparation of alcohols

Method	Reagent and conditions	Gives
Hydration of an alkene	dil. H_2SO_4	Markovnikov product (2° or 3°)
Hydroboration–oxidation	B_2H_6 , then H_2O_2/OH^-	anti -Markovnikov, so a 1° alcohol
From a Grignard reagent + $HCHO$	dry ether, then H_2O/H^+	1° alcohol
From a Grignard reagent + $RCHO$	as above	2° alcohol
From a Grignard reagent + ketone	as above	3° alcohol
Hydrolysis of an alkyl halide	aqueous KOH	alcohol
Reduction of a carbonyl compound	$LiAlH_4$, $NaBH_4$ or H_2/Ni	1° from an aldehyde, 2° from a ketone
From a primary amine	HNO_2 ($NaNO_2$ + dil. HCl), 273–278 K	alcohol + $N_2\uparrow$
Bosch process (manufacture)	$CO + 2H_2$, $ZnO-Cr_2O_3$, 573–673 K, 200–300 atm	methanol
Fermentation (manufacture)	sugar, invertase then zymase, 298–303 K	ethanol

R6 · Preparation of phenol and of ethers

Starting material	Reagent and conditions	Name / note
Cumene	O_2 , then H^+/H_2O	manufacture ; acetone is the valuable by-product
Chlorobenzene	$NaOH$, 623 K, 300 atm, then dil. HCl	Dow's process
Benzene sulphonic acid	$NaOH$, then $NaOH$ fusion at 573–623 K, then H^+	alkali fusion
Benzene diazonium chloride	warm water, 283 K then boil	N_2 is evolved
Sodium alkoxide + 1° alkyl halide	330 K	Williamson's synthesis — any ether (M4)
2 mol alcohol	conc. H_2SO_4 , 413 K	dehydration — symmetrical ethers only

R7 · Distinguishing tests

Pair	Reagent	Observation
Alcohol / ether	sodium metal	alcohol: effervescence of H_2 ; ether: no reaction
Ethanol / phenol	neutral $FeCl_3$	phenol: violet ; ethanol: no colour
Phenol / benzoic acid	$NaHCO_3$ solution	acid: effervescence of CO_2 ; phenol: nothing
Phenol / ethanol	bromine water	phenol: white ppt. of 2,4,6-tribromophenol

Pair	Reagent	Observation
Methanol / ethanol	$I_2 + NaOH$	ethanol: yellow CHI_3 ; methanol: nothing
Propan-1-ol / propan-2-ol	$I_2 + NaOH$	propan-2-ol: yellow CHI_3
2° / 3° alcohol	Lucas reagent	3°: turbidity at once; 2°: in about 5 min
Alcohol / aldehyde	Tollens' or Fehling's	aldehyde: silver mirror / red ppt.

R8 · Named reactions index

Name	What it does
Williamson's synthesis	sodium alkoxide + 1° alkyl halide → ether
Kolbe's (Kolbe–Schmidt) reaction	sodium phenoxide + CO_2 → salicylic acid
Reimer–Tiemann reaction	phenol + $CHCl_3/NaOH$ → salicylaldehyde
Dow's process	chlorobenzene + $NaOH$ → phenol
Bosch process	$CO + 2H_2$ → methanol
Esterification	acid + alcohol → ester
Haloform / iodoform reaction	CH_3CO- or $CH_3CH(OH)-$ + $I_2/NaOH$ → CHI_3
Lucas test	distinguishes 1°, 2° and 3° alcohols
Victor Meyer test	distinguishes 1°, 2°, 3° by colour — red, blue, colourless
Azo dye (coupling) test	phenol + diazonium salt in mild alkali → orange dye

R9 · The orders and trends you may be asked to arrange

Boiling point (isomeric alcohols)

3° < 2° < 1°

Boiling point (same carbon number)

alkane < ether < alcohol < carboxylic acid

Acidic strength

3° alcohol < 2° < 1° < water < **phenol** < carbonic acid < $RCOOH$

Acidic strength (substituted phenols)

p-cresol < phenol < p-chlorophenol < **p-nitrophenol**

Acidic strength (substituted ethanols)

CH_3CH_2OH < CCl_3CH_2OH < CF_3CH_2OH

Reactivity to Lucas reagent / HX (S_N1)

1° < 2° < **3°**

Ease of dehydration ($E1$)

1° < 2° < **3°**

Rate of esterification (steric)

3° < 2° < **1°**

Solubility in water

hexan-1-ol < butan-1-ol < phenol < ethanol < methanol

Memory hook · Two families of trend, and they run **opposite ways**. Anything that goes through a **carbocation** (Lucas, HX , dehydration) favours 3°. Anything in which the alcohol acts as a **nucleophile** or is **oxidised** (esterification, oxidation) favours 1°. Acidity also favours 1°.

T1 Classify and name**RECOGNISE IT BY**

“Classify the following as 1°, 2° or 3°”, or “write the IUPAC name of ...”.

METHOD

Count the alkyl groups on the carbon carrying the –OH (C1). For an ether, name the smaller group as an alkoxy prefix on the larger. Check whether an aromatic –OH is on the ring (phenol) or on a side chain (ordinary alcohol).

WORKED EXAMPLE

Classify and name: (a) $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ (b) $(\text{CH}_3)_3\text{COH}$ (c) $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ (d) $\text{CH}_3\text{OCH}(\text{CH}_3)_2$

- (a) **butan-2-ol**, secondary
- (b) **2-methylpropan-2-ol**, tertiary
- (c) **phenylmethanol** (benzyl alcohol), **primary** — not a phenol
- (d) **2-methoxypropane**, an unsymmetrical ether

T2 Write and classify all the isomers of a given formula**RECOGNISE IT BY**

“Write all the isomers of $\text{C}_4\text{H}_{10}\text{O}$ and classify them.”

METHOD

Work systematically: **first all the alcohols** (draw each carbon skeleton, then move the –OH along it), **then all the ethers** (split the carbons about the oxygen in every possible way). Classify each as you write it.

WORKED EXAMPLE

All isomers of $\text{C}_4\text{H}_{10}\text{O}$.

Alcohols (4)	Class	Ethers (3)	Type
butan-1-ol	1°	ethoxyethane	symmetrical
butan-2-ol	2° (chiral)	1-methoxypropane	unsymmetrical
2-methylpropan-1-ol	1°	2-methoxypropane	unsymmetrical
2-methylpropan-2-ol	3°	—	—

Answer · Seven isomers in all — four alcohols and three ethers. The alcohols and ethers are **functional** isomers of one another; the three ethers are **metamers**.

T3 Arrange in order of boiling point

RECOGNISE IT BY

A list mixing alcohols, ethers and hydrocarbons, or a set of isomeric alcohols.

METHOD

Ask two questions in order. (1) **Can it hydrogen bond?** No O–H means low boiling point. (2) Among those that can, **how branched is it, and how many –OH groups?** More branching lowers the boiling point; more –OH raises it sharply.

WORKED EXAMPLE

Arrange in increasing order: butan-1-ol, butan-2-ol, 2-methylpropan-2-ol, diethyl ether, butane.

Answer

butane < diethyl ether < 2-methylpropan-2-ol < butan-2-ol < butan-1-ol

Butane and diethyl ether cannot hydrogen bond; among the three alcohols, branching decides.

T4 Arrange in order of acidic strength

RECOGNISE IT BY

Alcohols, phenols, substituted phenols or halogen-substituted alcohols.

METHOD

Judge the **stability of the anion** (C5). Electron-**withdrawing** groups stabilise it and strengthen the acid; electron-**releasing** groups do the reverse. On a ring, o/p substituents act by resonance as well as induction, m- only by induction.

WORKED EXAMPLE

Arrange in increasing acidic strength: ethanol, phenol, water, p-cresol, p-nitrophenol.

Answer

ethanol < water < p-cresol < phenol < p-nitrophenol

p-Cresol falls just below phenol because $-\text{CH}_3$ releases electrons; p-nitrophenol rises well above it because $-\text{NO}_2$ withdraws them.

T5 Arrange in order of reactivity

RECOGNISE IT BY

“...towards Lucas reagent”, “...ease of dehydration”, “...rate of esterification”.

METHOD

Decide first **which family the reaction belongs to**. Carbocation reactions run $3^\circ > 2^\circ > 1^\circ$; nucleophilic or oxidation reactions run $1^\circ > 2^\circ > 3^\circ$ (R9).

WORKED EXAMPLE

Arrange in decreasing reactivity towards Lucas reagent: 1-butanol, 2-butanol, 2-methyl-2-propanol.

Answer

2-methyl-2-propanol (3°) > 2-butanol (2°) > 1-butanol (1°)

The primary alcohol gives no turbidity at room temperature at all.

T6 “How will you obtain X from Y?” — conversions

RECOGNISE IT BY

A named starting material and a named product, with “give chemical equations”.

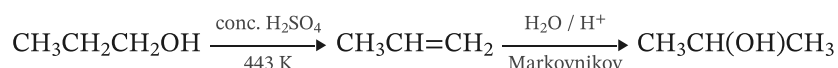
METHOD

Compare the two structures and ask **what has to change**: the functional group, the carbon number, or the position. To lengthen a chain use KCN or a Grignard reagent; to shorten it use decarboxylation; to move a functional group go via the alkene. Write conditions over every arrow.

WORKED EXAMPLE

Convert *propan-1-ol* into *propan-2-ol*.

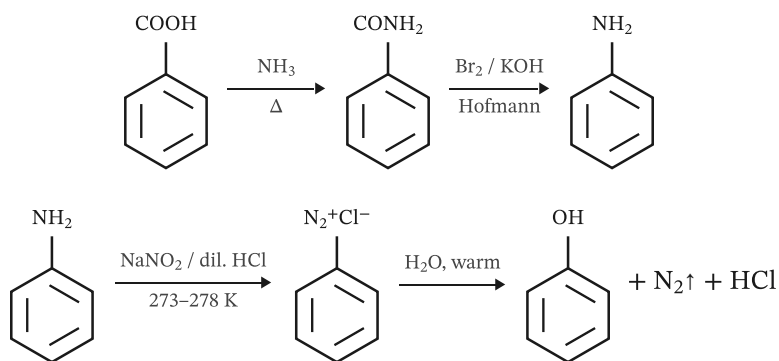
1. The carbon number is unchanged; only the **position** of the –OH moves. Go via the alkene, which destroys the position information, then add water back by Markovnikov's rule.



SECOND WORKED EXAMPLE — THE LONG ROUTE

Convert *benzoic acid* into *phenol*. (Heritage 2016–17)

1. There is no direct way to replace –COOH by –OH on a ring. The only practicable route runs through the **diazonium salt**, which is the universal gateway to almost any aromatic substituent — so first get to the amine.
2. Acid → amide → amine (losing one carbon by Hofmann degradation) → diazonium salt → phenol.



Memory hook · Any aromatic conversion that looks impossible — –COOH to –OH, –NO₂ to –OH, –NH₂ to –Cl, –Br or –CN — goes **through the diazonium salt**. Reduce or degrade whatever you have to the amine first, diazotise it, then displace N₂ with the group you want.

T7 "What happens when ...?"

RECOGNISE IT BY

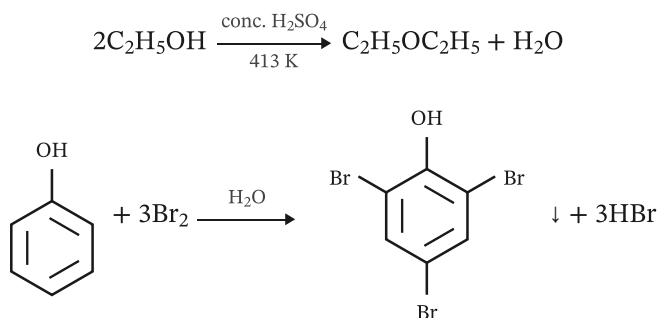
A reagent and conditions are given; the product is wanted.

METHOD

For an alcohol, identify which of the **four handles** (C6) the reagent attacks. For phenol, decide which of the **two faces** (C9) it attacks. Then read the answer off table R1, R2, R3 or R4. Always give the balanced equation with conditions.

WORKED EXAMPLE

What happens when (a) ethanol is warmed with conc. H_2SO_4 at 413 K, (b) phenol is treated with bromine water?



T8 Write the equation for a named reaction

RECOGNISE IT BY

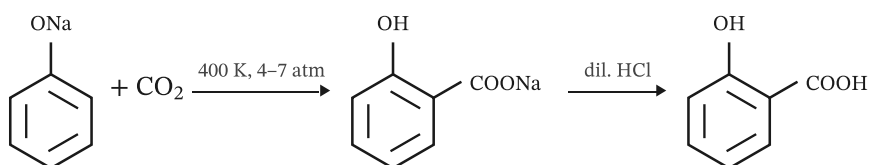
"Write chemical equations to illustrate the following named reactions."

METHOD

Learn table R8 as a two-column list: name $\leftrightarrow$ equation. The marks are for the **balanced equation with conditions over the arrow**, not for the name.

WORKED EXAMPLE

Write the equation for Kolbe's reaction.



Common slip · Starting from phenol instead of sodium phenoxide loses the mark — the phenoxide ion is essential (M6).

T9 Distinguish between a pair of compounds

RECOGNISE IT BY

“Give one chemical test to distinguish between ...”.

METHOD

Name the **reagent**, then state what **each** member does. A test that only says what one of them does is incomplete. Table R7 covers every pair that has ever been set.

WORKED EXAMPLE

Distinguish between phenol and ethanol.

Answer · Add **neutral ferric chloride solution**. Phenol gives an intense **violet colouration** (a coloured iron(III)–phenoxide complex); ethanol gives **no colour change**.

Alternative: bromine water gives a white precipitate of 2,4,6-tribromophenol with phenol only.

T10 Identify [A], [B], [C] in a reaction sequence

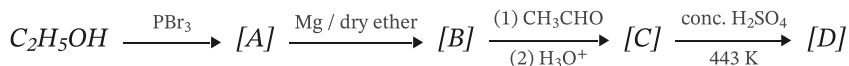
RECOGNISE IT BY

A chain of arrows with reagents written above them.

METHOD

Work **forwards** one arrow at a time, and **check the carbon count at every step**. Reagents that change the carbon number are the signposts: KCN and Grignard **add** one carbon; decarboxylation **removes** one.

WORKED EXAMPLE



- [A] = **C₂H₅Br** (bromoethane)
- [B] = **C₂H₅MgBr** (a Grignard reagent)
- [C] = a 2° alcohol from Grignard + aldehyde: CH₃CH(OH)C₂H₅, **butan-2-ol**
- [D] = dehydration, and by Saytzeff's rule the more substituted alkene: **but-2-ene**

T11 Structure elucidation from a molecular formula plus tests

RECOGNISE IT BY

“An organic compound [A] with molecular formula ... gives / does not give ...”.

METHOD

Treat each clue as a **filter** and apply them in this order — the first two usually settle the class outright:

WORKED EXAMPLE

1. **Reacts with Na?** Yes → it has an O–H, so alcohol or phenol. No → an ether.
2. **Violet with neutral FeCl₃?** Yes → phenol.
3. **Lucas timing?** Instant → 3°; about five minutes → 2°; nothing at room temperature → 1°.
4. **Iodoform?** Yes → there is a CH₃CH(OH)– or CH₃CO– group.
5. **Oxidation product?** Aldehyde or acid → 1°; ketone → 2°; nothing → 3°.
6. **Finally, check your structure against the molecular formula.** It is given so that you can verify the answer.

[A], C₇H₈O, gives a violet colour with neutral FeCl₃. It dissolves in NaOH but not in NaHCO₃. On vigorous oxidation with alkaline KMnO₄ followed by acidification it gives 4-hydroxybenzoic acid. Identify [A].

1. Violet with FeCl₃ ⇒ a **phenol**. Soluble in NaOH but not NaHCO₃ ⇒ confirms a phenol and rules out a carboxylic acid.
2. C₇H₈O is phenol plus one CH₂ ⇒ a **methylphenol (cresol)**.
3. Oxidation converts the methyl group into –COOH. The product is the **4**-hydroxy acid, so the methyl group was at position 4.

Answer · [A] = **p-cresol** (4-methylphenol).

T12 Identify an ether from its behaviour

RECOGNISE IT BY

The compound **fails** to react with sodium — the clue that there is no O–H.

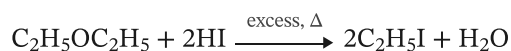
METHOD

No reaction with Na ⇒ an ether. Then: a **single** product with excess HI ⇒ the ether is **symmetrical**; two products ⇒ unsymmetrical, and C₁₃ tells you which fragment takes the iodine.

WORKED EXAMPLE

[A], C₄H₁₀O, does not react with sodium. With excess hot HI it gives a single organic product [B]; with PCl₅ it gives a single product [C]. Identify [A], [B] and [C].

- No reaction with Na ⇒ [A] is an **ether**.
- One product only ⇒ **symmetrical** ⇒ [A] = **diethyl ether**, C₂H₅OC₂H₅.
- [B] = **iodoethane**, C₂H₅I; [C] = **chloroethane**, C₂H₅Cl.



T13 Predict the products of ether cleavage**RECOGNISE IT BY**

“What are the products when ... is heated with HI?”

METHOD

Apply the C13 rule: two 1° groups → S_N2, smaller group takes the iodine; one 3° group → S_N1, tertiary group takes the iodine; aryl alkyl → phenol plus alkyl iodide, always.

WORKED EXAMPLE

Predict the products when the following are heated with HI: (a) methoxyethane, (b) 2-methoxy-2-methylpropane, (c) anisole.

Ether	Mechanism	Products
CH ₃ OC ₂ H ₅	S _N 2	CH ₃ I + C ₂ H ₅ OH
(CH ₃) ₃ COCH ₃	S _N 1	(CH ₃) ₃ CI + CH ₃ OH
C ₆ H ₅ OCH ₃	S _N 2 at the alkyl carbon	C ₆ H ₅ OH + CH ₃ I

T14 “Give a reason” / “account for the following”**RECOGNISE IT BY**

A one-sentence statement to be explained, usually worth one mark each.

METHOD

Every one of these reduces to a named effect: **hydrogen bonding, resonance, the +I or –I effect, steric hindrance, or carbocation stability**. Name the effect explicitly — that is what earns the mark.

WORKED EXAMPLE

Statement	The reason, in one line
Phenol is more acidic than ethanol	the phenoxide ion is resonance stabilised; the alkoxide is not
Alcohols boil far above ethers	alcohols have an O–H and form intermolecular hydrogen bonds
1° alcohols boil above isomeric 3° alcohols	branching reduces the surface area and so the van der Waals forces
Phenol is less soluble in water than ethanol	the large hydrophobic benzene ring outweighs the single –OH
Ethoxyethane does not react with Na, but ethanol does	the ether has no acidic O–H hydrogen for Na to displace
Ethanol gives the iodoform test but methanol does not	methanol has only one carbon, so no CH ₃ CH(OH)– group
Ethers should not be distilled to dryness	explosive peroxides, being less volatile, concentrate in the residue
Anisole with HI gives phenol, never iodobenzene	the sp ² C–O bond has partial double bond character and resists attack
Phenol is nitrated only after sulphonation	conc. HNO ₃ would otherwise oxidise the activated ring
Phenol does not liberate CO ₂ from NaHCO ₃	phenol is a weaker acid than carbonic acid
But 2,4,6-trinitrophenol is soluble in NaHCO ₃	three –NO ₂ groups stabilise the anion so strongly that it becomes a stronger acid than H ₂ CO ₃
Phenol turns blue litmus red but ethanol does not	phenol ionises appreciably; an alcohol is neutral to litmus
Williamson's synthesis needs a 1° halide	a 3° halide is too hindered for S _N 2 and gives E2 elimination
3° alcohols resist oxidation	there is no hydrogen on the carbon bearing the –OH
Grignard reagents must be made in dry ether	water or alcohol has an acidic H that destroys the reagent, giving an alkane
o-Nitrophenol is slightly less acidic than the p-isomer	intramolecular hydrogen bonding stabilises the un-ionised o-isomer

T15 Mechanism questions**RECOGNISE IT BY**

“Explain the mechanism of ...”, worth three marks.

METHOD

Three marks means **three steps**, each with its own equation, plus the label *slow / rate determining* on the right one. Name the mechanism (E1, S_N1, S_N2) at the end. The four that are set are M1 (dehydration), M2 (esterification), M4 (Williamson) and M7 (ether cleavage).

WORKED EXAMPLE

See **M1** for the fully worked dehydration of ethanol — the most frequently asked of the four.

T16 Passage / case-study questions

RECOGNISE IT BY

A paragraph about phenol or alcohols, then five one-mark parts.

METHOD

The passage is scaffolding, not information — the answers come from your own knowledge. Read the parts **first**, then skim the passage for the numbers or names it supplies. The five parts are almost always: one ordering, one reason, one equation, one preparation and one test or use.

WORKED EXAMPLE

Work **T4** (ordering), **T14** (reason), **T8** (equation), **R6** (preparation) and **R7** (test) and you have covered every part such a question can ask.

T17 Uses and identification of common compounds

RECOGNISE IT BY

“State one use of ...”, “name the compound used as an antiseptic”.

METHOD

Learn table C15. Two facts are asked disproportionately often: phenol is an **antiseptic at 0.2% and a disinfectant at 1%**, and **methanol is toxic** — it causes blindness and then death.

WORKED EXAMPLE

In what concentration does phenol act as an antiseptic, and in what concentration as a disinfectant?

Answer • A **0.2%** solution acts as an antiseptic; a **1%** solution acts as a disinfectant.

5 • Organic recall sheet

revise from this page alone

The four handles on an alcohol

- 1 O–H → Na, acylation
- 2 C–O → HX, PCl_3 , SOCl_2
- 3 β C–H → dehydration
- 4 carbinol C–H → oxidation

The two faces of phenol

–OH: NaOH, Na, Zn, acylation, PCl_5 , FeCl_3

Ring: Br_2 , HNO_3 , H_2SO_4 , Kolbe, Reimer–Tiemann, azo coupling — all **o/p**

Temperature decides

- 413 K → diethyl ether
- 443 K → ethene
- Br_2/CS_2 → mono
- $\text{Br}_2/\text{H}_2\text{O}$ → tri

Lucas timing

- 3° → instant
 - 2° → ~5 minutes
 - 1° → only on heating
- Reagent: conc. HCl + anhyd. ZnCl_2

Iodoform needs

$\text{CH}_3\text{--CO--}$ or $\text{CH}_3\text{--CH(OH)--}$

Yes: ethanol, propan-2-ol, acetone

No: methanol, propan-1-ol, benzaldehyde

Grignard gives

- + HCHO → 1° alcohol
- + RCHO → 2° alcohol
- + ketone → 3° alcohol
- + alcohol/water → alkane (destroyed)

Oxidation gives

- 1° → aldehyde → acid
- 2° → ketone
- 3° → no reaction

Ether + HI

S_N2 (1°) → **small** group gets I

S_N1 (3°) → **big** group gets I

Aryl → always phenol + alkyl iodide

Phenol acidity: the two limits

Dissolves in **NaOH** ✓

No CO_2 from **NaHCO₃** ✗

⇒ weaker than H_2CO_3 , stronger than water

Four preparations of phenol

Cumene + O_2 , H^+ (+ acetone)

Chlorobenzene + NaOH, 623 K (Dow)

Benzene sulphonic acid, NaOH fusion

Diazonium salt + warm water

Carbocation vs nucleophile

3° wins: Lucas, HX, dehydration

1° wins: acidity, esterification, oxidation, boiling point

Colours and precipitates

Violet — phenol + neutral $FeCl_3$

White ppt. — phenol + Br_2 water

Yellow ppt. — iodoform

Orange dye — azo coupling

Part VI · Aldehydes, Ketones and Carboxylic Acids

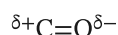
ISC unit 8 · Preparation of carbonyl compounds · nucleophilic addition and its mechanism · reactions of the α -hydrogen · oxidation and reduction · aldol and Cannizzaro · benzaldehyde · acidity of carboxylic acids and the effect of substituents · decarboxylation and HVZ · the distinguishing tests. Seventeen concepts, six mechanisms, sixteen problem types.

1 · The concepts, in the order they build

C1 – C17

C1 Why the carbonyl carbon is attacked by nucleophiles

The C=O bond is strongly **polar**: oxygen is far more electronegative than carbon, and the π electrons are pulled towards it. The carbon therefore carries a substantial δ^+ charge and is **electrophilic**.



Hold on to this · Everything in the first half of this unit follows from that one polarity. The carbon is attacked by **nucleophiles** — and because the carbon is sp^2 and planar, attack converts it to sp^3 , which is why the characteristic reaction is **nucleophilic addition** and not substitution.

C2 Preparation of aldehydes and ketones

Method	Reagents	Gives
From an alcohol	Cu at 573 K, or acidified $\text{K}_2\text{Cr}_2\text{O}_7$	$1^\circ \rightarrow$ aldehyde; $2^\circ \rightarrow$ ketone
From an alkene	O_3 , then $\text{Zn}/\text{H}_2\text{O}$ (ozonolysis)	aldehyde and/or ketone, by cleaving C=C
From an alkyne	H_2O , dil. H_2SO_4 , HgSO_4	ethyne gives acetaldehyde; others give ketones
From an acid chloride — Rosenmund's reduction	H_2 , Pd on BaSO_4 poisoned with S or quinoline	aldehyde
From an acid chloride — dialkyl cadmium	R_2Cd	ketone
From a calcium salt	dry distillation of the calcium carboxylate	calcium acetate $\rightarrow$ acetone; calcium formate $\rightarrow$ formaldehyde; a mixture of the two $\rightarrow$ acetaldehyde
From a nitrile — Stephen reduction	SnCl_2/HCl then H_2O	aldehyde
From a nitrile — Grignard	RMgX then H_3O^+	ketone
From an ester	DIBAL-H, or partial reduction	aldehyde
Benzaldehyde from toluene — Étard reaction	CrO_2Cl_2 (chromyl chloride) in CS_2 , then hydrolysis	benzaldehyde

Hold on to this · Rosenmund gives the aldehyde, dialkyl cadmium gives the ketone — both from the same acid chloride. And note why R_2Cd rather than a Grignard reagent: a Grignard reagent is so reactive that it would attack the ketone as soon as it formed and carry on to a **tertiary alcohol**. Dialkyl cadmium is milder and stops at the ketone.



C3 Physical properties

- Aldehydes and ketones boil **above** the corresponding alkanes and ethers, because of dipole–dipole attraction, but **below** alcohols of comparable mass, because they have no O–H hydrogen and cannot hydrogen bond to one another.
- The lower members are **soluble in water**, since the carbonyl oxygen can *accept* a hydrogen bond from water; solubility falls as the chain grows.
- Formaldehyde is a gas, acetaldehyde a volatile liquid, and benzaldehyde a liquid with a characteristic **bitter almond** odour.

C4 Reactivity: why an aldehyde beats a ketone

- Steric** — an aldehyde has only one alkyl group beside the carbonyl carbon, a ketone has two. The nucleophile finds the aldehyde easier to reach.
- Electronic** — an alkyl group releases electron density (+I) and so reduces the $\delta+$ on the carbon. Two alkyl groups reduce it twice over, making the ketone less electrophilic.
- An **aromatic** aldehyde is less reactive still, because the ring donates electron density into the carbonyl group by resonance.

Reactivity towards nucleophilic addition



Hold on to this · **HCHO is the most reactive of all** — it has no alkyl group at all, so nothing to hinder the approach and nothing to reduce the positive charge.

C5 Nucleophilic addition — the standard reagents

Reagent	Product	Note
HCN	cyanohydrin, $\text{R}-\text{CH}(\text{OH})-\text{CN}$	hydrolysis gives an α -hydroxy acid — see T2
NaHSO₃	bisulphite addition product	crystalline; used to purify aldehydes and methyl ketones, since the reaction is reversible
RMgX then H_3O^+	alcohol	$\text{HCHO} \rightarrow 1^\circ$, $\text{RCHO} \rightarrow 2^\circ$, ketone $\rightarrow 3^\circ$
alcohol / dry HCl	hemiacetal, then acetal	protects a carbonyl group

With **ammonia derivatives** the addition is followed by loss of water — an **addition–elimination** — and the product contains $\text{C}=\text{N}$:

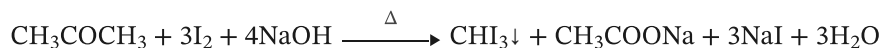
Reagent	Product
hydroxylamine, NH_2OH	oxime
hydrazine, NH_2NH_2	hydrazone
phenylhydrazine, $\text{C}_6\text{H}_5\text{NHNH}_2$	phenylhydrazone
2,4-dinitrophenylhydrazine (2,4-DNP , Brady's reagent)	2,4-dinitrophenylhydrazone — an orange-red precipitate , the test for a carbonyl group
semicarbazide	semicarbazone
excess NH_3 with HCHO	urotropine , hexamethylenetetramine, $(\text{CH}_2)_6\text{N}_4$



Urotropine is used as a urinary antiseptic; its nitration gives the explosive RDX.

C6 The α -hydrogen, and the iodoform reaction

A hydrogen on the carbon **next to** the carbonyl group is acidic, because the anion left behind is stabilised by resonance with C=O. That acidity drives the aldol condensation (M2) and the haloform reaction.



Hold on to this • The iodoform test needs a **CH₃CO–** group (or a CH₃CH(OH)– that oxidises to it). So acetaldehyde, acetone and acetophenone give it; **formaldehyde, benzaldehyde and propanal do not**. This is the same rule as in Part V.

C7 Oxidation — and the three reagents that distinguish

Reagent	Aldehyde	Ketone	Observation
Tollens' reagent — ammoniacal AgNO ₃	oxidised to the acid	no reaction	silver mirror or black precipitate
Fehling's solution — alkaline Cu ²⁺ tartrate	oxidised to the acid	no reaction	red-brown precipitate of Cu ₂ O
Benedict's solution	oxidised	no reaction	red precipitate of Cu ₂ O
2,4-DNP	orange-red precipitate	orange-red precipitate	detects <i>any</i> carbonyl — does not distinguish

- An aldehyde has a **hydrogen on the carbonyl carbon**, which can be removed to give the acid without breaking a C–C bond. A ketone has none, so it resists mild oxidation.
- **Aromatic aldehydes give Tollens' but not Fehling's**. That is the standard way of telling benzaldehyde from an aliphatic aldehyde.

Hold on to this • So the pair of tests is a two-step sieve. **2,4-DNP** answers “is there a carbonyl group?”; **Tollens'** answers “is it an aldehyde?”; **Fehling's** answers “is it an *aliphatic* aldehyde?”

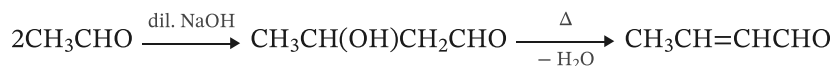
C8 Reduction — three different destinations

Reagent	Product	Name
LiAlH ₄ , NaBH ₄ or H ₂ /Ni	alcohol (C=O → CH–OH)	—
Zn amalgam + conc. HCl	alkane (C=O → CH ₂)	Clemmensen reduction
NH ₂ NH ₂ then KOH/glycol, Δ	alkane	Wolff–Kishner reduction
red phosphorus + HI	alkane	—

Hold on to this • Clemmensen uses **acid** and Wolff–Kishner uses **base** — so you choose between them by what else is in the molecule. Both take the carbonyl all the way to CH₂; the hydride reagents stop at the alcohol.

C9 The three base-catalysed reactions

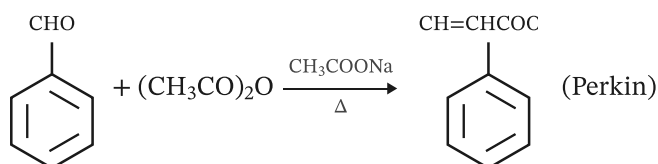
Reaction	Requires	Product
Aldol condensation	an α -hydrogen; dil. NaOH	a β -hydroxy aldehyde or ketone, which on heating loses water to give an α,β -unsaturated compound
Cross aldol	two different carbonyl compounds, at least one with an α -H	a mixture of up to four products
Cannizzaro reaction	no α-hydrogen ; conc. NaOH	a disproportionation — half is reduced to the alcohol, half oxidised to the salt of the acid



Memory hook · α -hydrogen present $\Rightarrow$ aldol. α -hydrogen absent $\Rightarrow$ Cannizzaro. The two reactions are mutually exclusive, and one sentence decides which applies. Formaldehyde and benzaldehyde have no α -H, so they Cannizzaro; acetaldehyde and acetone have α -H, so they aldol.

C10 Benzaldehyde — the reactions peculiar to it

Reaction	Reagents	Product
Cannizzaro	conc. NaOH	benzyl alcohol + sodium benzoate
Benzoin condensation	2 C ₆ H ₅ CHO + alcoholic KCN	benzoin , C ₆ H ₅ CH(OH)COC ₆ H ₅
Perkin reaction	acetic anhydride + sodium acetate, Δ	cinnamic acid , C ₆ H ₅ CH=CHCOOH
Oxidation	air, or KMnO ₄	benzoic acid
Electrophilic substitution	–CHO is meta directing and deactivating	m-nitrobenzaldehyde on nitration



Benzaldehyde is slowly oxidised by air to benzoic acid — which is why an old sample shows white crystals round the stopper. It is **stored in dark, well-filled bottles** for the same reason chloroform is (Part IV).

C11 The distinguishing tests, collected

Pair	Reagent	Observation
Aldehyde / ketone	Tollens' reagent	aldehyde gives a silver mirror; ketone no reaction
Formaldehyde / acetaldehyde	$I_2 + NaOH$ (iodoform)	acetaldehyde gives yellow CHI_3 ; formaldehyde does not
Aliphatic / aromatic aldehyde	Fehling's solution	aliphatic gives a red-brown precipitate; aromatic gives none
Acetaldehyde / acetone	Tollens' or Fehling's	only the aldehyde reduces them
Benzaldehyde / benzoic acid	$NaHCO_3$	the acid effervesces CO_2
Benzoic acid / phenol	$NaHCO_3$	the acid effervesces; phenol does not
Formic acid / acetic acid	Tollens', Fehling's, or $HgCl_2$	only formic acid reacts — silver mirror, red precipitate, or a grey precipitate of Hg . It also decolourises $KMnO_4$

Hold on to this · Formic acid is the only carboxylic acid that is also a reducing agent, because it alone contains a $-CHO$ group fused to the $-COOH$. That is why it reduces Tollens', Fehling's and mercury(II) chloride while acetic acid does nothing.

C12 Carboxylic acids — structure and nomenclature

- The $-COOH$ group; the carbon is sp^2 and the group is planar. The two $C-O$ bonds in the **carboxylate ion** are of **equal length**, because the charge is delocalised over both oxygens — that resonance is the whole reason for the acidity (M4).
- Monocarboxylic**: formic (methanoic), acetic (ethanoic), propionic, butyric, benzoic. **Dicarboxylic**: oxalic (ethanedioic), malonic, succinic, adipic, phthalic.
- They boil **higher than alcohols** of comparable mass, because they form a **cyclic dimer** held by two hydrogen bonds — which is also why their molar mass appears doubled in benzene (Part I, T13).

C13 Acidity of carboxylic acids, and the effect of substituents

A carboxylic acid is far stronger than an alcohol or phenol because the **carboxylate ion is resonance stabilised over two equivalent oxygens** (M4).

Acidic strength



- Electron-withdrawing** groups stabilise the anion and **strengthen** the acid; **electron-releasing** groups weaken it.
- Among the aliphatic acids, alkyl groups release electrons, so $HCOOH > CH_3COOH > CH_3CH_2COOH$: formic acid is the strongest, having no alkyl group at all.
- Halogens withdraw: $CCl_3COOH > CHCl_2COOH > CH_2ClCOOH > CH_3COOH$, and the effect weakens with distance from the $-COOH$.
- On a benzene ring, $-NO_2$ **strengthens** and $-CH_3$ or $-OCH_3$ **weakens**: 4-nitrobenzoic acid $>$ benzoic acid $>$ 4-methylbenzoic acid.

Hold on to this · The **ortho effect**: any ortho-substituted benzoic acid is **stronger** than benzoic acid itself, whether the substituent releases or withdraws electrons. The substituent twists the $-COOH$ group out of the plane of the ring, breaking its conjugation with the ring — steric inhibition of resonance.

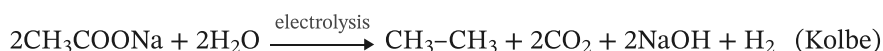
Remember which way pK_a runs: a **lower** pK_a means a **stronger** acid. So “the pK_a of 4-nitrobenzoic acid is lower than that of benzoic acid” is simply a restatement of “the nitro compound is the stronger acid”.

C14 Preparation of carboxylic acids

From	Reagents
A primary alcohol or an aldehyde	acidified KMnO_4 or $\text{K}_2\text{Cr}_2\text{O}_7$
A nitrile or an amide	acid or alkaline hydrolysis
A Grignard reagent	CO_2 (dry ice), then H_3O^+ — gives one carbon more
An alkylbenzene	KMnO_4/KOH , then H^+ — the whole side chain becomes $-\text{COOH}$
An acid derivative	hydrolysis of the ester, anhydride or acid chloride

C15 Reactions of carboxylic acids

Reagent	Product	Note
Na , NaOH , Na_2CO_3 , NaHCO_3	the salt, + H_2 / H_2O / CO_2	the NaHCO_3 effervescence is the test for a carboxylic acid
An alcohol + conc. H_2SO_4	ester	esterification (Part V, M2)
PCl_5 , PCl_3 or SOCl_2	acid chloride	
NH_3 then Δ	amide	via the ammonium salt
P_2O_5 , Δ	anhydride	
LiAlH_4	primary alcohol	the acid is reduced
Soda lime, Δ	alkane	decarboxylation , losing CO_2
Electrolysis of the sodium salt	alkane at the anode	Kolbe's electrolytic reaction
Cl_2 / red P	α -halo acid	Hell-Volhard-Zelinsky (HVZ) reaction
conc. $\text{HNO}_3/\text{H}_2\text{SO}_4$ on benzoic acid	m -nitrobenzoic acid	$-\text{COOH}$ is deactivating and meta directing



Common slip · Do not confuse **Kolbe's electrolytic reaction** (here — electrolysis of a carboxylate to give an alkane) with **Kolbe's reaction** in Part V (sodium phenoxide + CO_2 to give salicylic acid). Two quite different reactions share the name.

C16 Tests and uses of the three named acids

Acid	Test	Uses
Formic acid	effervescence with NaHCO_3 , and it reduces Tollens', Fehling's and HgCl_2 (grey Hg)	reducing agent; in leather tanning, rubber coagulation, textile dyeing
Acetic acid	effervescence with NaHCO_3 ; neutral FeCl_3 gives a blood-red colour; does not reduce Tollens'	vinegar; solvent; making esters and acetic anhydride
Benzoic acid	effervescence with NaHCO_3 ; neutral FeCl_3 gives a buff-coloured precipitate ; sublimes readily on warming	food preservative (as sodium benzoate); in medicine and dyes

C17 The named reactions of this unit, collected

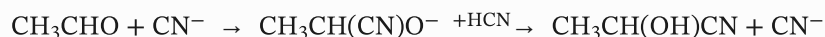
Name	What it does
Rosenmund's reduction	acid chloride + $\text{H}_2/\text{Pd}-\text{BaSO}_4(\text{poisoned}) \rightarrow$ aldehyde
Stephen reduction	nitrile + $\text{SnCl}_2/\text{HCl} \rightarrow$ aldehyde
Étard reaction	toluene + $\text{CrO}_2\text{Cl}_2 \rightarrow$ benzaldehyde
Aldol condensation	two carbonyls with $\alpha\text{-H}$ + dil. $\text{NaOH} \rightarrow \beta\text{-hydroxy carbonyl}$
Cannizzaro reaction	aldehyde without $\alpha\text{-H}$ + conc. $\text{NaOH} \rightarrow$ alcohol + carboxylate
Benzoin condensation	2 benzaldehyde + alc. $\text{KCN} \rightarrow$ benzoin
Perkin reaction	benzaldehyde + acetic anhydride/ $\text{CH}_3\text{COONa} \rightarrow$ cinnamic acid
Clemmensen reduction	$\text{C}=\text{O} + \text{Zn}-\text{Hg}/\text{conc. HCl} \rightarrow \text{CH}_2$
Wolff-Kishner reduction	$\text{C}=\text{O} + \text{N}_2\text{H}_4/\text{KOH} \rightarrow \text{CH}_2$
HVZ reaction	$\text{RCOOH} + \text{Cl}_2/\text{red P} \rightarrow \alpha\text{-chloro acid}$
Kolbe's electrolytic reaction	electrolysis of $\text{RCOONa} \rightarrow \text{R}-\text{R}$
Iodoform (haloform) reaction	$\text{CH}_3\text{CO}- + \text{I}_2/\text{NaOH} \rightarrow \text{CHI}_3$

2 · The mechanisms

M1 – M6

M1 Nucleophilic addition, worked with HCN

1. The nucleophile CN^- attacks the $\delta+$ **carbonyl carbon**. The π electrons shift on to oxygen, which becomes negatively charged, and the carbon changes from sp^2 (planar) to sp^3 (tetrahedral).
2. The alkoxide oxygen then takes a proton from HCN , regenerating CN^- and giving the **cyanohydrin**.

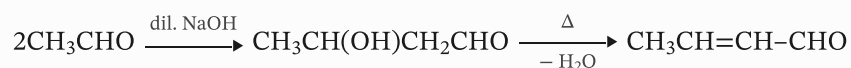


The reaction is **base catalysed**, because the attacking species is CN^- , not HCN — so a trace of alkali speeds it up. Hydrolysis of the cyanohydrin gives an **α -hydroxy acid**: acetaldehyde cyanohydrin gives **lactic acid**, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$.

■ mechanism complete

M2 Aldol condensation

1. The base removes an **α -hydrogen**, giving a **carbanion** (an enolate) which is stabilised by resonance with the carbonyl group.
2. That carbanion is itself a nucleophile, and it attacks the carbonyl carbon of a **second** molecule.
3. Protonation gives the **β -hydroxy aldehyde** — the aldol.
4. On heating, the aldol **loses water** to give an α,β -unsaturated aldehyde.



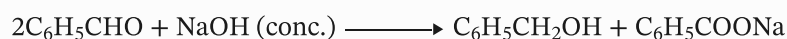
The product is named 3-hydroxybutanal, and after dehydration but-2-enal (crotonaldehyde). An α -hydrogen is **essential** — this is why formaldehyde and benzaldehyde cannot give an aldol.

■ mechanism complete

M3 Cannizzaro reaction — a disproportionation

Used when the aldehyde has **no α -hydrogen**, so no aldol is possible.

1. Hydroxide adds to the carbonyl carbon, giving a tetrahedral intermediate.
2. That intermediate transfers a **hydride ion** (H^-) to the carbonyl carbon of a **second** molecule of the aldehyde.
3. The molecule that lost the hydride is **oxidised** to the carboxylate; the one that gained it is **reduced** to the alcohol.



Hold on to this · **Half is oxidised and half is reduced** — that is what disproportionation means, and it is the phrase the examiner looks for. Formaldehyde and benzaldehyde are the two standard substrates, because neither has an α -hydrogen.

■ mechanism complete

M4 Why a carboxylic acid is so much more acidic than an alcohol or phenol

1. Ionisation gives the **carboxylate ion**, RCOO^- .
2. In that ion the negative charge is delocalised over **two equivalent oxygen atoms**. The two resonance contributors are of **equal energy**, which makes the delocalisation exceptionally effective, and the two C–O bonds are found experimentally to be of **equal length**.
3. In a **phenoxide** ion the charge is delocalised too, but on to **carbon** atoms of the ring, which are less electronegative than oxygen and hold it less comfortably. In an **alkoxide** ion there is no delocalisation at all.
4. The more stable the anion, the stronger the acid. Hence **$\text{RCOOH} > \text{H}_2\text{CO}_3 > \text{phenol} > \text{water} > \text{alcohol}$** .

*This is also why a carboxylic acid **liberates CO_2 from NaHCO_3** and phenol does not — the single most-used test in the organic part of the paper.*

■ proved

M5 Hell–Volhard–Zelinsky (HVZ) reaction

1. Red phosphorus and the halogen generate PCl_3 , which converts a little of the acid into its **acid chloride**.
2. The acid chloride enolises far more readily than the acid itself, and the enol is halogenated at the **α -carbon**.
3. Exchange with more of the parent acid releases the α -halo acid and regenerates the acid chloride.



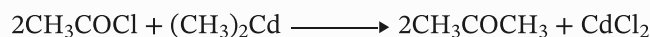
*Only the α -hydrogen is replaced, and the reaction fails for acids with no α -hydrogen — so **formic acid and benzoic acid do not undergo HVZ**. Repeating it gives the di- and tri-halo acids, hence the route to trichloroacetic acid.*

■ mechanism complete

M6 Why dialkyl cadmium and not a Grignard reagent

Both reagents will convert an acid chloride into a ketone. The difference is what happens **next**.

1. A **Grignard reagent** is extremely reactive. As soon as the ketone forms, the Grignard reagent still present attacks it again, adding a second alkyl group. The product on hydrolysis is a **tertiary alcohol**, not the ketone.
2. **Dialkyl cadmium**, R_2Cd , is a much **milder** alkylating agent. It is reactive enough to attack the acid chloride, which is highly electrophilic, but **not** reactive enough to attack the far less electrophilic ketone. The reaction therefore stops cleanly at the ketone.



Hold on to this · The general principle is worth carrying: when a reaction can run twice, you stop it after one step by using a **less reactive reagent**, not by using less of a reactive one.

■ **proved**

3 · Every type of problem that is set

T1 – T16

T1 Preparation — “how will you obtain ...?”

RECOGNISE IT BY

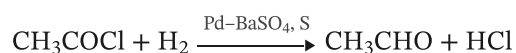
A named carbonyl compound or acid from a named starting material.

METHOD

Match against the table in C2 or C14. The two that are asked most are **Rosenmund** (acid chloride → aldehyde) and **calcium salt distillation** (→ ketone). Give conditions over the arrow.

WORKED EXAMPLE

Write equations for (a) Rosenmund's reduction (b) the dry distillation of calcium acetate. (Heritage 2024–25)



T2 Cyanohydrin and the acid it hydrolyses to

RECOGNISE IT BY

“The cyanohydrin of which compound gives lactic acid on hydrolysis?”

METHOD

Work **backwards**. Write the target α -hydroxy acid, replace its $-\text{COOH}$ by $-\text{CN}$ to get the cyanohydrin, then remove HCN to leave the carbonyl compound.

WORKED EXAMPLE

The cyanohydrin of which compound gives lactic acid on hydrolysis? (Heritage 2024–25)

1. lactic acid = $\text{CH}_3\text{--CH(OH)--COOH}$
2. so the cyanohydrin was $\text{CH}_3\text{--CH(OH)--CN}$
3. removing HCN leaves $\text{CH}_3\text{--CHO}$

Answer • **Acetaldehyde**. $\text{CH}_3\text{CHO} + \text{HCN} \rightarrow \text{CH}_3\text{CH(OH)CN}$, and hydrolysis of the $-\text{CN}$ to $-\text{COOH}$ gives 2-hydroxypropanoic acid, i.e. lactic acid. Formaldehyde would give glycolic acid and acetone would give 2-hydroxy-2-methylpropanoic acid.

T3 Products with ammonia derivatives

RECOGNISE IT BY

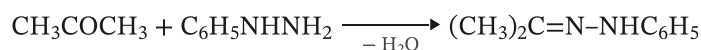
“What happens when acetone reacts with phenylhydrazine?”

METHOD

All of these are **addition–elimination**: add across C=O , then lose water, leaving C=N . Name the product from the reagent using the table in C5.

WORKED EXAMPLE

Write the equation for $\text{CH}_3\text{COCH}_3 + \text{C}_6\text{H}_5\text{NHNH}_2$. (Heritage 2024–25)



The product is **acetone phenylhydrazone**. With 2,4-DNP the analogous product is an orange-red precipitate — the general test for a carbonyl group.

T4 Which reagent distinguishes this pair?

RECOGNISE IT BY

“Give one chemical test to distinguish ...”

METHOD

Use the sieve in C7: 2,4-DNP for *any* carbonyl, **Tollens'** for an aldehyde, **Fehling's** for an *aliphatic* aldehyde, **iodoform** for $\text{CH}_3\text{CO-}$, **NaHCO_3** for an acid. Always state what **both** members do.

WORKED EXAMPLE

Distinguish between (i) benzoic acid and phenol (ii) formic acid and acetic acid. (Heritage 2024–25)

- (i) **Sodium bicarbonate**. Benzoic acid gives brisk **effervescence of CO_2** ; phenol, being weaker than carbonic acid, gives none.
- (ii) **Tollens' reagent** (or Fehling's, or HgCl_2). **Formic acid** reduces it, giving a **silver mirror** — it alone among the acids contains a $-\text{CHO}$ group. Acetic acid gives no reaction.

T5 Reduction — which reagent gives which product

RECOGNISE IT BY

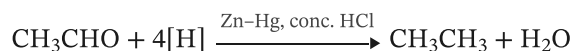
“What is obtained when acetaldehyde is reduced with Zn/Hg and conc. HCl?”

METHOD

Decide the destination first from C8: hydride reagents stop at the **alcohol**; Clemmensen, Wolff–Kishner and red P/HI go all the way to the **alkane**.

WORKED EXAMPLE

Acetaldehyde is reduced with hydrogen in the presence of Zn/Hg and conc. HCl. Write the equation and name the reaction. (Heritage 2024–25)



Answer · Ethane, by **Clemmensen reduction**. Note that C=O has become CH₂, not CH–OH — with LiAlH₄ the product would have been ethanol instead.

T6 Aldol or Cannizzaro?

RECOGNISE IT BY

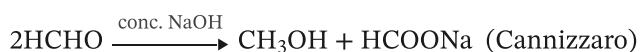
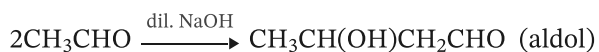
An aldehyde with dilute or concentrated alkali.

METHOD

One question settles it: **does the aldehyde have an α-hydrogen?** Yes → aldol. No → Cannizzaro. Formaldehyde and benzaldehyde have none.

WORKED EXAMPLE

What happens when (a) acetaldehyde (b) formaldehyde is treated with sodium hydroxide?



T7 Named reaction equations

RECOGNISE IT BY

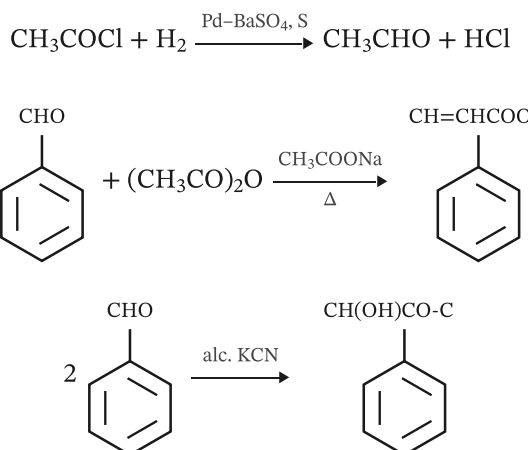
“Give balanced equations with conditions for the following name reactions.”

METHOD

Learn C17 as a two-column list. This unit contributes more named reactions than any other, and the three set most often together are **Rosenmund, Perkin and Benzoin**.

WORKED EXAMPLE

Give balanced equations for (i) Rosenmund's reduction (ii) Perkin reaction (iii) Benzoin condensation. (Heritage 2024–25)



T8 Identify [A], [B], [C] in a sequence

RECOGNISE IT BY

A chain of arrows through carbonyl chemistry.

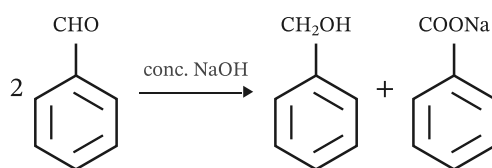
METHOD

Track the carbon count and watch for the **reagent that reveals the class** — Tollens' means aldehyde, iodoform means $\text{CH}_3\text{CO}-$, NaHCO_3 means acid.

WORKED EXAMPLE

An organic compound [A] has a characteristic odour. On treatment with NaOH it forms [B] and [C]. Compound [B], $\text{C}_7\text{H}_8\text{O}$, on partial oxidation gives back [A]. Compound [C] is the sodium salt of an acid; with soda lime it gives an aromatic hydrocarbon [D]. Deduce [A] to [D]. (Heritage 2024–25)

- One aldehyde giving both an alcohol and a carboxylate with NaOH is a **Cannizzaro** reaction, so [A] has **no α -hydrogen**.
- [B] = $\text{C}_7\text{H}_8\text{O}$ and partial oxidation returns [A] $\Rightarrow$ [B] is a primary alcohol, **benzyl alcohol**, $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$.
- So [A] = **benzaldehyde**, $\text{C}_6\text{H}_5\text{CHO}$ — and the characteristic odour (bitter almonds) confirms it.
- [C] = **sodium benzoate**, $\text{C}_6\text{H}_5\text{COONa}$.
- Soda lime decarboxylates it: [D] = **benzene**.



T9 Structure elucidation from a formula plus tests

RECOGNISE IT BY

“An organic compound C_3H_6O gives a positive iodoform test but a negative Tollens' test.”

METHOD

Apply the tests as filters, in this order: **2,4-DNP** (is there a carbonyl?), **Tollens'** (aldehyde or ketone?), **iodoform** (is there a methyl next to it?), **$NaHCO_3$** (is it an acid?). Then check the molecular formula.

WORKED EXAMPLE

An organic compound [A], C_3H_6O , gives a positive iodoform test but does not respond to Tollens' reagent. On reduction with $LiAlH_4$ it gives [B]. Identify [A] and [B].

1. negative Tollens' $\Rightarrow$ not an aldehyde $\Rightarrow$ a ketone
2. positive iodoform $\Rightarrow$ contains CH_3CO-
3. C_3H_6O with both $\Rightarrow$ [A] = **propanone (acetone)**
4. $LiAlH_4$ reduces $C=O$ to $CH-OH \Rightarrow$ [B] = **propan-2-ol**

T10 Acidity ordering

RECOGNISE IT BY

“Arrange in increasing order of acid strength”, or a pK_a comparison.

METHOD

Judge the **stability of the carboxylate anion**. Electron-withdrawing groups strengthen, electron-releasing weaken, and the effect falls off with distance. Remember that a **lower pK_a means a stronger acid**.

WORKED EXAMPLE

The pK_a value of 4-nitrobenzoic acid is lower than that of benzoic acid. Explain. (Heritage 2024–25)

Answer • A lower pK_a means 4-nitrobenzoic acid is the **stronger** acid. The $-NO_2$ group is strongly **electron withdrawing** by both the $-I$ and the $-R$ effect, and from the **para** position its resonance withdrawal reaches the carboxylate group directly. It therefore **disperses and stabilises the negative charge** on the 4-nitrobenzoate ion, so the proton is released more readily. Benzoic acid has no such help.

Increasing acid strength

4-methylbenzoic acid < benzoic acid < 4-chlorobenzoic acid < 4-nitrobenzoic acid

T11 Aliphatic acidity and the halogen effect

RECOGNISE IT BY

A set of substituted acetic acids.

METHOD

Count the withdrawing atoms and note their distance from $-COOH$. More halogens and closer halogens mean a stronger acid; alkyl groups weaken it.

WORKED EXAMPLE

Arrange in increasing order of acid strength: CH_3COOH , $CH_2ClCOOH$, $CHCl_2COOH$, CCl_3COOH , $HCOOH$.

Increasing acid strength

CH_3COOH < $HCOOH$ < $CH_2ClCOOH$ < $CHCl_2COOH$ < CCl_3COOH

Formic acid beats acetic acid because it has **no** electron-releasing alkyl group; but even one chlorine outweighs that difference.

T12 Reactions of carboxylic acids

RECOGNISE IT BY

“What happens when acetic acid is treated with ...?”

METHOD

Read off C15. The four asked most are **NaHCO₃** (the test), **PCl₅** (acid chloride), **soda lime** (decarboxylation) and **Cl₂/red P** (HVZ).

WORKED EXAMPLE

Write equations for (a) $\text{CH}_3\text{COOH} + \text{Cl}_2/\text{red P}$ (b) $\text{CH}_3\text{COONa} + \text{soda lime}$. (Heritage 2024–25)



T13 Conversions

RECOGNISE IT BY

“How can the following conversion be brought about?”

METHOD

Compare carbon counts first. Grignard with CO₂ or with an aldehyde adds a carbon; decarboxylation and Hunsdiecker remove one; oxidation and reduction move up and down the same carbon skeleton.

WORKED EXAMPLE

Convert (a) ethanal to propan-2-ol (b) acetaldehyde to acetamide. (Heritage 2024–25)



T14 Why dialkyl cadmium rather than a Grignard reagent

RECOGNISE IT BY

A “give a reason” question comparing the two for making a ketone.

METHOD

Name the problem with the Grignard reagent — it **over-reacts**, attacking the ketone it has just made and giving a **tertiary alcohol** — and then say why R₂Cd does not.

WORKED EXAMPLE

Dialkyl cadmium is considered superior to a Grignard reagent for the preparation of a ketone from an acid chloride. Explain. (Heritage 2024–25)

Answer • A Grignard reagent is **too reactive**. It converts the acid chloride to the ketone, but then attacks that ketone as well, adding a second alkyl group; on hydrolysis the product is a **tertiary alcohol**, not the ketone.

Dialkyl cadmium is a much milder alkylating agent — reactive enough towards the highly electrophilic acid chloride, but not towards the much less electrophilic ketone. The reaction therefore stops cleanly at the ketone stage.

T15 Urotropine and other ammonia products of formaldehyde

RECOGNISE IT BY

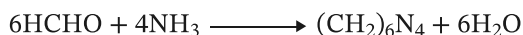
“Urotropine is formed by the action of _____ with _____.”

METHOD

Formaldehyde with **excess ammonia**. Six molecules of HCHO and four of NH₃ give hexamethylenetetramine.

WORKED EXAMPLE

Urotropine is formed by the action of _____ with _____. (Heritage 2024–25)



Answer · **Formaldehyde** with **ammonia**. Urotropine (hexamethylenetetramine) is used as a urinary antiseptic, and on nitration gives the explosive RDX.

T16 Benzaldehyde's peculiarities

RECOGNISE IT BY

Anything about benzaldehyde that an aliphatic aldehyde would not do.

METHOD

Three things set it apart: it gives **Tollens' but not Fehling's**; it **Cannizzaros** rather than aldols, having no α-H; and it undergoes **Benzoin** and **Perkin** reactions. Its –CHO group is **meta directing**.

WORKED EXAMPLE

Give one chemical test to distinguish between benzaldehyde and acetaldehyde.

Answer · **Fehling's solution**. Acetaldehyde, an aliphatic aldehyde, reduces it and gives a **red-brown precipitate of Cu₂O**. Benzaldehyde, an aromatic aldehyde, gives **no reaction** with Fehling's — although it does reduce Tollens', so Tollens' would *not* distinguish the pair. The iodoform test works as an alternative: acetaldehyde gives it, benzaldehyde does not.

4 · Aldehydes, ketones and carboxylic acids recall sheet

revise from this page alone

Reactivity to nucleophilic addition



steric + electronic (+I of alkyl) + resonance from a ring

The test sieve

2,4-DNP — any carbonyl (orange-red ppt.)

Tollens' — aldehyde (silver mirror)

Fehling's — **aliphatic** aldehyde only (red-brown)

Iodoform — CH₃CO– present

NaHCO₃ — carboxylic acid (CO₂)

α-hydrogen decides

Present → **aldol** (dil. NaOH)

Absent → **Cannizzaro** (conc. NaOH), a disproportionation

No α-H: HCHO, C₆H₅CHO

Reduction destinations

LiAlH₄/NaBH₄/H₂-Ni → **alcohol**

Zn-Hg + conc. HCl → alkane (**Clemmensen**)

N₂H₄/KOH → alkane (**Wolff-Kishner**)

Acid chloride, two fates

H₂/Pd-BaSO₄(poisoned) → **aldehyde** (Rosenmund)

R₂Cd → **ketone**

Grignard would over-react to a 3° alcohol

Acidity

RCOOH > H₂CO₃ > phenol > H₂O > alcohol

HCOOH > CH₃COOH > C₂H₅COOH

CCl₃ > CHCl₂ > CH₂Cl > CH₃

4-NO₂ > H > 4-CH₃ (benzoic)

lower pK_a = stronger acid

Ortho effect

Any ortho substituent makes benzoic acid **stronger** — steric inhibition of resonance, whether the group withdraws or releases.

Formic acid is special

The only acid that is also a **reducing agent** — it has a –CHO group

Reduces Tollens', Fehling's, HgCl_2 (grey Hg); decolourises KMnO_4

Directive effects

–CHO and –COOH are **deactivating, meta directing**
contrast –OH and –OCH₃: activating, o/p

Two different Kolbes

Kolbe's electrolytic — RCOONa electrolysed → alkane

Kolbe's reaction (Part V) — phenoxide + CO_2 → salicylic acid

HVZ

$\text{RCOOH} + \text{Cl}_2/\text{red P} \rightarrow \alpha\text{-chloro acid}$

Fails for HCOOH and $\text{C}_6\text{H}_5\text{COOH}$ — no $\alpha\text{-H}$

Benzaldehyde only

Tollens' yes, Fehling's **no**

Cannizzaro, **Benzoin** (alc. KCN), **Perkin**
 $((\text{CH}_3\text{CO})_2\text{O}/\text{CH}_3\text{COONa} \rightarrow \text{cinnamic acid})$

Urotropine

$6\text{HCHO} + 4\text{NH}_3 \rightarrow (\text{CH}_2)_6\text{N}_4$

urinary antiseptic; nitration gives RDX

Cyanohydrin → α -hydroxy acid

$\text{HCHO} \rightarrow \text{glycolic acid}$

$\text{CH}_3\text{CHO} \rightarrow \text{lactic acid}$

Part VII · Biomolecules

ISC unit 10 · Classification of carbohydrates · the reactions that establish the structure of glucose · anomers and mutarotation · disaccharides and polysaccharides · amino acids, the zwitter ion and the isoelectric point · the four levels of protein structure and denaturation · vitamins · DNA and RNA. Thirteen concepts, three structure proofs, eleven problem types.

1 · The concepts, in the order they build

C1 – C13

C1 Classification of carbohydrates

Class	On hydrolysis	Examples
Monosaccharides	cannot be hydrolysed further	glucose, fructose, galactose, ribose
Disaccharides	give two monosaccharide units	sucrose, maltose, lactose
Oligosaccharides	give 2 to 10 units	includes the disaccharides
Polysaccharides	give a very large number of units	starch, cellulose, glycogen

- A monosaccharide with an **aldehyde** group is an **aldose** (glucose); one with a **ketone** group is a **ketose** (fructose).
- Polysaccharides are **not sweet** and are not sugars, though they are carbohydrates.

Hold on to this · A carbohydrate is defined as an **optically active polyhydroxy aldehyde or ketone**, or a compound that gives such units on hydrolysis. The phrase “polyhydroxy aldehyde or ketone” is what the definition mark is for.

C2 Reducing and non-reducing sugars

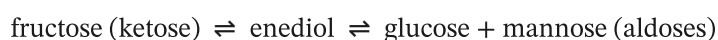
A **reducing sugar** reduces Tollens' reagent and Fehling's solution, because it has a **free aldehyde or ketone group** — that is, a free **anomeric carbon**.

Sugar	Reducing?	Why
Glucose, fructose, galactose	yes	free carbonyl group
Maltose	yes	only one of the two anomeric carbons is used in the linkage
Lactose	yes	same reason
Sucrose	no	both anomeric carbons are involved in the glycosidic linkage, so neither carbonyl group is free

Hold on to this · Sucrose is **the** non-reducing disaccharide, and the reason is always the same: its glycosidic bond ties up the C-1 of glucose *and* the C-2 of fructose, leaving no free carbonyl. Maltose and lactose leave one free, so they reduce.

C3 Why fructose reduces Tollens' although it is a ketone

A ketone should not reduce Tollens'. Fructose does, and the reason is worth knowing precisely: **the test is carried out in alkaline medium**, and in alkali fructose **tautomerises** through an **enediol** intermediate into glucose and mannose, both of which are aldoses.



Hold on to this · So fructose reduces Tollens' and Fehling's **indirectly**. This is also why those two reagents cannot be used to tell glucose from fructose — for that you need **bromine water**, which is not alkaline (C6).

C4 The open-chain structure of glucose, and the evidence for it

Glucose is $\text{CH}_2\text{OH}(\text{CHOH})_4\text{CHO}$ — a straight chain of six carbons with five hydroxyl groups and one aldehyde group. Each of the following reactions establishes one part of that:

Reaction	Product	What it proves
Heating with HI	n-hexane	the six carbons form a straight chain
Hydroxylamine , NH_2OH	glucose oxime	a carbonyl group is present
Bromine water	gluconic acid (a monocarboxylic acid)	the carbonyl group is an aldehyde , not a ketone
Acetic anhydride (excess)	glucose penta -acetate	there are five hydroxyl groups
Warm dilute HNO₃	saccharic acid (a dicarboxylic acid)	there is a primary alcoholic group ($-\text{CH}_2\text{OH}$) as well as the $-\text{CHO}$, both oxidised to $-\text{COOH}$
Phenylhydrazine (3 mol)	glucosazone	reaction at C-1 and C-2

Hold on to this · The **HNO₃** and **bromine water** results are the two that matter most and are most often confused. Bromine water gives a **monocarboxylic** acid, so only the $-\text{CHO}$ was oxidised: the carbonyl is an aldehyde. Nitric acid gives a **dicarboxylic** acid, so a second oxidisable group — the primary $-\text{CH}_2\text{OH}$ at the far end — must also be present.

C5 The cyclic structure, anomers and mutarotation

- The open-chain form does not explain everything — glucose does not give the Schiff's test, and it forms only a **monoacetal** rather than the diacetal a free $-\text{CHO}$ would give. The explanation is that glucose exists largely as a **six-membered cyclic hemiacetal**, formed by the C-5 hydroxyl attacking the C-1 carbonyl.
- C-1 thereby becomes a new chiral centre, called the **anomeric carbon**. The two forms that result are **anomers**: **α -D-glucose**, with the C-1 $-\text{OH}$ on the opposite side from the C-5 $-\text{OH}$ in the Fischer projection, and **β -D-glucose**, with it on the same side.
- Mutarotation** is the gradual change in the specific rotation of a freshly prepared solution of either anomer, until an equilibrium mixture of the two is reached.

Form	Specific rotation
pure α -D-glucose	$+111^\circ$
pure β -D-glucose	$+19^\circ$
equilibrium mixture (after mutarotation)	$+52.5^\circ$

Hold on to this · **Mutarotation** is the interconversion of the α and β anomers in solution, and it proves that the ring can open and close. The rapid conversion of α -D-glucose into β -D-glucose in solution is mutarotation — that is the one-mark objective.

D and L refer to the configuration of the **lowest chiral carbon** (C-5 in glucose) compared with D-glyceraldehyde, and have nothing to do with the direction of optical rotation. A D sugar may be either dextro- or laevorotatory.

C6 Distinguishing glucose from fructose

Tollens' and Fehling's will not do it, because fructose tautomerises in alkali (C3). The test that works is **bromine water**, which is **acidic and so cannot cause tautomerisation**.



Hold on to this · So **glucose decolourises bromine water and fructose does not**. This is the syllabus's named "test for glucose and fructose", and it must be given with the equation.

C7 The three disaccharides

Disaccharide	Units	Linkage	Reducing?
Sucrose	α -D-glucose + β -D-fructose	C-1 of glucose to C-2 of fructose	no
Maltose	two α -D-glucose	C-1 to C-4	yes
Lactose	β -D-galactose + β -D-glucose	C-1 to C-4	yes

Invert sugar. Sucrose has a specific rotation of $+66.5^\circ$. On hydrolysis it gives an equimolar mixture of glucose ($+52.5^\circ$) and fructose (-92.4°), whose net rotation is -20° . Because the sign of rotation is **inverted**, the product is called **invert sugar** and the process **inversion of cane sugar**.

*That inversion is also the reason the hydrolysis of sucrose is the standard example of a **pseudo first order** reaction in Part II — the rotation gives a convenient way of following the rate.*

C8 The three polysaccharides

Polysaccharide	Made of	Linkage	Note
Starch	α -D-glucose	amylose : linear, α -1,4 · amylopectin : branched, α -1,4 and α -1,6	the storage carbohydrate of plants; amylose gives the blue colour with iodine
Cellulose	β -D-glucose	linear, β-1,4	the structural material of plant cell walls; humans cannot digest it, having no enzyme for the β linkage
Glycogen	α -D-glucose	α -1,4 and α -1,6, more highly branched than amylopectin	"animal starch" — the storage carbohydrate of animals, in liver and muscle

Hold on to this · Starch and cellulose are both polymers of glucose and differ only in the α or β linkage — and that one difference is why one is food and the other is wood.

C9 Amino acids, the zwitter ion and the isoelectric point

An α -amino acid has both an **amino** and a **carboxyl** group on the same carbon: $\text{H}_2\text{N}-\text{CHR}-\text{COOH}$.

- Because the molecule contains both an acid and a base, a proton transfers internally to give a **zwitter ion** — $^+\text{H}_3\text{N}-\text{CHR}-\text{COO}^-$. It carries **both a positive and a negative charge but is electrically neutral overall**, which is why it is also called a **dipolar ion**.
- This explains their properties: amino acids are **crystalline solids of high melting point**, soluble in water and insoluble in organic solvents — they behave like salts, not like ordinary organic compounds.
- They are **amphoteric**: in acid the zwitter ion accepts a proton and becomes a cation, in alkali it loses one and becomes an anion.
- The **isoelectric point** is the pH at which the amino acid exists entirely as the zwitter ion, so that it has **no net charge and does not migrate** towards either electrode in an electric field.

Hold on to this · Glycine, $\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$, is the only naturally occurring amino acid that is not optically active, because its α -carbon carries **two hydrogen atoms** and so is not a chiral centre. Every other α -amino acid has four different groups on that carbon.

At a pH **below** the isoelectric point the amino acid is a cation and migrates to the **cathode**; **above** it, it is an anion and migrates to the **anode**.

C10 Peptides and proteins

- A **peptide bond** is the amide link $-\text{CO}-\text{NH}-$ formed between the $-\text{COOH}$ of one amino acid and the $-\text{NH}_2$ of the next, with loss of water.
- Two residues give a **dipeptide**, three a tripeptide, many a **polypeptide**; a polypeptide of more than about a hundred residues, with a definite shape, is a **protein**.

Classification by shape	Structure	Solubility	Examples
Fibrous	long, thread-like chains held side by side	insoluble in water	keratin (hair, nails), myosin (muscle)
Globular	chains coiled into a compact ball	soluble in water	insulin, albumin, haemoglobin, enzymes

C11 The four levels of protein structure, and denaturation

Level	What it is	Held together by
Primary	the sequence in which the amino acids are joined	peptide (covalent) bonds
Secondary	the local shape the chain takes up — the α-helix or the β-pleated sheet	hydrogen bonds between the C=O and N-H of the backbone
Tertiary	the overall folding of the whole chain into its three-dimensional shape	hydrogen bonds, disulphide bridges, ionic and van der Waals forces
Quaternary	the way two or more folded chains (subunits) assemble	the same non-covalent forces; e.g. haemoglobin, 4 subunits

Denaturation is the loss of the secondary, tertiary and quaternary structure — but **not** the primary — on heating or on a change of pH. The hydrogen bonds and other weak interactions are broken while the peptide bonds survive, so the chain unfolds into a random shape, loses its biological activity and usually coagulates.

Hold on to this · That is why the answer to “how can a secondary structure be converted into the primary structure?” is **by denaturation** — heat it, or change the pH, and the hydrogen bonds holding the helix or sheet give way, leaving the peptide sequence intact.

Everyday examples: boiling an egg (albumin coagulates) and curdling milk with acid.

C12 Vitamins

Class	Vitamins	Note
Fat soluble	A, D, E, K	stored in the liver and fatty tissue; an excess can be harmful
Water soluble	B complex and C	not stored, so they must be supplied regularly in the diet

Vitamin	Deficiency disease
A	night blindness, xerophthalmia
B ₁ (thiamine)	beri-beri
B ₂ (riboflavin)	cheilosis, digestive disorders
B ₆ (pyridoxine)	convulsions
B ₁₂	pernicious anaemia
C (ascorbic acid)	scurvy
D	rickets in children, osteomalacia in adults
E	sterility, muscular weakness
K	increased blood clotting time

Memory hook · The fat-soluble four are **A, D, E, K** — everything else is water soluble. Chemical names and structures are **not** required by the syllabus, only the classification and the deficiency diseases.

C13 Nucleic acids — DNA and RNA

- A **nucleotide** is the basic repeating unit: a nitrogenous **base**, a **pentose sugar** and a **phosphate** group. Without the phosphate it is a **nucleoside**.
- The bases are **purines** — adenine (A) and guanine (G), which have two fused rings — and **pyrimidines** — cytosine (C), thymine (T) and uracil (U), which have one ring.

	DNA	RNA
Sugar	2-deoxy -D-ribose	D-ribose
Bases	A, G, C, T	A, G, C, U
Structure	double helix , two strands held by hydrogen bonds between complementary bases (A–T, G–C)	usually a single strand
Location	chiefly in the nucleus	chiefly in the cytoplasm
Function	stores and transmits hereditary information	directs protein synthesis

The syllabus asks for the differences and the double-helical structure in words; **no chemical structures are required**.

2 • The structure proofs

P1 – P3

P1 Establishing the open-chain structure of glucose

The molecular formula $C_6H_{12}O_6$ alone settles nothing. Each observation below removes one possibility, and together they fix the structure as $CH_2OH(CHOH)_4CHO$.

1. **Prolonged heating with HI gives n-hexane**. All six carbons are therefore in one **unbranched chain**.
2. **It forms an oxime with hydroxylamine and adds HCN**. A **carbonyl** group is present.
3. **Bromine water oxidises it to gluconic acid, a monocarboxylic acid**. Only one group has been oxidised, and mild oxidation converts $-CHO$ to $-COOH$ but leaves a ketone alone. So the carbonyl group is an **aldehyde**, and it is at the end of the chain.
4. **Acetylation with excess acetic anhydride gives a penta-acetate**. There are **five** $-OH$ groups.
5. **Warm dilute nitric acid gives saccharic acid, a dicarboxylic acid**. A second group has been oxidised to $-COOH$, and it can only be a **primary alcoholic group** — so the far end of the chain is $-CH_2OH$.



Step 3 against step 5 is the pair that earns the marks: **monocarboxylic** from bromine water proves the aldehyde; **dicarboxylic** from nitric acid proves the primary alcohol at the other end.

■ **proved**

P2 Why the cyclic structure is needed, and what mutarotation proves

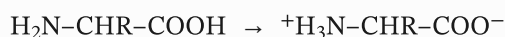
1. The open-chain structure predicts that glucose should give the **Schiff's test** and should react with **NaHSO₃**, as an aldehyde does. It does **neither**.
2. With methanol and dry HCl it forms a **monomethyl acetal**, whereas a free –CHO would give a dimethyl acetal.
3. Both observations are explained if the –CHO is not free but is tied up as a **cyclic hemiacetal**, formed by the C-5 hydroxyl group adding across the C-1 carbonyl to give a six-membered ring.
4. That makes C-1 a new chiral centre — the **anomeric carbon** — so two forms exist, α and β .
5. **Mutarotation** confirms it. A freshly made solution of pure α -D-glucose has a rotation of $+111^\circ$, which falls steadily to $+52.5^\circ$; pure β -D-glucose starts at $+19^\circ$ and rises to the same $+52.5^\circ$. Both are reaching the **same equilibrium mixture**, which can only happen if the ring opens and closes in solution, passing through the open-chain form.

■ proved

P3 The zwitter ion, and why glycine is not optically active

The zwitter ion.

1. An amino acid contains a basic –NH₂ group and an acidic –COOH group in the same molecule.
2. The acid protonates the base **internally**:



3. The result carries **one positive and one negative charge** and is therefore neutral overall — a **dipolar ion**. This is why amino acids are high-melting crystalline solids, water soluble, and amphoteric.

Why glycine is the exception.

1. Optical activity requires a carbon with **four different** groups.
2. In a general α -amino acid the α -carbon carries –H, –NH₂, –COOH and –R: four different groups, so it is chiral.
3. In **glycine**, R = H. The α -carbon therefore carries –H, –H, –NH₂ and –COOH — **two of its groups are identical**, so it is not a chiral centre and the molecule has a plane of symmetry.

Glycine is therefore the only naturally occurring α -amino acid that is optically inactive.

■ proved

T1 Classify a carbohydrate, or say whether it reduces**RECOGNISE IT BY**

“Classify sucrose”, or “is maltose a reducing sugar? Give a reason.”

METHOD

Two questions. **How many units on hydrolysis?** — that gives the class. **Is an anomeric carbon free?** — that decides whether it reduces. Sucrose is the only one of the three disaccharides in which both are tied up.

WORKED EXAMPLE

Why is sucrose a non-reducing sugar while maltose is reducing?

Answer · In **sucrose** the glycosidic linkage joins the **C-1 of glucose to the C-2 of fructose** — that is, it involves the anomeric carbon of **both** units. No free aldehyde or ketone group remains, so sucrose cannot reduce Tollens' or Fehling's. In **maltose** the linkage is C-1 to C-4, so the anomeric carbon of the second glucose unit stays **free** and can open to the aldehyde form, which reduces.

T2 What does this reaction of glucose prove?**RECOGNISE IT BY**

A reagent and its product are given, and the structural conclusion is wanted.

METHOD

Match against the table in C4 — but reason it rather than recall it: ask what **changed**, and what feature that change requires. A monocarboxylic acid means one oxidisable group; a dicarboxylic acid means two.

WORKED EXAMPLE

Glucose gives gluconic acid with bromine water but saccharic acid with warm dilute nitric acid. What does each result establish?

- **Bromine water** → **gluconic acid**, a **monocarboxylic acid**. Bromine water is a mild oxidant: it oxidises an aldehyde to an acid but leaves a ketone and a primary alcohol untouched. So the carbonyl group of glucose is an **aldehyde**.
- **Nitric acid** → **saccharic acid**, a **dicarboxylic acid**. A stronger oxidant has converted a second group into –COOH, and the only candidate is a **primary alcoholic (–CH₂OH)** group at the other end of the chain.

T3 Distinguish glucose from fructose

RECOGNISE IT BY

“Give a chemical test to distinguish glucose and fructose.”

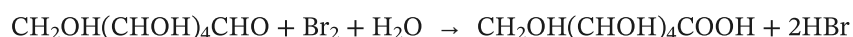
METHOD

Do **not** offer Tollens' or Fehling's — both are alkaline, and fructose tautomerises in alkali and reduces them too (C3). Use **bromine water**, which is acidic.

WORKED EXAMPLE

How would you distinguish between glucose and fructose?

Answer · Shake each with **bromine water**. **Glucose**, being an aldose, is oxidised to gluconic acid and **decolourises** the bromine water. **Fructose**, a ketose, gives **no reaction** and the brown colour remains. Bromine water works because it is **acidic**, so fructose cannot tautomerise to an aldose as it does in the alkaline Tollens' and Fehling's tests.



T4 Mutarotation and anomers

RECOGNISE IT BY

“What is mutarotation?”, or the rapid interconversion of the two anomers.

METHOD

Define it as the **change in specific rotation of a freshly prepared solution until equilibrium is reached**, and say *why*: the ring opens and closes, so the α and β anomers interconvert.

WORKED EXAMPLE

The rapid conversion of α -D-glucose and β -D-glucose in solution is known as: (Heritage 2024–25)

Answer · **Mutarotation**. A fresh solution of α -D-glucose has a specific rotation of $+111^\circ$ and one of β -D-glucose $+19^\circ$; both drift to **$+52.5^\circ$** , the value of the equilibrium mixture. The interconversion is possible because the cyclic hemiacetal ring can **open to the open-chain form and close the other way round**.

Common slip · It is **not** racemisation. Racemisation gives an optically inactive 50:50 mixture of enantiomers; the α and β anomers are diastereomers, and the equilibrium mixture is still optically active at $+52.5^\circ$.

T5 Invert sugar

RECOGNISE IT BY

“Why is the hydrolysis product of sucrose called invert sugar?”

METHOD

Quote the three rotations and show that the **sign** changes. That change of sign is the whole point of the name.

WORKED EXAMPLE

Explain why the hydrolysis of sucrose is called inversion of cane sugar.

1. sucrose $+66.5^\circ$
2. hydrolysis gives glucose $(+52.5^\circ)$ and fructose (-92.4°) , equimolar
3. net rotation $= \frac{1}{2}(+52.5 - 92.4) \approx -20^\circ$

Answer · The rotation changes from **$+66.5^\circ$ to about -20°** — the **sign is inverted**, because fructose is strongly laevorotatory and outweighs the dextrorotatory glucose. The mixture is therefore called **invert sugar**.

T6 Linkages in disaccharides and polysaccharides

RECOGNISE IT BY

“Name the linkage in maltose”, or “how do starch and cellulose differ?”

METHOD

Give the **units**, the **positions** joined, and whether the linkage is **α or β** . All three carry marks, and the α/β distinction is the one that matters biologically.

WORKED EXAMPLE

How do starch and cellulose differ, although both are polymers of glucose?

Answer · **Starch** is built from **α -D-glucose**, joined **α -1,4** in amylose (linear) and **α -1,4** with **α -1,6** branches in amylopectin. **Cellulose** is built from **β -D-glucose**, joined **β -1,4** in long unbranched chains that pack side by side into fibres. The difference in the linkage is why starch is digestible by humans while cellulose is not — we have no enzyme that hydrolyses the **β -1,4** bond.

T7 The zwitter ion

RECOGNISE IT BY

“What is a zwitter ion?”, or “why is it also called a dipolar ion?”

METHOD

Give the structure, then the two properties that follow: it carries **both charges but is neutral overall**, and it makes the amino acid behave like a **salt** — high melting, water soluble, amphoteric.

WORKED EXAMPLE

Why is a zwitter ion also known as a dipolar ion? (Heritage 2024–25)



Answer · Because it carries **two opposite charges on the same molecule** — a positive charge on the protonated amino group and a negative charge on the carboxylate — and is therefore a **dipole**, while being electrically **neutral overall**. The internal proton transfer from **$-\text{COOH}$ to $-\text{NH}_2$** is what creates it.

T8 Optical activity of amino acids

RECOGNISE IT BY

“Except glycine, all naturally occurring α -amino acids are optically active. Give a reason.”

METHOD

State the requirement for chirality — **four different groups** — then show that glycine fails it because $R = H$ gives it two hydrogens on the α -carbon.

WORKED EXAMPLE

Except glycine, all other naturally occurring α -amino acids are optically active. Explain. (Heritage 2024–25)

Answer · Optical activity needs a carbon with **four different groups**. In a general α -amino acid the α -carbon carries $-H$, $-NH_2$, $-COOH$ and $-R$, all different, so it is a **chiral centre**. In **glycine** R is itself a hydrogen atom, so the α -carbon carries **two identical hydrogens**; it is not a chiral centre, the molecule has a plane of symmetry, and glycine is therefore optically inactive.

T9 Protein structure, and converting one level into another

RECOGNISE IT BY

A diagram of an α -helix or β -sheet, with “which structure is this?” and “how can it be converted to the primary structure?”

METHOD

An α -**helix** or a β -**pleated sheet** is a **secondary** structure, held by **hydrogen bonds** in the backbone. To reduce it to the primary structure, break those hydrogen bonds — that is, **denature** it — while leaving the peptide bonds intact.

WORKED EXAMPLE

(a) Which type of structure do an α -helix and a β -pleated sheet represent? (b) How can these structures be converted into the primary structure? (Heritage 2024–25)

- (a) Both are **secondary** structures — the local shape taken up by the polypeptide chain, held in place by **hydrogen bonds** between the $C=O$ and $N-H$ groups of the backbone.
- (b) By **denaturation** — heating the protein or changing the pH. This breaks the **hydrogen bonds** that maintain the helix or sheet, so the chain uncoils into a random shape and loses its biological activity. The **peptide bonds are covalent and are not broken**, so the primary structure — the sequence of amino acids — survives unchanged.

Familiar examples of denaturation: the coagulation of egg white on boiling, and the curdling of milk on adding acid.

T10 Vitamins

RECOGNISE IT BY

“Classify the following vitamins”, or a deficiency disease.

METHOD

Only two things are examinable: the **fat-soluble four (A, D, E, K)** against the water-soluble rest, and the **deficiency diseases**. No structures are needed.

WORKED EXAMPLE

Name the vitamin whose deficiency causes (a) scurvy (b) rickets (c) night blindness, and classify each as fat or water soluble.

Disease	Vitamin	Class
Scurvy	C	water soluble
Rickets	D	fat soluble
Night blindness	A	fat soluble

T11 DNA against RNA

RECOGNISE IT BY

“Give three differences between DNA and RNA.”

METHOD

Pick from the four reliable axes: the **sugar**, the **bases**, the **number of strands**, and the **function**. Give the difference on both sides — a statement about DNA alone is only half an answer.

WORKED EXAMPLE

State three points of difference between DNA and RNA.

	DNA	RNA
Sugar	2-deoxy-D-ribose	D-ribose
Base present	thymine	uracil
Strands	double helix	usually single stranded
Function	stores hereditary information	directs protein synthesis

4 • Biomolecules recall sheet

revise from this page alone

Glucose structure proofs

HI → n-hexane ⇒ **straight chain**

NH₂OH → oxime ⇒ **carbonyl**

Br₂ water → gluconic acid (**mono**) ⇒ **aldehyde**

(CH₃CO)₂O → **penta**-acetate ⇒ **5 -OH**

HNO₃ → saccharic acid (**di**) ⇒ **-CH₂OH**

Glucose vs fructose

Bromine water — glucose decolourises, fructose does not
Tollens'/Fehling's fail: alkaline, so fructose tautomerises to an aldose

Mutarotation

$\alpha + 111^\circ \rightarrow +52.5^\circ \leftarrow \beta + 19^\circ$

ring opens and closes; $\alpha \rightleftharpoons \beta$

Not racemisation

Reducing or not

Reducing: glucose, fructose, maltose, lactose

Non-reducing: sucrose — both anomeric carbons tied up

The disaccharides

Sucrose α -glucose + β -fructose, C1–C2, non-reducing

Maltose 2 α -glucose, C1–C4, reducing

Lactose β -galactose + β -glucose, C1–C4, reducing

Invert sugar

sucrose $+66.5^\circ \rightarrow$ glucose $+52.5^\circ$ and fructose -92.4°

net $\approx -20^\circ$, so the sign inverts

The polysaccharides

Starch α -glucose: amylose α -1,4 linear; amylopectin branched

Cellulose β -glucose, β -1,4 linear — indigestible

Glycogen α -glucose, most branched, animal starch

Zwitter ion

$^+\text{H}_3\text{N}-\text{CHR}-\text{COO}^-$

Both charges, neutral overall $\Rightarrow$ **dipolar**

Isoelectric point: no net migration in a field

Glycine

The **only** optically inactive natural α -amino acid

R = H, so the α -carbon has **two hydrogens** — not chiral

Protein structure

Primary — sequence, **peptide** bonds

Secondary — α -helix / β -sheet, **hydrogen** bonds

Tertiary — overall folding

Quaternary — assembly of subunits

Denaturation

Heat or pH change breaks the **hydrogen bonds**

2°, 3°, 4° lost; **1° survives**

Boiled egg; curdled milk

Vitamins

Fat soluble: **A, D, E, K**. Water soluble: B complex, C

A night blindness · B₁ beri-beri · B₁₂ pernicious anaemia · C scurvy · D rickets · K slow clotting

DNA vs RNA

deoxyribose vs ribose

T vs **U**

double helix vs single strand

heredity vs protein synthesis

The whole First Term on two pages

The last thing to read. Everything below is drawn from the seven recall sheets; if a line here means nothing to you, go back to that part.

I Solutions

$$M = 1000K_f w_2 / (\Delta T_f w_1) \cdot M = w_2 RT / (\pi V)$$

$$K_f(\text{water}) \ 1.86, K_b \ 0.52, K_f(\text{benzene}) \ 5.12$$

$$i = \text{observed/calculated} = \text{normal } M / \text{observed } M$$

$$\text{Dissociation } \alpha = (i-1)/(n-1) \cdot \text{dimer } \alpha = 2(1-i)$$

$$p = K_H x; \text{big } K_H = \text{low solubility}$$

+ve deviation → minimum-boiling azeotrope (ethanol+water)

Molality is temperature-independent; molarity is not

II Chemical Kinetics

$$k = (2.303/t) \log[a/(a-x)] \cdot t_{1/2} = 0.693/k$$

$$\log(k_2/k_1) = [E_a/19.147][(T_2-T_1)/T_1 T_2]$$

$$\text{Units of } k: \text{mol}^{1-n} \text{L}^n \text{s}^{-1}$$

$$k \propto 1/t_{1/2} \text{ — invert the times}$$

$$t_{50}:t_{75}:t_{87.5} = 1:2:3; t_{90} = 3.32 t_{1/2}$$

Catalyst lowers E_a , not ΔH or K_c

Order experimental; molecularity theoretical, never fractional

III Coordination Compounds

$$\mu = \sqrt{n(n+2)}: 0, 1.73, 2.83, 3.87, 4.90, 5.92$$

d^2sp^3 inner/low spin · sp^3d^2 outer/high spin

dsp^2 square planar · sp^3 tetrahedral

$\Delta_o > P$ low spin; $\Delta_o < P$ high spin

Strong field: CN^- , CO , en , NH_3 . Weak: halides, H_2O

$[Ni(CN)_4]^{2-}$ square planar diamagnetic · $[NiCl_4]^{2-}$ tetrahedral paramagnetic

Ions outside the brackets = conductivity and $AgCl$ count

Colour needs unpaired d electrons + **d-d transition**

MA_5B shows **no** geometrical isomerism

IV Haloalkanes and Haloarenes

S_N2 : 1 step, $\text{rate} = k[RX][Nu]$, $1^\circ > 2^\circ > 3^\circ$, **inversion**

S_N1 : 2 steps, $\text{rate} = k[RX]$, $3^\circ > 2^\circ > 1^\circ$, **racemisation**

$R-I > R-Br > R-Cl > R-F$

aq. $KOH \rightarrow$ alcohol; **alc.** $KOH \rightarrow$ alkene

Silver sends the nitrogen in: $AgCN \rightarrow$ isocyanide, $AgNO_2 \rightarrow$ nitroalkane

Finkelstein $NaI/\text{acetone}$ · Swarts AgF · Hunsdiecker $-1^\circ C$

Wurtz · Wurtz-Fittig · Fittig · Sandmeyer · Gattermann

Halogen on a ring: **-I deactivating, +R o/p directing**

$CHCl_3 \rightarrow$ phosgene in air/light

V Alcohols, Phenols and Ethers

Four handles: $O-H$ · $C-O$ · $\beta C-H$ · carbinol $C-H$

Two faces of phenol: at $-OH$, and on the ring (all o/p)

413 K → ether; 443 K → alkene

Br_2/CS_2 mono; Br_2/H_2O tri (white ppt)

Lucas: 3° instant, $2^\circ \sim 5$ min, 1° on heating

Iodoform needs CH_3CO- or $CH_3CH(OH)-$

Ether+HI: S_N2 small gets I; S_N1 big gets I; aryl → phenol

Phenol dissolves in $NaOH$, **not** $NaHCO_3$

Grignard: $HCHO \rightarrow 1^\circ$, $RCHO \rightarrow 2^\circ$, ketone → 3°

VI Aldehydes, Ketones, Carboxylic Acids

Reactivity $HCHO > RCHO > R_2CO > ArCHO$

2,4-DNP any carbonyl · Tollens' aldehyde · Fehling's **aliphatic** aldehyde

α -H present → aldol; absent → Cannizzaro

Clemmensen (acid) and Wolff-Kishner (base) → CH_2

Rosenmund → aldehyde; $R_2Cd \rightarrow$ ketone

$HCOOH > CH_3COOH$; $CCl_3 > CHCl_2 > CH_2Cl$

Ortho effect: any o-substituent strengthens benzoic acid

Formic acid is the only acid that reduces Tollens'

$-CHO$ and $-COOH$ are **meta** directing

Benzoin (alc. KCN) · Perkin (cinnamic acid) · HVZ (α -Cl acid)

VII Biomolecules

$HI \rightarrow$ hexane · $Br_2/\text{water} \rightarrow$ **monoacid** (aldehyde) · $HNO_3 \rightarrow$ **diacid** ($-CH_2OH$) · penta-acetate (5 OH)

Bromine water tells glucose from fructose

Mutarotation: $\alpha + 111^\circ \rightarrow +52.5^\circ \leftarrow \beta + 19^\circ$

Sucrose non-reducing (both anomeric C used); invert sugar $\approx -20^\circ$

Starch α -1,4 · cellulose β -1,4 · glycogen most branched

Zwitter ion neutral but dipolar; isoelectric point = no migration

Glycine alone is optically inactive (two H on α -C)

1° peptide · 2° H-bond helix/sheet · denaturation loses $2^\circ, 3^\circ, 4^\circ$ only

Fat soluble A D E K; C scurvy, D rickets, K slow clotting

DNA deoxyribose/T/double; RNA ribose/U/single

Endnote

Scope taken from the Heritage School First Term paper, 2024–25, and checked against the ISC 2027 syllabus in Chemistry/Syllabus/. Chemical Kinetics (Part II) and Alcohols, Phenols and Ethers (Part V) are the same text as in *Chemistry-Concepts-Proofs-Problem-Types.pdf*, generated from the same source modules, so the two documents cannot disagree. Every numerical worked example was recomputed independently before release — see `verify_first_term.py`.